

Find the fourier series for $f(x) = e^{-ax}$, $-\pi < x < \pi$ and derive the series from $\frac{\pi}{2 \sinh \pi a}$

Let $f(x) = e^{-ax}$, $-\pi < x < \pi$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

$$a_0 = a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{-a} \right]_{-\pi}^{\pi} = \frac{e^{-a\pi} - e^{a\pi}}{-a}$$

$$a_0 = \frac{1}{a} [e^{a\pi} - e^{-a\pi}]$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$
$$= \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \cos nx dx$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[\frac{e^{-ax}}{a^2+n^2} \left[-a \cos nx + n \sin nx \right] \right]_{-\pi}^{\pi} \\
&= \frac{1}{\pi} \left[\frac{e^{-a\pi}}{a^2+n^2} \left[-a \cos n\pi + n \sin n\pi \right] - \frac{e^{a\pi}}{a^2+n^2} \left[-a \cos(-n\pi) + n \sin(-n\pi) \right] \right] \\
&= \frac{1}{\pi} \left[\frac{e^{-a\pi}}{a^2+n^2} \left[-a(-1)^n + n(0) \right] - \frac{e^{a\pi}}{a^2+n^2} \left[-a(-1)^n + 0 \right] \right] \\
&= \frac{1}{\pi} \left[\frac{e^{-a\pi}}{a^2+n^2} \left[-a(-1)^n \right] + \frac{e^{a\pi}}{a^2+n^2} \left[a(-1)^n \right] \right]
\end{aligned}$$

$$a_n = \frac{a(-1)^n}{\pi(a^2+n^2)} \left[-e^{-a\pi} + e^{a\pi} \right] = \frac{a(-1)^n}{\pi(a^2+n^2)} \left(e^{a\pi} - e^{-a\pi} \right)$$

b_n's $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \sin nx \, dx$

$$\begin{aligned}
&= \frac{1}{\pi} \left[\frac{e^{-ax}}{a^2+n^2} \left[-a \sin nx - n \cos nx \right] \right]_{-\pi}^{\pi} \\
&= \frac{1}{\pi} \left[\frac{e^{-a\pi}}{a^2+n^2} \left[-a \sin n\pi - n \cos n\pi \right] - \frac{e^{a\pi}}{a^2+n^2} \left[-a \sin(-n\pi) - n \cos(-n\pi) \right] \right]
\end{aligned}$$

$$= \frac{1}{\pi} \left[\frac{e^{-a\pi}}{a^2+n^2} \left[-n(-1)^n \right] + \frac{e^{a\pi}}{a^2+n^2} \left[n(-1)^n \right] \right]$$

$$b_n = \frac{n}{\pi} \frac{(-1)^n}{a^2+n^2} \left(e^{a\pi} - e^{-a\pi} \right)$$

Substitute a_0 , a_n , and b_n in $f(x)$.

$$e^{-ax} = \frac{1}{2a\pi} \left[e^{a\pi} - e^{-a\pi} \right] \left[\frac{1}{2a\pi} + \sum_{n=1}^{\infty} \frac{a(-1)^n}{\pi(a^2+n^2)} \left[e^{a\pi} - e^{-a\pi} \right] \cos nx \right. \\ \left. + \sum_{n=1}^{\infty} \frac{n(-1)^n}{\pi(a^2+n^2)} \left[e^{a\pi} - e^{-a\pi} \right] \sin nx \right]$$

$$e^{-ax} = \left[e^{a\pi} - e^{-a\pi} \right] \left[\frac{1}{2a\pi} + \sum_{n=1}^{\infty} \frac{a(-1)^n}{\pi(a^2+n^2)} \cos nx + \sum_{n=1}^{\infty} \frac{n(-1)^n}{\pi(a^2+n^2)} \sin nx \right]$$

$$= 2 \sinh a\pi \left[\frac{1}{2a\pi} + \sum_{n=1}^{\infty} \frac{a(-1)^n}{\pi(a^2+n^2)} \cos nx + \sum_{n=1}^{\infty} \frac{n(-1)^n}{\pi(a^2+n^2)} \sin nx \right]$$

$$\frac{e^{-ax}}{2 \sinh a\pi} = \frac{1}{2a\pi} + \sum_{n=1}^{\infty} \frac{a(-1)^n}{\pi(a^2+n^2)} \cos nx + \sum_{n=1}^{\infty} \frac{n(-1)^n}{\pi(a^2+n^2)} \sin nx$$

Sub $a=1, x=0$

$$\frac{1}{2 \sinh \pi} = \frac{1}{\pi} \left[\frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{(n^2+1)} \cos 0 + \sum_{n=1}^{\infty} \frac{n(-1)^n}{(n^2+1)} \sin 0 \right]$$

$$\Rightarrow \frac{\pi}{2 \sinh \pi} = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2+1}$$

$$= \frac{1}{2} + \frac{-1}{1+1} + \frac{1}{5} + \frac{-1}{10} + \dots$$

4. Find the Fourier Series for $f(x) = x + x^2$, $-\pi < x < \pi$ and hence deduce that

$$1) \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{1}{4} \left[\pi + \frac{2\pi^2}{3} \right]$$

$$2) \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} = \frac{\pi^2}{12}$$

$$f(x) = x + x^2, -\pi < x < \pi$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$a_0: a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) dx$$

$$= \frac{1}{\pi} \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} + \frac{\pi^3}{3} - \left(\frac{\pi^2}{2} + \frac{\pi^3}{3} \right) \right]$$

$$= \frac{1}{\pi} \left[\frac{2\pi^2}{3} \right]$$

$$\boxed{a_0 = \frac{2}{3} \pi^2}$$

$$a_n \text{ or } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \cos nx dx$$

$$= \frac{1}{\pi} \left[(x+x^2) \frac{\sin nx}{n} - \int (1+2x) \frac{\sin nx}{n} dx \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(x+x^2) \frac{\sin nx}{n} - \left[(1+2x) \left[\frac{-\cos nx}{n^2} \right] - \int 2 \left[\frac{-\cos nx}{n^2} \right] dx \right] \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(x+x^2) \frac{\sin nx}{n} + (1+2x) \frac{\cos nx}{n^2} - \frac{2 \sin nx}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\left[(\pi+\pi^2) \frac{\sin n\pi}{n} + (1+2\pi) \frac{\cos n\pi}{n^2} - \frac{2 \sin n\pi}{n^3} \right] - \left[(\pi+\pi^2) \frac{\sin(-n\pi)}{n} - (1-2\pi) \frac{\cos(-n\pi)}{n^2} + \frac{2 \sin(-n\pi)}{n^3} \right] \right]$$

$$= \frac{1}{\pi} \left[(1+2\pi) \frac{(-1)^n}{n^2} - (1-2\pi) \frac{(-1)^n}{n^2} \right]$$

$$a_n = \frac{(-1)^n}{n^2 \pi} \left[\cancel{1+2\pi} - \cancel{1+2\pi} \right]$$

$$= \frac{4\pi(-1)^n}{n^2 \cancel{\pi}}$$

$$\boxed{a_n = \frac{4(-1)^n}{n^2}}$$

$$b_n \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$\Rightarrow b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (x+x^2) \sin nx \, dx$$

$$= \frac{1}{\pi} \left[(x+x^2) \left[-\frac{\cos nx}{n} \right] - \int (1+2x) \left[-\frac{\cos nx}{n} \right] dx \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[-(x+x^2) \frac{\cos nx}{n} + \left[(1+2x) \frac{\sin nx}{n^2} - \int 2 \frac{\sin nx}{n^2} dx \right] \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(-x-x^2) \frac{\cos nx}{n} + (1+2x) \frac{\sin nx}{n^2} + \frac{2 \cos nx}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(-\pi-\pi^2) \frac{\cos n\pi}{n} + (1+2\pi) \frac{\sin n\pi}{n^2} + \frac{2 \cos n\pi}{n^3} - \left[(\pi-\pi^2) \frac{\cos(-n\pi)}{n} + (1-2\pi) \frac{\sin(-n\pi)}{n^2} + \frac{2 \cos(-n\pi)}{n^3} \right] \right]$$

$$= \frac{1}{\pi} \left[(-\pi-\pi^2) \frac{(-1)^n}{n} + \frac{2(-1)^n}{n^3} - (\pi-\pi^2) \frac{(-1)^n}{n} - \frac{2(-1)^n}{n^3} \right]$$

$$= \frac{1}{\pi} \left[(-\pi-\pi^2) \frac{(-1)^n}{n} - (\pi-\pi^2) \frac{(-1)^n}{n} \right]$$

$$= \frac{(-1)^n}{n\pi} \cdot [-\pi - \cancel{\pi^2} - \pi + \cancel{\pi^2}]$$

$$= \frac{-2\pi(-1)^n}{n\pi} \Rightarrow b_n = \frac{-2(-1)^n}{n}$$

Substitute in f(x)

Sub a_0, a_n, b_n in $f(x)$

$$(x+x^2) = \frac{2}{6}\pi^2 + \sum_{n=1}^{\infty} \frac{4(-1)^n}{n^2} \cos nx + \sum_{n=1}^{\infty} \frac{-2(-1)^n}{n} \sin nx$$

$$\begin{aligned} x(x+x^2) &= \frac{\pi^2}{3} + 4 \left[-\frac{1}{1^2} \cos x + \frac{1}{2^2} \cos 2x - \frac{1}{3^2} \cos 3x + \dots \right] \\ &\quad - 2 \left[-\frac{1}{1} \sin x + \frac{1}{2} \sin 2x - \frac{1}{3} \sin 3x + \dots \right] \end{aligned}$$

(i) Put $x = \pi$

$$\pi + \pi^2 - \frac{\pi^2}{3} = 4 \left[-\frac{1}{1^2} \cos \pi + \frac{1}{2^2} \cos 2\pi - \frac{1}{3^2} \cos 3\pi + \dots \right] - 2[0]$$

$$\pi + \frac{2\pi^2}{3} = 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\frac{1}{4} \left[\pi + \frac{2\pi^2}{3} \right] = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

(ii) Put $x = 0$

$$0 = \frac{\pi^2}{3} + 4 \left[\frac{1}{1^2} \cos 0 + \frac{1}{2^2} \cos 2 \cdot 0 - \frac{1}{3^2} \cos 3 \cdot 0 + \dots \right] - 2[0]$$

$$\frac{\pi^2}{3} = 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right]$$

$$\frac{\pi^2}{12} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

* Expand the fourier Series $f(x) = \begin{cases} -4, & -\pi < x < 0 \\ 4, & 0 < x < \pi \end{cases}$

Given $f(x) = \begin{cases} -4, & -\pi < x < 0 \\ 4, & 0 < x < \pi \end{cases}$

The fourier series is defined as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{--- (1)}$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -4 dx + \int_0^{\pi} 4 dx \right]$$

$$= \frac{1}{\pi} \left[\left[-4x \right]_{-\pi}^0 + \left[4x \right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\left[-4(0) + 4(-\pi) \right] + \left[4(\pi) - 4(0) \right] \right]$$

$$= \frac{1}{\pi} \left[-4\pi + 4\pi \right]$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -4 \cos nx dx + \int_0^{\pi} 4 \cos nx dx \right]$$

$$= \frac{1}{\pi} \left[-4 \int_{-\pi}^0 \cos nx \, dx + 4 \int_0^{\pi} \cos nx \, dx \right]$$

$$= \frac{1}{\pi} \left[-4 \left[\frac{\sin nx}{n} \right]_{-\pi}^0 + 4 \left[\frac{\sin nx}{n} \right]_0^{\pi} \right]$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^0 -4 \sin nx \, dx + \int_0^{\pi} 4 \sin nx \, dx \right]$$

$$= \frac{1}{\pi} \left[-4 \left[-\frac{\cos nx}{n} \right]_{-\pi}^0 + 4 \left[-\frac{\cos nx}{n} \right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[4 \left[\frac{\cos nx}{n} \right]_{-\pi}^0 - 4 \left[\frac{\cos nx}{n} \right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[4 \left[\frac{\cos 0}{n} - \frac{\cos(-n\pi)}{n} \right] - 4 \left[\frac{\cos n\pi}{n} - \frac{\cos 0}{n} \right] \right]$$

$$= \frac{1}{\pi} \left[\frac{4}{n} - \frac{(-1)^n 4}{n} - \frac{4(-1)^n}{n} + \frac{4}{n} \right]$$

$$= \frac{1}{\pi} \left[\frac{8}{n} - \frac{8(-1)^n}{n} \right]$$

$$b_n = \frac{8}{n\pi} [1 - (-1)^n]$$

Substitute a_0, a_n, b_n values in Eq (i):

$$f(x) = \frac{0}{2} + \sum_{n=1}^{\infty} 0 \cos nx + \sum_{n=1}^{\infty} \frac{8}{n\pi} [1 - (-1)^n] \sin nx$$

$$f(x) = \sum_{n=1}^{\infty} \frac{8}{n\pi} [1 - (-1)^n] \sin nx$$

$$= \frac{8}{\pi(1)} [1 - (-1)^1] \sin x + \frac{8}{\pi(2)} [1 - (-1)^2] \sin 2x$$

$$+ \frac{8}{3\pi} [1 - (-1)^3] \sin 3x$$

$$= \frac{8}{\pi(1)} [2 \sin x] + \frac{8}{(2)\pi} [0] + \frac{8}{3\pi} [2 \sin 3x] + \dots$$

$$= \frac{8}{\pi} \left[\frac{2}{1} \sin x + \frac{2}{3} \sin 3x + \dots \right]$$

$$= \frac{16}{\pi} \left[\frac{\sin x}{1} + \frac{\sin 3x}{3} + \dots \right]$$

Change of Interval:

The fourier series is defined as in the interval as $[c, c+2L]$ as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{\cos n\pi x}{L} + \sum_{n=1}^{\infty} b_n \frac{\sin n\pi x}{L}$$

where

$$1) a_0 = \frac{1}{L} \int_c^{c+2L} f(x) dx$$

$$2) a_n = \frac{1}{L} \int_c^{c+2L} f(x) \cos \frac{n\pi x}{L} dx$$

$$3) b_n = \frac{1}{L} \int_c^{c+2L} f(x) \sin \frac{n\pi x}{L} dx$$

* Find the fourier series Expansion for the function $f(t) = \{1-t^2, -1 \leq t \leq 1\}$

Given

$$f(t) = \{1-t^2; -1 \leq t \leq 1\}$$

$f(t)$ is defined as

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{\cos n\pi t}{L} + \sum_{n=1}^{\infty} b_n \frac{\sin n\pi t}{L}$$

Here $L=1$

$$a_0 = \frac{1}{L} \int_{-1}^1 f(t) dt$$

$$= \frac{1}{1} \int_{-1}^1 (1-t^2) dt$$

$$= \left[t - \frac{t^3}{3} \right]_{-1}^1$$

$$= \left[1 - \frac{1}{3} + 1 - \frac{1}{3} \right]$$

$$= 2 - \frac{2}{3}$$

$$\boxed{a_0 = \frac{4}{3}}$$

$$a_n = \frac{1}{L} \int_{-1}^1 f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{1}{1} \int_{-1}^1 (1-t^2) \cos n\pi t dt$$

$$= \left[(1-t) \frac{\sin n\pi t}{n\pi} - \int (-2t) \frac{\sin n\pi t}{n\pi} dt \right]_{-1}^1$$

$$= \left[(1-t^2) \frac{\sin n\pi t}{n\pi} + \left[2t \left[\frac{-\cos n\pi t}{n^2 \pi^2} \right] - \int 2 \left[\frac{-\cos n\pi t}{n^2 \pi^2} \right] dt \right] \right]_{-1}^1$$

$$= \left[(1-t^2) \frac{\sin n\pi t}{n\pi} - 2t \frac{\cos n\pi t}{n^2 \pi^2} + 2 \frac{\sin n\pi t}{n^3 \pi^3} \right]_{-1}^1$$

$$= \left[0 - \frac{2(-1)^n}{n^2 \pi^2} + 0 - \left(0 - 2(-1) \frac{\cos(-n\pi)}{n^2 \pi^2} + \frac{2 \sin(-n\pi)}{n^3 \pi^3} \right) \right]$$

$$a_n = \left[\frac{-2(-1)^n}{n^2 \pi^2} - \frac{2(-1)^n}{n^2 \pi^2} \right]$$

$$a_n = \frac{-4(-1)^n}{n^2 \pi^2}$$

$$b_n = \frac{1}{L} \int_{-1}^1 (1-t^2) \sin \frac{n\pi}{L} t dt$$

$$= \frac{1}{1} \int_{-1}^1 (1-t^2) \sin n\pi t dt$$

$$= \left[(1-t^2) \left[\frac{-\cos n\pi t}{n\pi} \right] - \int (-2t) \left[\frac{-\cos n\pi t}{n\pi} \right] dt \right]_{-1}^1$$

$$= \left[(t^2-1) \frac{\cos n\pi t}{n\pi} - \int 2t \frac{\cos n\pi t}{n\pi} dt \right]_{-1}^1$$

$$\Rightarrow (t^2-1) \frac{\cos n\pi t}{n\pi} - \left[2t \frac{\sin n\pi t}{n^2 \pi^2} - \int 2 \frac{\sin n\pi t}{n^2 \pi^2} dt \right]_{-1}^1$$

$$\Rightarrow \left[(t^2-1) \frac{\cos n\pi t}{n\pi} - 2t \frac{\sin n\pi t}{n^2 \pi^2} - \frac{2 \cos n\pi t}{n^3 \pi^3} \right]_{-1}^1$$

$$b_n = \left[\frac{-2(-1)^n}{n^3 \pi^3} + \frac{2(-1)^{n+1}}{n^3 \pi^3} \right]$$

$$b_n = 0$$

$$\therefore 1 - t^2 = \frac{4}{6} + \sum_{n=1}^{\infty}$$

* Find the Fourier series ~~for~~ :

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 2, & 1 < x < 3 \end{cases}$$

Given: $f(x) = \begin{cases} 1, & 0 < x < 1 \\ 2, & 1 < x < 3 \end{cases}$

$$2L = 3.$$

$$\boxed{L = 3/2}$$

The Fourier series is defined as.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$a_0: a_0 = \frac{1}{L} \int_0^3 f(x) dx$$

$$= \frac{1}{3/2} \left[\int_0^1 1 dx + \int_1^3 2 dx \right]$$

$$= \frac{2}{3} \left[(x)_0^1 + (2x)_1^3 \right]$$

$$= \frac{1}{2} \left[[1-0] + [6-2] \right]$$

$$\boxed{a_0 = \frac{10}{3}}$$

$$a_n = \frac{1}{L} \int_0^3 f(x) \cos \frac{n\pi}{L} x dx$$

$$= \frac{1}{3/2} \left[\int_0^1 \cos \frac{n\pi}{3/2} x dx + \int_1^3 2 \cdot \frac{\cos \frac{n\pi}{3/2} x dx}{3/2} \right]$$

$$= \frac{2}{3} \left[\int_0^1 \cos \frac{2n\pi}{3} x dx + 2 \int_1^3 \cos \frac{2n\pi}{3} x dx \right]$$

$$= \frac{2}{3} \left[\left[\frac{\sin \frac{2n\pi}{3} x}{\frac{2n\pi}{3}} \right]_0^1 + 2 \left[\frac{\sin \frac{2n\pi}{3} x}{\frac{2n\pi}{3}} \right]_1^3 \right]$$

$$= \frac{2}{3} \left[\left[\frac{\sin \frac{2n\pi}{3}}{\frac{2n\pi}{3}} - \frac{\sin 0}{\frac{2n\pi}{3}} \right] + 2 \left[\frac{\sin 2n\pi}{\frac{2n\pi}{3}} - \frac{\sin \frac{2n\pi}{3}}{\frac{2n\pi}{3}} \right] \right]$$

$$= \frac{2}{3} \left[\frac{\sin \frac{2n\pi}{3}}{\frac{2n\pi}{3}} - \frac{2 \sin \frac{2n\pi}{3}}{\frac{2n\pi}{3}} \right]$$

$$a_n = \frac{2}{3} \left[\frac{-\sin \frac{2n\pi}{3}}{\frac{2n\pi}{3}} \right]$$

$$b_n = \frac{1}{3/2} \int_0^3 f(x) \sin \frac{n\pi}{3/2} x dx.$$

$$= \frac{2}{3} \left[\int_0^1 1 \cdot \frac{\sin 2n\pi}{3} x dx + \int_1^3 2 \cdot \sin \frac{2n\pi}{3} x dx \right]$$

$$= \frac{2}{3} \left[\left[-\frac{\cos \frac{2n\pi}{3} x}{2n\pi/3} \right]_0^1 + 2 \left[-\frac{\cos \frac{2n\pi}{3} x}{2n\pi/3} \right]_1^3 \right]$$

$$= \frac{2}{3} \left[\left(\frac{-\cos \frac{2n\pi}{3}}{2n\pi/3} + \frac{\cos 0}{2n\pi/3} \right) - 2 \left(\frac{\cos 2n\pi}{2n\pi/3} - \frac{\cos \frac{2n\pi}{3}}{2n\pi/3} \right) \right]$$

$$\Rightarrow \frac{2}{3} \left[\frac{-\cos \frac{2n\pi}{3}}{\frac{2n\pi}{3}} + \frac{3}{2n\pi} - \frac{2 \cdot 3}{2n\pi} + 2 \frac{\cos \frac{2n\pi}{3}}{\frac{2n\pi}{3}} \right]$$

$$= \frac{2}{3} \left[\frac{\cos \frac{2n\pi}{3}}{\frac{2n\pi}{3}} - \frac{3}{2n\pi} \right]$$

Sub a_0, a_n, b_n in $f(x)$

$$f(x) = \frac{10}{6} + \sum_{n=1}^{\infty} \frac{2}{3} \left[\frac{-\sin \frac{2n\pi}{3}}{2n\pi/3} \right] \cos \frac{2n\pi}{3} x$$

$$+ \sum_{n=1}^{\infty} \frac{2}{3} \left[\frac{\cos \frac{2n\pi}{3}}{\frac{2n\pi}{3}} - \frac{3}{2n\pi} \right] \sin \frac{2n\pi}{3} x$$

Fourier Series for Even and odd functions

A function $f(x)$ is said to be Even and

if $\boxed{f(-x) = f(x)}$

A function $f(x)$ is said to be odd

if $\boxed{f(-x) = -f(x)}$

* $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$; if $f(x)$ is Even

* $\int_{-a}^a f(x) dx = 0$; if $f(x)$ is odd.

Fourier Series for Even function in $[-\pi, \pi]$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

In the interval $[-l, l]$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{\cos nx}{L} x$$

$$a_0 = \frac{2}{L} \int_0^L f(x) dx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

Fourier Series for Odd function in $[-\pi, \pi]$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

In the interval $[-l, l]$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

* Find the Fourier series Expansion of :

$f(x) = x \sin x, -\pi < x < \pi$. Hence reduce that

$$\frac{1}{1 \times 3} - \frac{1}{3 \times 5} + \frac{1}{5 \times 7} - \frac{1}{7 \times 9} + \dots = \frac{1}{4} [\pi - 2]$$

Ex Given $f(x) = x \sin x, -\pi < x < \pi$

$$f(-x) = (-x) \sin(-x)$$

$$= -x - \sin x$$

$$= x \sin x$$

$$= f(x)$$

$\therefore f(-x) = f(x), f(x)$ is Even function

Then the Fourier Series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} x \sin x dx$$

$$= \frac{2}{\pi} \left[-x \cos x - \int 1 - \cos x dx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-x \cos x + \sin x \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-\pi \cos \pi + \sin \pi + 0 - \sin 0 \right]$$

$$\cos(1+n)\pi = (-1)^{1+n}$$

$$\cos(1-n)\pi = (-1)^{1-n}$$

$$\sin(1+n)\pi = 0$$

$$\sin(1-n)\pi = 0$$

$$= \frac{2}{\pi} \left[-\pi(-1) + 0 \right]$$

$$a_0 = 2$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \sin x \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x 2 \sin x \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \left[\sin(1+n)x + \sin(1-n)x \right] dx$$

$$= \frac{1}{\pi} \left[\int_0^{\pi} x \sin(1+n)x \, dx + \int_0^{\pi} x \sin(1-n)x \, dx \right]$$

$$= \frac{1}{\pi} \left[\left(-x \frac{\cos(1+n)x}{1+n} - \int 1 \cdot \frac{-\cos(1+n)x}{1+n} \, dx \right) \Big|_0^{\pi} + \left(-x \frac{\cos(1-n)x}{1-n} - \int 1 \cdot \frac{-\cos(1-n)x}{1-n} \, dx \right) \Big|_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\left(\frac{-x \cos(1+n)x}{1+n} + \frac{\sin(1+n)x}{(1+n)^2} \right) \Big|_0^{\pi} + \left(\frac{-x \cos(1-n)x}{1-n} + \frac{\sin(1-n)x}{(1-n)^2} \right) \Big|_0^{\pi} \right]$$

$$\Rightarrow \frac{1}{\pi} \left[\frac{-\pi \cos(1+n)\pi}{1+n} + \frac{\sin(1+n)\pi}{(1+n)^2} + 0 - \frac{\sin(1+n)0}{1+n^2} + \left[\frac{-\pi \cos(1-n)\pi}{1-n} + \frac{\sin(1-n)\pi}{(1-n)^2} \right] - 0 \left[0 + \frac{\sin(1-n)0}{(1-n)^2} \right] \right]$$

$$\Rightarrow \frac{1}{\pi} \left[\frac{-\pi (-1)^{1+n}}{1+n} - \frac{\pi (-1)^{1-n}}{1-n} \right]$$

$$a_n = \frac{-\pi}{\pi} \left[\frac{(-1)^{1+n}}{1+n} + \frac{(-1)^{1-n}}{1-n} \right]$$

$$a_n = (-1) \left[\frac{(-1)^{1+n}}{1+n} + \frac{(-1)^{1-n}}{1-n} \right]$$

$$a_n = -1 \left[\frac{(1-n)(-1)^{1+n} + (-1)^{1-n}(1+n)}{1^2 - n^2} \right]$$

$$a_n = \left[\frac{(-1)^{1+n} - n(-1)^{1+n} + (-1)^{1-n} + (-1)^{1-n}n}{1^2 - n^2} \right]$$

Sub a_0, a_n in sub in $f(x)$.

$$x \sin x = \frac{2}{2} + \sum_{n=1}^{\infty} \left[\frac{(-1)^{1+n} - n(-1)^{1+n} + (-1)^{1-n} + (-1)^{1-n}n}{1^2 - n^2} \right] \cos nx$$

$$= 1 + \sum_{n=1}^{\infty} \left[\frac{(-1)^{1+n} - n(-1)^{1+n} + (-1)^{1-n} + (-1)^{1-n}n}{1^2 - n^2} \right] \cos nx$$

$$= 1 + \sum_{n=1}^{\infty} \left[\frac{(-1)^1(-1)^n - n(-1)^1(-1)^n + (-1)^1(-1)^n + (-1)^1(-1)^nn}{1^2 - n^2} \right] \cos nx$$

$$\Rightarrow 1 + \sum_{n=1}^{\infty} \left[\frac{(-1)^1 - n(-1)^n + (-1)^{-n} + n(-1)^{-n}}{1^2 - n^2} \right] \cos nx$$

for $n=1$; The value of a_1 is undefined so the series starts from $n=2$ to ∞ .

Now calculate a_1

$$a_1 = \frac{2}{\pi} \int_0^{\pi} f(x) \cos x dx.$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin x \cos x dx.$$

$$= \frac{1}{\pi} \int_0^{\pi} x (2 \sin x \cos x) dx$$

$$= \frac{1}{\pi} \left[x \frac{-\cos 2x}{2} - \int 1 \cdot \frac{-\cos 2x}{2} dx \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[-x \frac{\cos 2x}{2} + \frac{\sin 2x}{4} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left[-\pi \frac{\cos 2\pi}{2} + \frac{\sin 2\pi}{4} + 0 \right]$$

$$= \frac{1}{\pi} \left[-\frac{\pi(1)}{2} + 0 \right]$$

$$\boxed{a_1 = -\frac{1}{2}}$$

$$\therefore x \sin x = 1 + \frac{1}{2} \cos x + \sum_{n=1}^{\infty} \left[\frac{(-1)^n - n(-1)^n + (-1)^{-n} + n(-1)^{-n}}{(1-n)(1+n)} \right]$$

$\cos nx$

$$= 1 - \frac{1}{2} \cos x + \sum_{n=2}^{\infty} \frac{(1-n)(-1)^n + (1+n)(-1)^{-n}}{(1-n)(1+n)} \cos nx$$

$$= 1 - \frac{1}{2} \cos x + \left(\frac{-2}{1 \cdot 3} \right) \cos 2x + \frac{1}{4} \cos 3x + \frac{2}{-3 \cdot 5} \cos 4x + \dots$$

sin x

$$= 1 - \frac{1}{2} \cos x - \frac{2}{1 \cdot 3} \cos 2x + \frac{1}{4} \cos 3x - \frac{2}{3 \cdot 5} \cos 4x + \dots$$

Put $x = \pi/2$

$$\frac{\pi}{2} \sin \frac{\pi}{2} = 1 - \frac{1}{2} \cos \frac{\pi}{2} - \frac{2}{1 \cdot 3} \cos \pi + \frac{1}{4} \cos \frac{3\pi}{2} - \frac{2}{3 \cdot 5} \cos 2\pi + \dots$$

$$\frac{\pi}{2} (1) = 1 - \frac{1}{2} (0) - \frac{2}{1 \cdot 3} (-1) + \frac{1}{4} (0) - \frac{2}{3 \cdot 5} (1) + \dots$$

$$\frac{\pi}{2} - 1 = \frac{2}{1 \cdot 3} - \frac{2}{3 \cdot 5} + \dots$$

$$\frac{1}{2} \left(\frac{\pi}{2} - 1 \right) = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \dots$$

$$\frac{1}{4} [\pi - 2] = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \dots$$

* Find the Fourier Series for the function

$$f(x) = \begin{cases} x, & -\pi < x < 0 \\ -x, & 0 < x < \pi \end{cases} \quad \text{Hence deduce that}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$$

Let: $f(x) = x$

$$f(-x) = -x$$

$$= f(x)$$

$f(x)$ is Even function

$$f(x) = -x$$

$$f(-x) = -(-x)$$

$$= x$$

$$= f(x)$$

$f(x)$ is an Even function.

The Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (-x) dx \Rightarrow \frac{2}{\pi} \left[-\frac{x^2}{2} \right]_0^{\pi}$$

$$\Rightarrow -\frac{1}{\pi} [\pi^2 - 0]$$

$$\boxed{a_0 = -\pi}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \Rightarrow \frac{2}{\pi} \int_0^{\pi} -x \cos nx dx$$

$$\Rightarrow -\frac{2}{\pi} \int_0^{\pi} x \cos nx dx$$

$$= \frac{-2}{\pi} \left[x \cdot \frac{\sin nx}{n} - \int 1 \cdot \frac{\sin nx}{n} dx \right]_0^{\pi}$$

$$= \frac{-2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$= \frac{-2}{\pi} \left[\frac{\pi \sin n\pi}{n} + \frac{\cos n\pi}{n^2} - 0 - \frac{\cos 0}{n^2} \right]$$

$$= \frac{-2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right]$$

$$= \frac{-2}{\pi} \left[\frac{(-1)^n - 1}{n^2} \right]$$

$$= \frac{2[1 - (-1)^n]}{\pi n^2}$$

$$\therefore f(x) = \frac{-\pi}{2} + \sum_{n=1}^{\infty} \frac{2[1 - (-1)^n]}{\pi n^2} \cos nx$$

$$f(x) = \frac{-\pi}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2} \cos nx$$

$$= \frac{-\pi}{2} + \frac{2}{\pi} \left[\frac{2}{1} \cos x + 0 + \frac{2}{3} \cos 3x + 0 + \frac{2}{5} \cos 5x + \dots \right]$$

$$f(x) = \frac{-\pi}{2} + \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$$

$$x=0$$

$$0 = -\frac{\pi}{2} + \frac{4}{\pi} \left[\frac{\cos 0}{1^2} + \frac{\cos 0}{3^2} + \frac{\cos 0}{5^2} + \dots \right]$$

$$= \frac{\pi}{2} \cdot \frac{\pi}{4} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$\Rightarrow \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

* Expand the function $f(x) = x^3$ as Fourier series $-\pi < x < \pi$.

$$\text{Let } f(x) = x^3$$

$$f(-x) = (-x)^3$$

$$= -x^3$$

$$= -f(x).$$

$f(x)$ is odd function.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx.$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^3 \sin nx$$

$$= \frac{2}{\pi} \left[-x^3 \frac{\cos nx}{n} + \int \frac{3x^2}{n} \cos nx dx \right]_0^{\pi}$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} x^3 \sin nx \, dx$$

$$b_n = \frac{2}{\pi} \left[x^3 \int \sin nx \, dx - \int 3x^2 \int \sin nx \, dx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[x^3 \left[-\frac{\cos nx}{n} \right] + \int 3x^2 \frac{\cos nx}{n} \, dx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-x^3 \frac{\cos nx}{n} + \int 3x^2 \frac{\cos nx}{n} \, dx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[-x^3 \frac{\cos nx}{n} + \left[3x^2 \frac{\sin nx}{n} - \int 6x \frac{\sin nx}{n^2} \, dx \right]_0^{\pi} \right]$$

$$= \frac{2}{\pi} \left[-x^3 \frac{\cos nx}{n} + 3x^2 \frac{\sin nx}{n} + \left[6x \frac{\cos nx}{n^2} + \int 6 \frac{\cos nx}{n^2} \, dx \right]_0^{\pi} \right]$$

$$= \frac{2}{\pi} \left[-x^3 \frac{\cos nx}{n} + 3x^2 \frac{\sin nx}{n} + 6x \frac{\cos nx}{n^2} - 6 \frac{\sin nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{-\pi^3 \cos n\pi}{n} + 0 + \frac{6\pi \cos n\pi}{n^2} - 0 - \left[-0 + 0 + 0 - 6(0) \right] \right]$$

$$\Rightarrow \frac{2}{\pi} \left[\frac{-\pi^3 (-1)^n}{n} + \frac{6\pi}{n^2} (-1)^n \right]$$

$$\Rightarrow \frac{-2\pi^2 (-1)^n}{n} + \frac{12}{n^2} (-1)^n$$

$$b_n = -2(-1)^n \left[\frac{\pi^2}{n} - \frac{6}{n^2} \right]$$

$$f(x) = \sum_{n=1}^{\infty} -2(-1)^n \left[\frac{\pi^2}{n} - \frac{6}{n^2} \right] \sin nx$$

$$\Rightarrow \frac{2}{\pi} \left[-x^3 \frac{\cos nx}{n} + 3x^2 \frac{\sin nx}{n} + 6x \frac{\cos nx}{n^2} - 6 \frac{\sin nx}{n^3} \right]_0^{\pi}$$

Half-Range Fourier Series:

Half-Range Fourier series $[0, \pi]$.

Half Range Cosine Series

$$* f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where

$$* a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

where

$$* a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

Half-Range Sine Series:

$$* f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$* b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

Half-Range Fourier series in $[0, l]$

Half Range cosine Series.

$$* f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{\cos n\pi x}{L}$$

where

$$* a_0 = \frac{2}{L} \int_0^L f(x) dx$$

where

$$* a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

Half-Range Sine Series:

$$* f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

where

$$* b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

~~Q.1~~

* Obtain the half range Cosine ~~Sine~~ Series

for $f(x) = x$, $0 < x < 2$

Given $f(x) = x$, $0 < x < 2$.

Here $L = 2$.

Half Range cosine Series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}$$

$$a_0 = \frac{2}{L} \int_0^L f(x) dx$$

$$= \frac{2}{2} \int_0^2 x dx \Rightarrow \left[\frac{x^2}{2} \right]_0^2 \Rightarrow \left[\frac{4}{2} - 0 \right] = 2$$

$$\boxed{a_0 = 2}$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

$$= \frac{2}{2} \int_0^2 x \cos \frac{n\pi x}{2} dx$$

$$= \int_0^2 \frac{\cos \frac{n\pi}{2} x}{\frac{n\pi}{2}} dx$$

$$= \left[x \frac{\sin \frac{n\pi}{2} x}{\frac{n\pi}{2}} - \int 1 \cdot \frac{\sin \frac{n\pi}{2} x}{\frac{n\pi}{2}} dx \right]_0^2$$

$$= \left[x \cdot \frac{\sin \frac{n\pi}{2} x}{\frac{n\pi}{2}} + \frac{\cos \frac{n\pi}{2} x}{\frac{n^2 \pi^2}{4}} \right]_0^2$$

$$= \left[2 \frac{\sin n\pi}{\frac{n\pi}{2}} + \frac{\cos n\pi}{\frac{n^2 \pi^2}{4}} - 0 - \frac{\cos 0}{\frac{n^2 \pi^2}{4}} \right]$$

$$a_n = \frac{4(-1)^n}{n^2 \pi^2} - \frac{4}{n^2 \pi^2}$$

$$a_n = \frac{4}{n^2 \pi^2} [(-1)^n - 1]$$

Sub a_0, a_n in $f(x)$

$$f(x) = \frac{2}{2} + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [(-1)^n - 1] \cos \frac{n\pi}{2} x$$

$$= 1 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [(-1)^n - 1] \cos \frac{n\pi}{2} x$$

* Find the half range sine series for

$$f(x) = ax + b, \quad 0 < x < 1$$

Given, $f(x) = ax + b, \quad 0 < x < 1.$

$$\boxed{L = 1}$$

Half Range Sine Series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

$$= 2 \int_0^1 (ax + b) \sin \frac{n\pi x}{L} dx$$

$$= 2 \left[(ax + b) \left[\frac{-\cos n\pi x}{n\pi} \right] - \int a \cdot \left[\frac{-\cos n\pi x}{n\pi} \right] dx \right]_0^1$$

$$= 2 \left[-(ax + b) \frac{\cos n\pi x}{n\pi} + a \frac{\sin n\pi x}{n^2 \pi^2} \right]_0^1$$

$$= 2 \left[-(a+b) \frac{\cos n\pi}{n\pi} + a \frac{\sin n\pi}{n^2 \pi^2} + b \frac{\cos 0}{n\pi} - \frac{a \sin 0}{n^2 \pi^2} \right]$$

$$= 2 \left[-(a+b) \frac{(-1)^n}{n\pi} + \frac{b}{n\pi} \right]$$

~~$$b_n = \frac{2}{n\pi} \left[(a+b) \frac{(-1)^n}{n\pi} + \frac{b}{n\pi} \right]$$~~

$$b_n = \frac{2}{n\pi} \left[b - (a+b)(-1)^n \right]$$

Sub b_n in $f(x)$

$$f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left[b - (a+b)(-1)^n \right] \sin n\pi x$$

Obtain the half range sine series $f(x) = e^x$; $0 < x < \pi$

* Find half range cosine series is defined by

$$f(x) = \pi x - x^4, \quad 0 < x < \pi$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi x - x^4) dx$$

$$= \frac{2}{\pi} \left[\frac{\pi x^2}{2} - \frac{x^5}{5} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[\frac{\pi^3}{2} - \frac{\pi^5}{5} \right]$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} (\pi x - x^4) \cos nx dx$$

$$= \frac{2}{\pi} \left[(\pi x - x^4) \frac{\sin nx}{n} - \int (\pi - 4x^3) \frac{\sin nx}{n} dx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(\pi x - x^4) \frac{\sin nx}{n} - \left[(\pi - 4x^3) \left(-\frac{\cos nx}{n^2} \right) - \int -12x^2 \left(-\frac{\cos nx}{n^3} \right) dx \right]_0^{\pi} \right]$$

$$= \frac{2}{\pi} \left[(\pi x - x^4) \frac{\sin nx}{n} + (\pi - 4x^3) \frac{\cos nx}{n^2} + \int 12x^2 \frac{\cos nx}{n^3} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(\pi x - x^4) \frac{\sin nx}{n} + (\pi - 4x^3) \frac{\cos nx}{n^2} + \left[12x^2 \frac{\sin nx}{n^3} - \int 24x \frac{\sin nx}{n^3} dx \right] \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(\pi x - x^4) \frac{\sin nx}{n} + (\pi - 4x^3) \frac{\cos nx}{n^2} + 12x^2 \frac{\sin nx}{n^3} + 24x \frac{\cos nx}{n^4} - \int 24 \left[\frac{-\cos nx}{n^4} \right] dx \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(\pi x - x^4) \frac{\sin nx}{n} + (\pi - 4x^3) \frac{\cos nx}{n^2} + 12x^2 \frac{\sin nx}{n^3} + 24x \frac{\cos nx}{n^4} + 24 \frac{\sin nx}{n^5} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[(\pi^2 - \pi^4) \frac{\sin n\pi}{n} + (\pi - 4\pi^3) \frac{\cos n\pi}{n^2} + 12\pi^2 \frac{\sin n\pi}{n^3} + 24\pi \frac{\cos n\pi}{n^4} - 24 \frac{\sin n\pi}{n^5} - 0 - \frac{\pi \cos 0}{n^2} - 0 - 0 + 24 \frac{\sin 0}{n^5} \right]$$

$$a_n = \frac{2}{\pi} \left[\frac{\pi - 4\pi^3(-1)^n}{n^2} + 24\pi \frac{(-1)^n}{n^4} - \frac{\pi}{n^2} \right]$$

Sub a_0, a_n in $f(x)$

$$(\pi x - x^4) = \frac{2/\pi \left[\frac{\pi^3}{2} - \frac{\pi^5}{5} \right]}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi} \left[\frac{\pi - 4\pi^3(-1)^n}{n^2} + 24\pi \frac{(-1)^n}{n^4} - \frac{\pi}{n^2} \right] \cos nx$$

* Obtain the half range sine series $f(x) = e^x$;

$$0 < x < 1;$$

Given; $f(x) = e^x$; $0 < x < 1$

Half range sine series {here $L=1$ }

$$f(x) = \sum_{n=0}^{\infty} b_n \sin \frac{n\pi}{L} x$$

$$b_n = \frac{2}{L} \int_0^1 f(x) \sin \frac{n\pi}{L} x dx$$

$$= \frac{2}{1} \int_0^1 e^x \sin \frac{n\pi}{1} x dx$$

$$= 2 \int_0^1 e^x \sin n\pi x dx$$

$$= 2 \left[\frac{e^x}{a^2 + b^2} (a \sin bx - b \cos bx) \right]_0^1$$

$$= 2 \left[\frac{e^x}{1 + n^2 \pi^2} (\sin n\pi x - n\pi \cos n\pi x) \right]_0^1$$

$$= 2 \left[\frac{e^1}{1 + n^2 \pi^2} (\sin n\pi - n\pi \cos n\pi) \right] - \frac{e^0}{1 + n^2 \pi^2} (\sin 0 - n\pi \cos 0)$$

$$= 2 \left[\frac{e}{1 + n^2 \pi^2} (-n\pi (-1)^n) - \frac{1}{1 + n^2 \pi^2} (-n\pi) \right]$$

$$= \frac{2}{1 + n^2 \pi^2} [-n\pi (-1)^n e + n\pi]$$

$$\sin n\pi = 0, \sin 0 = 0$$

$$\cos n\pi = (-1)^n; \cos 0 = 1$$

$$\int e^{ax} \sin bx =$$

$$\frac{e^x}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$= \frac{2n\pi}{1+n^2\pi^2} \left[-e(-1)^n + 1 \right]$$

$$b_n = \frac{2n\pi}{1+n^2\pi^2} \left[1 - e(-1)^n \right]$$

* Now Sub b_n in $f(x)$

$$e^x = \sum_{n=0}^{\infty} \frac{2n\pi}{1+n^2\pi^2} \left[1 - e(-1)^n \right] \sin n\pi x.$$

UNIT - III

Z-Transform

* If the function u_n is defined for $n \geq 0$, then its Z-transform is defined as $Z(u_n)$.

$$* Z(u_n) = U(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

* The inverse of Z-transform is defined as

$$Z^{-1}[U(z)] = u_n$$

Some Standard Reasons:

$$1) Z(a^n) = \frac{z}{z-a}$$

$$2) Z(n^p) = -z \frac{d}{dz} \left\{ Z(n^{p-1}) \right\}$$

$$3) Z(1) = \frac{z}{z-1}$$

$$4) Z(n) = \frac{z}{(z-1)^2}$$

$$5) Z(n^2) = \frac{z^2+z}{(z-1)^3}$$

$$6) Z(n^3) = \frac{z^3+4z^2+z}{(z-1)^4}$$

$$7) Z(n^4) = \frac{z^4+11z^3+11z^2+z}{(z-1)^5}$$

$$\star \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

$$\star \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\star \frac{1}{1-x} = (1-x)^{-1} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$$

$$\star \frac{1}{1+x} = (1+x)^{-1} = \sum_{n=0}^{\infty} (-x)^n = 1 - x + x^2 - x^3 + \dots$$

$$\star \frac{1}{(1-x)^2} = (1-x)^{-2} = \sum_{n=0}^{\infty} (n+1)x^n = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$\star \frac{1}{(1+x)^2} = (1+x)^{-2} = \sum_{n=0}^{\infty} (n+1)(-x)^n = 1 - 2x + 3x^2 - 4x^3 + \dots$$

$$\star e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\star e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

Linearity property

If a, b, c are constants and u_n, v_n, w_n are discrete function then

$$Z[au_n + bv_n - cw_n] = aZ[u_n] + bZ[v_n] - cZ[w_n]$$

Standard Sequences:

1) If $u_n = 1$; for all 'n', then $Z[u_n] = ?$

Sol Given

$$u_n = 1$$

$$Z[u_n] = U(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$Z[1] = \sum_{n=0}^{\infty} 1 \cdot z^{-n}$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots$$

$$= 1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 + \dots$$

$$Z[1] = \frac{1}{1 - \frac{1}{z}} = \frac{z}{z-1}$$

Hence proved

2) If $u_n = C$; C is a constant for every 'n', then

$$Z(u_n) = U(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$Z(C) = \sum_{n=0}^{\infty} C z^{-n}$$

$$= C \cdot \sum_{n=0}^{\infty} z^{-n}$$

$$= C \cdot \frac{z}{z-1}$$

$$Z(C) = \frac{Cz}{z-1}$$

3) If $u_n = a^n$; $n \geq 0$ then

$$Z(u_n) = U(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$Z(a^n) = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= 1 + \frac{a}{z} + \left(\frac{a}{z}\right)^2 + \left(\frac{a}{z}\right)^3 + \dots$$

$$= \frac{1}{1 - \frac{a}{z}} = \frac{z}{z-a}$$

$$Z(a^n) = \frac{z}{(z-a)}$$

4) If $u_n = (-1)^n$, $n \geq 0$ then

$$Z(u_n) = U(z) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$Z(-1)^n = \sum_{n=0}^{\infty} (-1)^n z^{-n}$$

$$= 1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots$$

$$= 1 - \frac{1}{z} + \left(\frac{1}{z}\right)^2 - \left(\frac{1}{z}\right)^3 + \dots$$

$$= \frac{1}{1 + \frac{1}{z}} = \frac{z}{z+1}$$

$$z[(-1)^n] = \frac{z}{z+1}$$

5) If $U_n = \frac{1}{n!}$ then

$$z[U_n] = U(z) = \sum_{n=0}^{\infty} U_n z^{-n}$$

$$z\left[\frac{1}{n!}\right] = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$$

$$= 1 + \frac{1}{1!} \frac{1}{z} + \frac{1}{2!} \frac{1}{z^2} + \frac{1}{3!} \frac{1}{z^3} + \dots$$

$$= 1 + \frac{(1/z)}{1!} + \frac{(1/z)^2}{2!} + \frac{(1/z)^3}{3!} + \dots$$

$$\therefore z\left[\frac{1}{n!}\right] = e^{1/z}$$

6)

If $U_n = \frac{1}{n+1}$ then $z\left(\frac{1}{n+1}\right) = -z \log\left(\frac{z-1}{z}\right)$

$$z[U_n] = U(z) = \sum_{n=0}^{\infty} U_n z^{-n}$$

$$z\left[\frac{1}{n+1}\right] = \sum_{n=0}^{\infty} \left[\frac{1}{n+1}\right] z^{-n}$$

$$= 1 + \frac{1}{2} z^{-1} + \frac{1}{3} z^{-2} + \frac{1}{4} z^{-3} + \dots$$

$$= 1 + \frac{1/z}{2} + \frac{(1/z)^2}{3} + \frac{(1/z)^3}{4} + \dots$$

$$= z \left[\frac{1}{z} + \frac{(1/z)^2}{2} + \frac{(1/z)^3}{3} + \dots \right]$$

$$= z \left[-\log\left(1 - \frac{1}{z}\right) \right]$$

$$z\left[\frac{1}{n+1}\right] = -z \log\left(\frac{z-1}{z}\right)$$

If $U_n = n^p$ then $z[n^p] = -z \frac{d}{dz} [z(n^{p-1})]$.

$$z[U_n] = U[z] = \sum_{n=0}^{\infty} U_n z^{-n}$$

$$z[n^p] = \sum_{n=0}^{\infty} n^p z^{-n} \quad \text{--- (1)}$$

$$z[n^{p-1}] = \sum_{n=0}^{\infty} n^{p-1} z^{-n} \quad \text{--- (2)}$$

differentiate (2) w.r.t 'z' on both sides.

$$\frac{d}{dz} [z(n^{p-1})] = \frac{d}{dz} \left[\sum_{n=0}^{\infty} n^{p-1} z^{-n} \right]$$

$$= \sum_{n=0}^{\infty} n^{p-1} (-n) z^{-n-1}$$

$$= - \sum_{n=0}^{\infty} n^{p-1} n \frac{z^{-n}}{z}$$

$$= - \frac{1}{z} \sum_{n=0}^{\infty} n^p z^{-n}$$

$$= -\frac{1}{z} z[n^p]$$

$$\therefore z[n^p] = -z \frac{d}{dz} [z[n^{p-1}]]$$

If $p=1$

$$z[n^1] = -z \frac{d}{dz} [z(1)]$$

$$= -z \frac{d}{dz} \left[\frac{z}{z-1} \right]$$

$$d \frac{u}{v} = \frac{v u' - u v'}{v^2}$$

$$z[n^1] = \frac{z}{(z-1)^2}$$

If $p=2$

$$z[n^2] = -z \frac{d}{dz} [z(n)]$$

$$= -z \frac{d}{dz} \left[\frac{z}{(z-1)^2} \right]$$

$$\frac{z^2 + z}{(z-1)^3}$$

$$* \mathcal{Z}[na^n]$$

Here $U_n = na^n$.

$$\mathcal{Z}[U_n] = \sum_{n=0}^{\infty} U_n z^{-n}$$

$$\mathcal{Z}[na^n] = \sum_{n=0}^{\infty} na^n z^{-n}$$

$$= \sum_{n=0}^{\infty} n \left(\frac{a}{z}\right)^n$$

$$= 0 + 1 \cdot \left(\frac{a}{z}\right)^1 + 2 \left(\frac{a}{z}\right)^2 + 3 \left(\frac{a}{z}\right)^3 + \dots$$

$$+ \left(\frac{a}{z}\right)^1 + 2 \left(\frac{a}{z}\right)^2 + 3 \left(\frac{a}{z}\right)^3 + \dots$$

$$= \left(\frac{a}{z}\right) \cdot \left[1 + 2 \left(\frac{a}{z}\right) + 3 \left(\frac{a}{z}\right)^2 + \dots \right]$$

$$= \frac{a}{z} \cdot \frac{1}{\left(1 - \frac{a}{z}\right)^2}$$

$$= \frac{a}{z} \left[\frac{z^2}{(z-a)^2} \right]$$

$$\mathcal{Z}[na^n] = \frac{az}{(z-a)^2}$$

$$Z\left[\frac{1}{n}\right]$$

$$u_n = \frac{1}{n}$$

We know

$$Z[u_n] = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$Z\left[\frac{1}{n}\right] = \sum_{n=0}^{\infty} \frac{1}{n} z^{-n}$$

$$= 0 + \frac{1}{1} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{3} z^{-3} + \dots$$

$$= 0 + \frac{1}{z} + \frac{1}{2} \left(\frac{1}{z}\right)^2 + \frac{1}{3} \left(\frac{1}{z}\right)^3 + \dots$$

$$= \frac{1}{z} + \frac{(1/z)^2}{2} + \frac{(1/z)^3}{3} + \dots$$

$$Z\left[\frac{1}{n}\right] = -\log\left(1 - \frac{1}{z}\right)$$

$$= -\log\left[\frac{z-1}{z}\right]$$

* Damping root Rule:-

If $z[un] = U[z]$ then $z[a^{-n}un] = U[az]$

and $z[a^n un] = U[\frac{z}{a}]$

Some Standard Results:-

$$(i) z[na^n] = \frac{az}{(z-a)^2}$$

$$(ii) z[n^2 a^n] = \frac{az^2 + a^2 z}{(z-a)^3}$$

$$(iii) z[\cos n\theta] = \frac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}$$

$$(iv) z[\sin n\theta] = \frac{z \sin\theta}{z^2 - 2z\cos\theta + 1}$$

$$(v) z[a^n \cos n\theta] = \frac{z(z - a\cos\theta)}{z^2 - 2az\cos\theta + a^2}$$

$$(vi) z[a^n \sin n\theta] = \frac{az \sin\theta}{z^2 - 2az\cos\theta + a^2}$$

$$(vii) z[a^n] = \frac{z}{z-a}$$

$$(viii) z[n^p] = -z \frac{d}{dz} [z(n^{p-1})]$$

$$(ix) z[n] = \frac{z}{(z-1)^2} ; (x) z[1] = \frac{z}{z-1}$$

$$(xi) z[n^2] = \frac{z^2 + z}{(z-1)^3} ; (xii) z[n^3] = \frac{z^3 + 4z^2 + z}{(z-1)^4}$$

① Find the Z-Transforms

① $3n - 4 \sin \frac{n\pi}{4} + 5a$

Given $U_n = 3n - 4 \sin \frac{n\pi}{4} + 5a$

$$Z[3n - 4 \sin \frac{n\pi}{4} + 5a] = Z[3n] - Z[4 \sin \frac{n\pi}{4}] + Z[5a]$$

$$= 3Z[n] - 4Z[\sin \frac{n\pi}{4}] + 5a[Z(1)]$$

$$= 3 \frac{Z}{(Z-1)^2} - 4 \frac{Z \sin \frac{\pi}{4}}{Z^2 - 2Z \cos \frac{\pi}{4} + 1} + 5a \frac{Z}{Z-1}$$

$$= \frac{3Z}{(Z-1)^2} + \frac{5aZ}{Z-1} - \frac{4Z \cdot \frac{1}{\sqrt{2}}}{Z^2 - 2Z \frac{1}{\sqrt{2}} + 1}$$

$$\Rightarrow \frac{3Z + 5aZ(Z-1)}{(Z-1)^2} - \frac{4Z \cdot \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}} [\sqrt{2} Z^2 - 2Z + \sqrt{2}]}$$

$$\Rightarrow \frac{3Z + 5aZ^2 - 5aZ}{(Z-1)^2} - \frac{4Z}{\sqrt{2} [Z^2 - \sqrt{2}Z + 1]}$$

$$\Rightarrow \frac{(3-5a)Z + 5aZ^2}{(Z-1)^2} - \frac{2\sqrt{2}Z}{Z^2 - \sqrt{2}Z + 1}$$

$$(ii) (n+1)^2$$

$$Z[U_n] = Z[(n+1)^2]$$

$$= Z[n^2 + 2n + 1]$$

$$= Z[n^2] + 2Z[n] + Z[1]$$

$$= \frac{Z^2 + Z}{(Z-1)^3} + 2 \frac{Z}{(Z-1)^2} + \frac{Z}{(Z-1)}$$

$$= \frac{Z^2 + Z + 2Z(Z-1) + Z(Z-1)^2}{(Z-1)^3}$$

$$= \frac{Z^2 + \cancel{Z} + 2Z^2 - 2\cancel{Z} + Z^3 - 2\cancel{Z}^2 + \cancel{Z}}{(Z-1)^3}$$

$$= \frac{Z^2 + Z^3}{(Z-1)^3}$$

$$Z[U_n] = \frac{Z^2(Z+1)}{(Z-1)^3}$$

$$(iii) \sin(3n+5)$$

$$Z[\sin(3n+5)] = Z[\sin 3n \cos 5 + \cos 3n \sin 5]$$

$$= Z[\sin 3n \cos 5] + Z[\cos 3n \sin 5]$$

$$= \cos 5 \cdot Z[\sin 3n] + \sin 5 \cdot Z[\cos 3n]$$

$$= \cos 5 \cdot \frac{Z \sin 3}{Z^2 - 2Z \cos 3 + 1} + \sin 5 \cdot \frac{Z[Z - \cos 3]}{Z^2 - 2Z \cos 3 + 1}$$

$$= \frac{\cos 5 \cdot z \sin 3 + z^2 \sin 5 - z \sin 5 \cos 3}{z^2 - 2z \cos 3 + 1}$$

$$= \frac{z [z \sin 5 + \cos 5 \sin 3 - \sin 5 \cos 3]}{z^2 - 2z \cos 3 + 1}$$

$$= \frac{z [z \sin 5 - [\sin 5 \cos 3 - \cos 5 \sin 3]]}{z^2 - 2z \cos 3 + 1}$$

$$= \frac{z [z \sin 5 - \sin(5-3)]}{z^2 - 2z \cos 3 + 1}$$

$$= \frac{z [z \sin 5 - \sin 2]}{z^2 - 2z \cos 3 + 1}$$

* Prove that $z[na^n] = \frac{az}{(z-a)^2}$

By damping rule

$$\text{Let } z[u_n] = U(z) \text{ then}$$

$$z[a^n u_n] = U\left(\frac{a}{z}\right)$$

$$z[a^{-n} u_n] = U(az)$$

Given, $u_n = n$

$$z[u_n] = z[n] = \frac{z}{(z-1)^2}$$

$$z[na^n] = z[un a^n] = u\left[\frac{a}{z}\right]$$

$$= \frac{\frac{a}{z}}{\left[\frac{a}{z} - 1\right]^2}$$

$$= \frac{z^2 \cdot a}{z[a-z]^2} = \frac{az^2}{z[a-z]^2}$$

$$= \frac{az}{(z-a)^2}$$

* Given $z[(\cos\theta + i\sin\theta)^n]$ Hence Evaluate $z(\cos\theta)$
and $z(\sin\theta)$

$$\text{Given, } z[(\cos\theta + i\sin\theta)^n] = z[(e^{i\theta})^n]$$

$$= \frac{z}{z - e^{i\theta}}$$

By ~~the binomial theorem~~

By the De Moivre's theorem

$$z[(\cos\theta + i\sin\theta)^n] = \frac{z}{z - e^{i\theta}} \times \frac{z - \bar{e}^{i\theta}}{z - \bar{e}^{i\theta}}$$

$$= \frac{z[z - \bar{e}^{i\theta}]}{(z - e^{i\theta})(z - \bar{e}^{i\theta})}$$

$$= \frac{z[z - \bar{e}^{i\theta}]}{z^2 - ze^{i\theta} - z\bar{e}^{i\theta} + e^{i\theta} \cdot e^{-i\theta}}$$

$$= \frac{z[z - e^{-i\theta}]}{z^2 - ze^{i\theta} - z\bar{e}^{i\theta} + 1}$$

$$z^2 - ze^{i\theta} - z\bar{e}^{i\theta} + 1$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$= \frac{z[z \cdot e^{i\theta}]}{z^2 - z[e^{i\theta} + e^{-i\theta}] + 1}$$

$$= \frac{z[z - [\cos\theta - i\sin\theta]]}{z^2 - z(2\cos\theta + 1)}$$

$$= \frac{(z^2 - z\cos\theta) + zis\theta}{z^2 - 2z\cos\theta + 1}$$

$$z[\cos\theta + i\sin\theta] = \frac{[z^2 - z\cos\theta]}{z^2 - 2z\cos\theta + 1} + \frac{[zis\theta]}{z^2 - 2z\cos\theta - 1}$$

Compare real and imaginary parts.

$$z[\cos\theta] = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1}$$

$$z[is\theta] = \frac{zis\theta}{z^2 - 2z\cos\theta - 1}$$

*

Find $z[\sin t]$

Given $u_n = \sin nt = \frac{e^{int} - e^{-int}}{2i}$

$$\sin nt = \frac{e^{int} - e^{-int}}{2i}$$

$$\cos nt = \frac{e^{int} + e^{-int}}{2}$$

$$e^{int} + e^{-int} = 2\cos nt$$

$$e^{int} - e^{-int} = 2is\theta nt$$

$$e^{int} = \cos nt + is\theta nt$$

$$e^{-int} = \cos nt - is\theta nt$$

$$z[u_n] = z[\sin nt] = z\left[\frac{e^{int} - e^{-int}}{2i}\right]$$

$$= \sum_{n=0}^{\infty} \left[\frac{e^{int} - e^{-int}}{2i} \right] z^{-n}$$

$$= \frac{1}{2i} \sum_{n=0}^{\infty} [e^{it} z^{-n} - e^{-it} z^{-n}]$$

$$= \frac{1}{2i} \left[\sum_{n=0}^{\infty} [e^{it} z^{-1}]^n - \sum_{n=0}^{\infty} [e^{-it} z^{-1}]^n \right]$$

$$= \frac{1}{2i} \left[1 + (e^{it} z^{-1}) + [e^{it} z^{-1}]^2 + [e^{it} z^{-1}]^3 + \dots \right]$$

$$- \left[1 + (e^{-it} z^{-1}) + [e^{-it} z^{-1}]^2 + \dots \right]$$

$$= \frac{1}{2i} \left[\frac{1}{1 - e^{it} z^{-1}} - \frac{1}{1 - e^{-it} z^{-1}} \right]$$

$$= \frac{1}{2i} \left[\frac{z}{z - e^{it}} - \frac{z}{z - e^{-it}} \right]$$

$$= \frac{1}{2i} \left[\frac{z}{z - e^{it}} \right] - \frac{1}{2i} \left[\frac{z}{z - e^{-it}} \right]$$

$$= \frac{1}{2i} \left[\frac{z}{z - e^{it}} \times \frac{z + e^{-it}}{z - e^{-it}} \right] - \frac{1}{2i} \left[\frac{z}{z - e^{-it}} \times \frac{z - e^{it}}{z - e^{it}} \right]$$

$$= \frac{1}{2i} \left[\frac{z(z - e^{-it})}{z^2 - ze^{it} - ze^{-it} + 1} \right] - \frac{1}{2i} \left[\frac{z(z - e^{it})}{z^2 - ze^{it} - ze^{-it} + 1} \right]$$

$$= \frac{1}{2i} \left[\frac{z^2 - ze^{it} - z^2 + ze^{it}}{z^2 - z(e^{it} + e^{-it}) + 1} \right]$$

$$= \frac{1}{2i} \left[\frac{z(e^{it} - e^{-it})}{z^2 - z[e^{it} + e^{-it}] + 1} \right]$$

$$= \frac{1}{2i} \left[\frac{z \cdot 2i \sin t}{z^2 - 2z \cos t + 1} \right]$$

$$= \left[\frac{z \sin t}{z^2 - 2z \cos t + 1} \right]$$

$$* z \left[\sin n \frac{\pi}{2} \right]$$

Here $\theta = \frac{\pi}{2}$.

$$z [\sin n \theta] = z \sin \theta$$

$$= \frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$$

$$z \left[\sin n \frac{\pi}{2} \right] = \frac{z \sin \pi/2}{z^2 - 2z \cos \frac{\pi}{2} + 1}$$

$$z \left[\sin n \frac{\pi}{2} \right] = \frac{z}{z^2 + 1}$$

$$\star \mathcal{Z}[a^n \cos nt]$$

By damping Rule ; $\mathcal{Z}[a^n u_n] = U\left[\frac{z}{a}\right]$
 we know that

$$\mathcal{Z}[u_n] = \mathcal{Z}[\cos nt] = \frac{z^2 - z \cos t}{z^2 - 2z \cos t + 1}$$

$$= U(z)$$

$$\therefore \mathcal{Z}[a^n \cos t] = \frac{\left(\frac{z}{a}\right)^2 - \frac{z}{a} \cos t}{\left(\frac{z}{a}\right)^2 - 2\left(\frac{z}{a}\right) \cos t + 1}$$

$$= \frac{z^2 - az \cos t}{z^2 - 2az \cos t + a^2}$$

$$= \frac{z^2 - az \cos t}{z^2 - 2az \cos t + a^2}$$

$$= \frac{z^2 - az \cos t}{z^2 - 2az \cos t + a^2}$$

$$= \frac{z^2 - az \cos t}{z^2 - 2az \cos t + a^2}$$

$$= \frac{z^2 - az \cos t}{z^2 - 2az \cos t + a^2}$$

$$= \frac{z[z - a \cos t]}{z^2 - 2az \cos t + a^2}$$

$$= \frac{z[z - a \cos t]}{z^2 - 2az \cos t + a^2}$$

$$* \mathcal{Z}[a^n \sin nt]$$

By damping rule, $\mathcal{Z}[a^n u_n] = U\left[\frac{z}{a}\right]$.

We know that

$$\mathcal{Z}[u_n] = \mathcal{Z}[\sin nt] = \frac{z \sin t}{z^2 - 2z \cos t + 1}$$

$$= U(z)$$

$$\therefore \mathcal{Z}[a^n \sin nt] = \frac{\left(\frac{z}{a}\right) \sin t}{\left(\frac{z}{a}\right)^2 - 2\left(\frac{z}{a}\right) \cos t + 1}$$

$$= \frac{\frac{z}{a} \sin t}{z^2 - 2az \cos t + a^2}$$

$$\mathcal{Z}[a^n \sin nt] = \frac{az \sin t}{z^2 - 2az \cos t + a^2}$$

$$* \mathcal{Z}\left[3^n \cos n \frac{\pi}{2}\right]$$

By damping rule, $\mathcal{Z}[a^n u_n] = U\left[\frac{z}{a}\right]$

We know th

$$\mathcal{Z}[u_n] = \mathcal{Z}[\cos n \theta] = \frac{z(z - a \cos \theta)}{z^2 - 2az \cos \theta + a^2}$$

$$Z \left[3^n \cos n \frac{\pi}{2} \right] = \frac{Z \left(Z - 3 \cos \frac{\pi}{2} \right)}{Z^2 - 2(3)Z \cos \frac{\pi}{2} + 3^2}$$

$$Z \left[3^n \cos n \frac{\pi}{2} \right] = \frac{Z^2}{Z^2 + 9}$$

* Find the Z-Transform of 2^{2k+3}

$$\text{Given } U_k = 2^{k+1} \Rightarrow 2^{2k} \cdot 2^3 = 8 \cdot 2^{2k}$$

$$Z[U_k] = \sum_{k=0}^{\infty} U_k z^{-k}$$

$$= \sum_{k=0}^{\infty} 8 \cdot 2^{2k} z^{-k}$$

$$= 8 \left[\sum_{k=0}^{\infty} 2^{2k} z^{-k} \right]$$

$$\Rightarrow 8 \left[1 + (2^2 z^{-1}) + (2^2 z^{-1})^2 + (2^2 z^{-1})^3 + \dots \right]$$

$$\Rightarrow 8 \cdot \frac{1}{1 - 2^2 z^{-1}}$$

$$= 8 \cdot \frac{z}{z - 4}$$

$$= \frac{8z}{z - 4}$$

Shifting properties:

(1) Shifting U_n to the right

$$\text{If } Z[U_n] = U[z] \text{ then } Z[U_{n-k}] = z^{-k} U[z]$$

(2) Shifting U_n to the left

If $z[u_n] = U(z)$, then

$$z[u_{n+1}] = z[u(z) - u(0)]$$

$$z[u_{n+2}] = z^2[u(z) - u(0) - u(1)z^{-1}]$$

$$z[u_{n+3}] = z^3[u(z) - u(0) - u(1)z^{-1} - u(2)z^{-2}]$$

⋮

$$z[u_{n+k}] = z^k[u(z) - u(0) - u(1)z^{-1} - u(2)z^{-2} - \dots - u(k-1)z^{-(k-1)}]$$

* Find $z[e^{an}]$

$$= z[(e^a)^n]$$

$$\left(\frac{z}{1-e^{-a}}\right) = \frac{z[u(z)]}{z-e^a}$$

* Find $z[ne^{an}]$

$$z[ne^{an}] = z[(e^a)^n \cdot n]$$

By damping rule

$$z[an] = u\left(\frac{z}{a}\right)$$

$$\text{if } z[u_n] = U(z)$$

We know that $z[u_n] = z[n]$

$$= \frac{z}{(z-1)^2}$$

$$= u(z)$$

$$Z[an^n] = Z[(e^a)^n \cdot n] = U\left(\frac{z}{e^a}\right) = \frac{z/e^a}{(z/e^a - 1)^2}$$

$$Z[ne^{an}] = \frac{\frac{z/e^a}{(z - e^a)^2}}{e^{az}}$$

$$= \frac{e^{az} z}{(z - e^a)^2}$$

* Find $Z[n^2 \cdot e^{an}]$

$$Z[n^2 e^{an}] = Z[(e^a)^n \cdot n^2]$$

$$\text{Let } U_n = n^2$$

$$Z[U_n] = Z[n^2]$$

$$= \frac{z^2 + z}{(z - 1)^3}$$

By damping rule $Z[a^n U_n] = U\left(\frac{z}{a}\right)$

$$Z[(e^a)^n \cdot n^2] = U\left(\frac{z}{e^a}\right)$$

$$= \frac{\left(\frac{z}{e^a}\right)^2 + \frac{z}{e^a}}{\left(\frac{z}{e^a} - 1\right)^3}$$

$$= \frac{z^2 + ze^a}{(z - e^a)^3}$$

$$= \frac{z^2 + ze^a}{(z - e^a)^3}$$

$$= \frac{e^a [z + ze^a]}{(z - e^a)^3}$$

$$= \frac{e^a [z + ze^a]}{(z - e^a)^3}$$

* Find $Z[e^t \sin 2t]$

Sol Let $u_n = \sin 2t$

$$Z[u_n] = Z[\sin 2t]$$

$$= \frac{Z \sin 2t}{Z^2 - 2Z \cos 2t + 1}$$

By damping rule

$$Z[e^t \sin 2t] = Z[e^t \sin 2t]$$

$$= u \left[\frac{Z}{c} \right]$$

$$= \frac{Z/e \sin 2t}{(Z/e)^2 - 2(Z/e) \cos 2t + 1}$$

$$= \frac{Z e \sin 2t}{Z^2 - 2Z \cos 2t + e^2}$$

* Find: $Z \left[\cos \left(\frac{n\pi}{2} + \frac{\pi}{2} \right) \right]$

Sol $Z \left[\cos \left(\frac{n\pi}{2} + \frac{\pi}{2} \right) \right] = Z \left[\cos \frac{n\pi}{2} \cos \frac{\pi}{2} - \sin \frac{n\pi}{2} \sin \frac{\pi}{2} \right]$

$$= Z[0 - \sin \frac{n\pi}{2} \cdot 1]$$

$$= -Z \left[\sin \frac{n\pi}{2} \right]$$

$$= -Z \left[\sin \frac{n\pi}{2} \right]$$

We know that

$$Z[\sin n\theta] = \frac{Z \sin \theta}{Z^2 - 2Z \cos \theta + 1}$$

$$\therefore Z\left[\sin n \frac{\pi}{2}\right] = \frac{-Z \sin \frac{\pi}{2}}{Z^2 - 2Z \cos \frac{\pi}{2} + 1}$$

$$= \frac{-Z}{Z^2 + 1}$$

* Find the $Z\left[\frac{1}{2}(n-1)(n+2)\right]$

Sol $Z\left[\frac{1}{2}(n-1)(n+2)\right]$

$$\Rightarrow \frac{1}{2} Z[(n-1)(n+2)]$$

$$\Rightarrow \frac{1}{2} Z[n^2 + n - 2]$$

$$= \frac{1}{2} [Z[n^2] + Z[n] - 2Z[1]]$$

$$= \frac{1}{2} \left[\frac{Z^3 + Z}{(Z-1)^3} + \frac{Z}{(Z-1)^2} - \frac{2Z}{Z-1} \right]$$

$$= \frac{1}{2} \left[\frac{Z^3 + Z + Z(Z-1) - 2Z(Z-1)^2}{(Z-1)^3} \right]$$

$$= \frac{1}{2} \left[\frac{Z^3 + Z + Z^2 - Z - 2Z^2 + 4Z - 2Z^2 + 2Z}{(Z-1)^3} \right]$$

$$= \frac{1}{2} \left[\frac{2Z^2 - 2Z^3 + 2Z + 4Z^2}{(Z-1)^3} \right]$$

$$= \frac{-Z^3 + 3Z^2 + 2Z}{(Z-1)^3}$$

① $z[(2n-1)^2]$ ② $z\left[\cos \frac{n\pi}{2}\right]$ ③ $z[a^n \cos n\frac{\pi}{2}]$

Given $z[(2n-1)^2]$

$$\Rightarrow z[4n^2 + 1 - 4n]$$

$$\Rightarrow 4z[n^2] + z[1] - 4[z]$$

$$\Rightarrow 4\left(\frac{z^2+z}{(z-1)^3}\right) + \left(\frac{z}{z-1}\right) - 4\left(\frac{z}{(z-1)^2}\right)$$

$$\Rightarrow \frac{4z^2+4z}{(z-1)^3} + \frac{z}{z-1} - \frac{4z}{(z-1)^2}$$

$$\Rightarrow \frac{4z^2+4z + z(z-1)^2 - 4z(z-1)}{(z-1)^3}$$

$$\Rightarrow \frac{4z^2+4z + z[z^2+1-2z] - 4z^2+4z}{(z-1)^3}$$

$$\Rightarrow \frac{8z^2+z^3+z-2z^2}{(z-1)^3}$$

$$\Rightarrow \frac{z^3-2z^2+9z}{(z-1)^3}$$

$$\Rightarrow \frac{z^3-2z^2+9z}{(z-1)^3}$$

② $z\left[\cos \frac{n\pi}{2}\right]$

Let $z[\cos n\theta] = \frac{z[z - \cos \theta]}{z^2 - 2z \cos \theta + 1}$

$$z\left[\cos n\frac{\pi}{2}\right] = \frac{z[z - \cos \frac{\pi}{2}]}{z^2 - 2z \cos \frac{\pi}{2} + 1}$$

$$= \frac{z^2 - 2z \cos \pi/2}{z^2 - 2z \cos \pi/2 + 1}$$

$$= \frac{z^2}{z^2 + 1}$$

$$* z[a^n \cos n \frac{\pi}{2}]$$

$$\text{Wk.T } z[a^n \cos n \theta] = \frac{z[z - a \cos \theta]}{z^2 - 2az \cos \theta + a^2}$$

$$z[a^n \cos n \frac{\pi}{2}] = \frac{z(z - a \cos \pi/2)}{z^2 - 2az \cos \pi/2 + a^2}$$

$$= \frac{z^2}{z^2 - 2az \cos \pi/2 + a^2}$$

$$= \frac{z^2}{z^2 + a^2}$$

Multiplication by n .

$$\text{If } z[u_n] = u(z) \text{ then } z[n u_n] = -z \frac{d}{dz} [u(z)]$$

$$\text{Similarly } z[n^2 u_n] = \left(-z \frac{d}{dz}\right)^2 u(z)$$

$$z[n^3 u_n] = \left(-z \frac{d}{dz}\right)^3 u(z)$$

$$z[n^n u_n] = \left(-z \frac{d}{dz}\right)^n u(z)$$

Initial value theorem:

$$\text{If } \mathcal{Z}[u_n] = u(z) \text{ then } u_0 = \lim_{z \rightarrow \infty} u(z)$$

$$\text{Similarly } u_1 = \lim_{z \rightarrow \infty} z [u(z) - u_0]$$

$$u_2 = \lim_{z \rightarrow \infty} \frac{1}{2} z^2 [u(z) - u_0 - u_1 z^{-1}]$$

$$u_3 = \lim_{z \rightarrow \infty} \frac{1}{6} z^3 [u(z) - u_0 - u_1 z^{-1} - u_2 z^{-2}]$$

Final value theorem:

$$\text{If } \mathcal{Z}[u_n] = u(z) \text{ then}$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{z \rightarrow 1} (z-1) u(z)$$

Show that $\mathcal{Z}\left[\frac{1}{n!}\right] = e^{1/z}$. Hence Evaluate $\mathcal{Z}\left[\frac{1}{(n+1)!}\right]$

and $\mathcal{Z}\left[\frac{1}{(n+2)!}\right]$

Sol) By z -transform

$$\mathcal{Z}[u_n] = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$\mathcal{Z}\left[\frac{1}{n!}\right] = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$$

$$= 1 + \frac{1}{z} + \frac{1}{2!} \left(\frac{1}{z}\right)^2 + \frac{1}{3!} \left(\frac{1}{z}\right)^3 + \dots$$

$$= e^{1/z}$$

$$\mathcal{Z}\left[\frac{1}{n!}\right] = e^{1/z}$$

$$= u(z)$$

Here $u_n = \frac{1}{n!}$

By shifting to the left one step

$$z \left[\frac{1}{(n+1)!} \right] = z [u(z) - u(0)]$$

$$\boxed{u(0) = \frac{1}{0!} = 1}$$

$$= z [e^{1/z} - 1]$$

By shifting to the left two steps

$$z \left[\frac{1}{(n+2)!} \right] = z^2 [u(z) - u(0) - u(1)z^{-1}]$$

$$= z^2 [e^{1/z} - 1 - 1 \cdot z^{-1}]$$

$$= z^2 \left[e^{1/z} - 1 - \frac{1}{z} \right]$$

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* Find $Z[2(3^n) + 5(n)]$ and hence Reduce.

$$Z[2 \cdot 3^{n+3} + 5(n+3)]$$

We know that

$$Z[2 \cdot 3^n + 5 \cdot n] = Z[2 \cdot 3^n] + Z[5 \cdot n]$$

$$= 2Z[3^n] + 5Z[n]$$

$$= 2 \cdot \frac{Z}{Z-3} + 5 \cdot \frac{Z}{(Z-1)^2}$$

$$= \frac{2Z(Z-1)^2 + 5Z(Z-3)}{(Z-3)(Z-1)^2}$$

$$= \frac{2Z^3 - 4Z^2 + 2Z + 5Z^2 - 15Z}{(Z-3)(Z-1)^2}$$

$$Z[u_n] = Z[2 \cdot 3^n + 5 \cdot n] = \frac{2Z^3 + Z^2 - 13Z}{(Z-3)(Z-1)^2}$$

$$\therefore u_0 = 2, u_1 = 11, u_2 = 28$$

By shifting properly to the left 3 steps.

$$Z[u_{n+3}] = Z^3 \left[u(Z) - u(0) - u(1)Z^{-1} - u(2)Z^{-2} \right]$$

$$= Z^3 \left[\frac{2Z^3 + Z^2 - 13Z}{(Z-3)(Z-1)^2} - 2 - 11 \frac{1}{Z} - 28 \frac{1}{Z^2} \right]$$

* If $f(z) = \frac{5z^2 + 3z + 12}{(z-1)^4}$; find the value of

$f(2)$ & $f(3)$.

$$f(2) = \frac{5z^2 + 3z + 12}{(z-1)^4}$$

$$= \frac{z^2 \left(5 + \frac{3}{z} + \frac{12}{z^2} \right)}{z^4 \left(1 - \frac{1}{z} \right)^4}$$

$$f(z) = \frac{5 + \frac{3}{z} + \frac{12}{z^2}}{z^2 \left(1 - \frac{1}{z} \right)^4} = U(z)$$

By initial value Theorem,

$$f(0) = U_0 = \lim_{z \rightarrow \infty} f(z) = \lim_{z \rightarrow \infty} \left(\frac{5 + \frac{3}{z} + \frac{12}{z^2}}{z^2 \left(1 - \frac{1}{z} \right)^4} \right)$$

$$f(1) = \lim_{z \rightarrow \infty} z [f(z) - f(0)]$$

$$= \lim_{z \rightarrow \infty} \left(z \left[\frac{5 + \frac{3}{z} + \frac{12}{z^2}}{z^2 \left(1 - \frac{1}{z} \right)^4} - 0 \right] \right)$$

$$f(1) = 0$$

$$f(2) = U_2 = \lim_{z \rightarrow \infty} z^2 \left[f(z) - f(0) - f(1)z^{-1} \right]$$

$$= \lim_{z \rightarrow \infty} \frac{1}{2} z^2 \left[\frac{5z^2 + 3z + 12}{(z-1)^4} - 0 - 0z^{-1} \right]$$

$$9 = \lim_{z \rightarrow \infty} z^2 \left[\frac{5 + \frac{3}{z} + \frac{12}{z^2}}{z^2 \left(1 - \frac{1}{z}\right)^4} - 0 - 0z^{-1} \right]$$

$$f(2) = 5$$

$$f(3) = 43 = \lim_{z \rightarrow \infty} z^3 \left[f(z) - f(0) - f(1)z^{-1} - f(2)z^{-2} \right]$$

$$= \lim_{z \rightarrow \infty} z^3 \left[\frac{5 + \frac{3}{z} + \frac{12}{z^2}}{z^2 \left(1 - \frac{1}{z}\right)^4} - 0 - 0z^{-1} - 5z^{-2} \right]$$

$$= \lim_{z \rightarrow \infty} z^3 \left[\frac{5z^2 + 3z + 12}{(z-1)^4} - \frac{5}{z^2} \right]$$

$$= \lim_{z \rightarrow \infty} z^3 \left[\frac{(5z^2 + 3z + 12)z^2 - 5(z-1)^4}{(z-1)^4 z^2} \right]$$

$$= \lim_{z \rightarrow \infty} z^3 \left[\frac{5z^4 + 3z^3 + 12z^2 - 5z^4 + 20z^3 - 30z^2 + 20z - 5}{(z-1)^4 z^2} \right]$$

$$= \lim_{z \rightarrow \infty} z^3 \left[\frac{23z^3 - 18z^2 + 20z - 5}{(z-1)^4 z^2} \right]$$

$$= \lim_{z \rightarrow \infty} \left[\frac{23 - \frac{18}{z} + \frac{20}{z^2} - \frac{5}{z^3}}{\left(1 - \frac{1}{z}\right)^4} \right]$$

$$= \lim_{z \rightarrow \infty} \frac{23 - \frac{18}{z} + \frac{20}{z^2} - \frac{5}{z^3}}{\left(1 - \frac{1}{z}\right)^4}$$

$$f(3) = 23$$

* Find z ~~to~~ $z[n \cos n\theta]$

Given $U_n = \cos n\theta$

$$z[U_n] = z[\cos n\theta] = \frac{z[z - (\cos \theta)]}{z^2 - 2z \cos \theta + 1} = U(z)$$

By multiplication by n^2 .

$$z[n \cdot U_n] = -z \frac{d}{dz} [U(z)]$$

$$= -z \frac{d}{dz} \left[\frac{z^2 - z \cos \theta}{z^2 - 2z \cos \theta + 1} \right]$$

$$= -z \left[\frac{(z^2 - 2z \cos \theta + 1)(2z - \cos \theta) - z(z^2 - z \cos \theta)(2z - 2 \cos \theta)}{(z^2 - 2z \cos \theta + 1)^2} \right]$$

$$= -z \left[\frac{2z^3 - z^2 \cos \theta - 4z^2 \cos \theta + 2z \cos^2 \theta + 2z - \cos \theta - 2z^3 + 2z^2 \cos \theta - 2z^2 \cos \theta + 2z \cos^2 \theta}{(z^2 - 2z \cos \theta + 1)^2} \right]$$

$$= -z \left[\frac{\cancel{2z^3} - z^2 \cos \theta - \cancel{4z^2 \cos \theta} + \cancel{2z \cos^2 \theta} + 2z - \cos \theta - \cancel{2z^3} + \cancel{2z^2 \cos \theta} + \cancel{2z^2 \cos \theta} - \cancel{2z \cos^2 \theta}}{(z^2 - 2z \cos \theta + 1)^2} \right]$$

$$= -z \left[\frac{2z - z^2 \cos \theta - \cos \theta}{(z^2 - 2z \cos \theta + 1)^2} \right]$$

INVERSE Z-TRANSFORMS

$$* \quad Z^{-1} \left[\frac{z}{z-1} \right] = 1$$

$$* \quad Z^{-1} \left[\frac{z}{z-a} \right] = a^n$$

$$* \quad Z^{-1} \left[\frac{z}{(z-1)^2} \right] = n$$

$$* \quad Z^{-1} \left[\frac{z^2+z}{(z-1)^3} \right] = n^2$$

$$* \quad Z^{-1} \left[\frac{z^3+4z^2+z}{(z-1)^4} \right] = n^3$$

$$* \quad Z^{-1} \left[\frac{az}{(z-a)^2} \right] = n a^n$$

$$* \quad Z^{-1} \left[e^{1/z} \right] = \frac{1}{n!}$$

$$* \quad Z^{-1} \left[\log \frac{z}{z-1} \right] = \frac{1}{n}$$

$$* \quad Z^{-1} \left[z \log \frac{z}{z-1} \right] = \frac{1}{n+1}$$

Definition! $Z[U_n] = U(z)$,

Which can also be return as $U_n = Z^{-1}[U(z)]$

then U_n is called the inverse z -transform of

$U(z)$. Thus finding the Sequences U_n from

$U(z)$ is defined as Inverse z -transform,

it is denoted by Z^{-1} .

→ They are 4 methods

(1) Integration Method

(2) Power series method

(3) Partial fractions Method

(4) Convolution Theorem Method

Partial fractions Method.

If $U(z)$ rational function then the partial fraction method is often useful method to find its Inverse. The method consisting of decomposing $U\left[\frac{z}{z}\right]$ into partial fraction multiply the result Expansion by z & then Apply Inverse.

① $\frac{U(z)}{z}$

② P.f (M)

③ multiply z

④ Apply Inverse to z -trans.

* find the Inverse z -transform of $U(z) = \frac{z^2 + 3z}{z^2 - 3z + 2}$

So! Given $U(z) = \frac{z^2 + 3z}{z^2 - 3z + 2}$

$$\Rightarrow \frac{U(z)}{z} = \frac{z^2 + 3z}{z[z^2 - 3z + 2]}$$

$$= \frac{z(z+3)}{z(z^2 - 3z + 2)}$$

$$\therefore \frac{U(z)}{z} = \frac{z+3}{z^2 - 3z + 2}$$

$$\frac{U(z)}{z} = \frac{z+3}{(z-1)(z-2)}$$

Now $\frac{z+3}{(z-1)(z-2)}$

resolve into partial fraction

$$\frac{z+3}{(z-1)(z-2)} = \frac{A}{(z-1)} + \frac{B}{z-2}$$

$$z+3 = A(z-2) + B(z-1)$$

Put $z=2$ $\Rightarrow$ $\boxed{B=5}$

Put $z=1$ $\Rightarrow$ $\boxed{A=-4}$

$$\frac{U(z)}{z} = \frac{z+3}{(z-1)(z-2)} = \frac{-4}{z-1} + \frac{5}{z-2}$$

$$\cancel{z} \frac{U(z)}{\cancel{z}} = \frac{-4z}{z-1} + \frac{5z}{z-2}$$

$$U(z) = \frac{-4z}{z-1} + \frac{5z}{z-2}$$

Apply z^{-1} on b.s

$$z^{-1}[U(z)] = z^{-1}\left[\frac{-4z}{z-1}\right] + z^{-1}\left[\frac{5z}{z-2}\right]$$

$$\textcircled{U} = -4z^{-1}\left[\frac{z}{z-1}\right] + 5z^{-1}\left[\frac{z}{z-2}\right]$$

$$U_n = -4(1) + 5(2^n)$$

$$U_n = -4 + 5(2^n)$$

And the z -transform $U(z) = \frac{z}{z^2 + 11z + 24}$.

Given

$$U(z) = \frac{z}{z^2 + 11z + 24}$$

$$\frac{U(z)}{z} = \frac{1}{z^2 + 11z + 24}$$

$$= \frac{1}{(z+3)(z+8)}$$

Now $\frac{1}{(z+3)(z+8)}$ Resolving into p.f.

$$\frac{1}{(z+3)(z+8)} = \frac{A}{z+3} + \frac{B}{z+8}$$

$$1 = A(z+8) + B(z+3)$$

$$\text{Put } z = -8 \quad \text{put } z = -3$$

$$B = \frac{-1}{5}$$

$$A = \frac{1}{5}$$

$$\frac{U(z)}{z} = \frac{1}{(z+3)(z+8)} = \frac{1}{5(z+3)} - \frac{1}{5(z+8)}$$

$$U(z) = \frac{z}{5(z+3)} - \frac{z}{5(z+8)}$$

Apply inverse on b.s.

$$z^{-1}[U(z)] = z^{-1}\left[\frac{z}{5(z+3)}\right] - z^{-1}\left[\frac{z}{5(z+8)}\right]$$

$$U_n = \frac{1}{5} z^{-1} \left[\frac{z}{z+3} \right] - \frac{1}{5} z^{-1} \left[\frac{z}{z+8} \right]$$

$$= \textcircled{5} \left(\cdot = \frac{1}{5} z^{-1} \left[\frac{z}{z-(-3)} \right] - \frac{1}{5} z^{-1} \left[\frac{z}{z-(-8)} \right] \right)$$

$$U_n = \frac{1}{5} (-3)^n - \frac{1}{5} (-8)^n$$

$$U_n = \frac{(-3)^n - (-8)^n}{5}$$

$$\therefore * U_n = \frac{4z^2 - 2z}{z^3 - 5z^2 + 8z - 4}$$

$$\frac{U_n}{2z} = \frac{(2z-1)}{(z-1)(z-2)^2}$$

Now $\frac{(2z-1)}{(z-1)(z-2)^2}$ resolve into P.F

$$\frac{2z-1}{(z-1)(z-2)^2} = \frac{A}{(z-1)} + \frac{B}{(z-2)} + \frac{C}{(z-2)^2}$$

~~$$2z-1 = A(z-2)(z-2)^2$$~~

~~$$2z-1 = A(z-2)(z-2)^2 + B(z-1)(z-2)^2$$~~

~~$$+ C(z-1)(z-2)$$~~

$$2z-1 = A(z-2)(z^2-4z+4) + B(z-1)(z^2-4z+4) + C(z-1)(z-2)$$

$$2z-1 = A[z^3-4z^2+4z-2z^2+8z-8] + B$$

$$B[z^3-4z^2+4z-z^2+4z-4] + C$$

$$[z^2-2z-z+2]$$

$$2z-1 = Az^3-4Az^2+4Az$$

$$2z-1 = A(z-2)^2 + B(z-1)(z-2) + C(z-1)$$

$$2z-1 = A(z^2+4-4z) + Bz^2-2Bz-Bz+2B$$

$$+ Cz - C$$

$$2z-1 = Az^2 + A4 - 4Az + Bz^2 - 2Bz - Bz + 2B$$

$$+ Cz - C$$

Compare z^2 , z and constants

$$z^2 \Rightarrow 0 = A + B \quad \text{--- (1)}$$

$$z \Rightarrow 2 = -4A - 3B + C \quad \text{--- (2)}$$

$$\text{Const} \Rightarrow -1 = 4A + 2B - C \quad \text{--- (3)}$$

Compare (2) and (3)

$$2 = -4A - 3B + C$$

$$-1 = 4A + 2B - C$$

$$1 = -B$$

$$B = -1$$

Sub B in (1)

$$0 = A - 1$$

$$A = 1$$

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$$-1 = 4 - 2 - 1$$

$$-1 = 2 - C$$

$$A(2+1-p) + Bx^2 - 2x + 1 = 2x^2 - 2x + 1$$

$$\frac{0(2)}{2(1)} = \frac{1}{2} + \frac{1}{(2-3)^2}$$

2. (continued)

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$$x^2 = 2x + AP$$

100 - 2 - 1

$$U_n = 2 - 2 \cdot 2^n + \dots 3 \cdot n \cdot 2^{n-1}$$

* Find inverse Z-transform of $U(z) = \frac{2z^2 + 3z}{(z+2)(z-4)}$

Given

$$U(z) = \frac{2z^2 + 3z}{(z+2)(z-4)}$$

$$\frac{U(z)}{z} = \frac{2z + 3}{(z+2)(z-4)}$$

Now $\frac{2z + 3}{(z+2)(z-4)}$ resolving into partial fraction

$$\frac{2z + 3}{(z+2)(z-4)} = \frac{A}{z+2} + \frac{B}{z-4}$$

$$2z + 3 = A(z-4) + B(z+2)$$

Put $z = 4$

$$11 = 6B$$

$$B = \frac{11}{6}$$

Put $z = -2$

$$-1 = -6A$$

$$A = \frac{1}{6}$$

$$\frac{2z + 3}{(z+2)(z-4)} = \frac{1}{6(z+2)} + \frac{11}{6(z-4)}$$

* Multiply with z on both sides.

$$U(z) = \frac{z}{6(z+2)} + \frac{11z}{6(z-4)}$$

Apply z^{-1}

$$z^{-1}[v(z)] = \frac{1}{6} z^{-1} \left[\frac{z}{z+2} \right] + \frac{11}{6} z^{-1} \left[\frac{z}{z-4} \right]$$

$$z^{-1}[v(z)] = \frac{1}{6} z^{-1} \left[\frac{z}{z-(-2)} \right] + \frac{11}{6} z^{-1} \left[\frac{z}{z-4} \right]$$

$$v_n = \frac{1}{6} (-2)^n + \frac{11}{6} 4^n$$

* ~~$z^2 + 2$~~

Convolution product

Let $f(n)$ & $g(n)$ is a sequences then
convolution product of $f(n)$ & $g(n)$ denoted by
 $f(n) * g(n)$ and defined by

$$\{ f(n) * g(n) = \sum_{m=0}^n f(m) g(n-m) \}$$

$f(n) * g(n) = f(0)g(n) + f(1)g(n-1) + f(2)g(n-2) + \dots + f(n)g(0)$
Convolution Theorem:

If $\mathcal{Z}^{-1}[F(z)] = f(n)$ & $\mathcal{Z}^{-1}[G(z)] = g(n)$;
then $\mathcal{Z}^{-1}[F(z) * G(z)] = f(n) * g(n) = \sum_{m=0}^n f(m) g(n-m)$

Where, $*$ denotes convolution operator

* Evaluate $\mathcal{Z}^{-1}\left[\frac{z^2}{(z-1)(z-3)}\right]$ using convolution theorem.

Given: $\left[\frac{z^2}{(z-1)(z-3)}\right]$

$$= \left[\frac{z}{z-1} \cdot \frac{z}{z-3} \right]$$

Let $F(z) = \frac{z}{z-1}$

$$\mathcal{Z}^{-1}[F(z)] = \mathcal{Z}^{-1}\left[\frac{z}{z-1}\right]$$

$$f(n) = 1^n$$

$G(z) = \frac{z}{z-3}$

$$\mathcal{Z}^{-1}[G(z)] = \mathcal{Z}^{-1}\left[\frac{z}{z-3}\right]$$

$$g(n) = 3^n$$

By convolution theorem

$$\mathcal{Z}^{-1}[F(z) \cdot G(z)] = f(n) * g(n) = \sum_{m=0}^n f(m) g(n-m)$$

$$= \sum_{m=0}^n 1^m 3^{n-m}$$

$$= \sum_{m=0}^n 1^m 3^n 3^{-m}$$

$$= 3^n \sum_{m=0}^n 1^m 3^{-m}$$

$$= 3^n \left[1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^n} \right]$$

$$= 3^n \left[\frac{1 - \frac{1}{3^{n+1}}}{1 - \frac{1}{3}} \right]$$

$$= \left(\frac{1 - \frac{1}{3^{n+1}}}{1 - \frac{1}{3}} \right)$$

$$= 3^n \left[\frac{3^{n+1} - 1}{3^{n+1}} \times \frac{3}{2} \right]$$

$$= 3^n \left[\frac{(3^{n+1} - 1)3}{2(3^{n+1})} \right]$$

$$= \frac{3^{n+1} - 1}{2}$$

* Evaluate $z^{-1} \left[\frac{z^2}{(z-2)(z-4)} \right]$

Given $\left[\frac{z^2}{(z-2)(z-4)} \right] = \left[\frac{z}{z-2} - \frac{z}{z-4} \right]$

$$F(z) = \frac{z}{z-2}$$

$$G(z) = \frac{z}{z-4}$$

$$z^{-1}[F(z)] = z^{-1} \left[\frac{z}{z-2} \right]$$

$$z^{-1}[G(z)] = z^{-1} \left[\frac{z}{z-4} \right]$$

$$f(n) = 2^n \quad g(n) = 4^n$$

$z^{-1}(n)$

$$z^{-1}[f(z) * G(z)] = f(n) * g(n)$$

$$= \sum_{m=0}^n f(m) g(n-m)$$

$$= \sum_{m=0}^n 2^m 4^{n-m}$$

$$\left(\begin{aligned} & (1 + 0.5 + 0.25 + 0.125 + \dots) \\ & = 0.5 \left(\frac{1 - 0.5^{n+1}}{1 - 0.5} \right) \end{aligned} \right) = \sum_{m=0}^n 2^m 4^n 4^{-m}$$

$$= 4^n \sum_{m=0}^n 2^m 4^{-m}$$

$$= 4^n \left[1 + 2 \cdot \frac{1}{4} + 2^2 \cdot \frac{1}{4^2} + 2^3 \cdot \frac{1}{4^3} + \dots + 2^n \cdot \frac{1}{4^n} \right]$$

$$= 4^n \left[1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n} \right]$$

$$= 4^n \left[\frac{1 - \frac{1}{2^{n+1}}}{1 - \frac{1}{2}} \right] \Rightarrow 4^n \left[\frac{\frac{2^{n+1} - 1}{2^{n+1}}}{\frac{2-1}{2}} \right]$$

$$= 4^n \left[\frac{2^{n+1} - 1}{2^{n+1}} \times \frac{2}{1} \right]$$

$$= 2^n \cancel{2^n} \left[\frac{2^{n+1} - 1}{2^{n+1}} \times \frac{2}{1} \right]$$

$$= 2^n [2^{n+1} - 1]$$

* Evaluate $z^{-1} \left[\frac{z^3}{(z+1)(z^2+4)} \right]$

Given $z^{-1} \left[\frac{z^3}{(z+1)(z^2+4)} \right] \Rightarrow \left[\frac{z}{(z+1)} \cdot \frac{z^2}{z^2+4} \right]$

~~$= \left[\frac{z}{z+1} \right]$~~

Let $F(z) = \frac{z}{z+1}$

$z^{-1}[F(z)] = z^{-1} \left[\frac{z}{z+1} \right]$

$= z^{-1} \left[\frac{z}{z - (-1)} \right]$

$\text{So } z^{-1} = (-1)^n$

$G(z) = \frac{z^2}{z^2+4}$

$z^{-1}[G(z)] = z^{-1} \left[\frac{z^2}{z^2+2^2} \right]$

$G(n) = 2^n \cos \frac{n\pi}{2}$

$\because z(a^n \cos \frac{n\pi}{2}) = \frac{z^2}{z^2+a^2}$
 $a^n \cos \frac{n\pi}{2} = z^{-1} \left[\frac{z^2}{z^2+a^2} \right]$

By convolution Theorem

$z^{-1}[F(z)G(z)] = f(n) * g(n) = \sum_{m=0}^n f(m)g(n-m)$

$= \sum_{m=0}^n (-1)^m 2^{n-m} \frac{\cos(n-m)\pi}{2}$

$= 2^n \sum_{m=0}^n (-1)^m 2^{-m} \frac{\cos(n-m)\pi}{2}$

USING CONVOLUTION THEOREM SHOW THAT

$$z^{-1} \left[\frac{1}{n!} * \frac{1}{n!} \right] = \frac{2^n}{n!}$$

Given $\left[\frac{1}{n!} * \frac{1}{n!} \right] = [f(n) * g(n)]$

By convolution theorem,

$$= f(n) * g(n) = \sum_{m=0}^n f(m) g(n-m)$$

So

$$= \sum_{m=0}^n \frac{1}{m!} \frac{1}{(n-m)!}$$

$$= \sum_{m=0}^n \frac{1}{m!} \frac{1}{(n-m)!}$$

$$= \frac{1}{n!} \sum_{m=0}^n \frac{n!}{m! (n-m)!}$$

$$= \frac{1}{n!} \left[1 + \frac{n}{1!} + \frac{n(n-1)}{2!} + \dots + \frac{n!}{n!} \right]$$

$$\Rightarrow \frac{1}{n!} + \frac{1}{1!} \frac{1}{(n-1)!} + \frac{1}{2!} \frac{1}{(n-2)!} + \dots + \frac{1}{n!}$$

$$\Rightarrow \frac{1}{n!} + \frac{1}{1!} \frac{n}{n(n-1)!} + \frac{1}{2!} \frac{n(n-1)}{n(n-1)(n-2)!} + \dots + \frac{1}{n!}$$

$$\Rightarrow \left[\frac{1}{n!} + \frac{1}{1!} \frac{n}{n!} + \frac{1}{2!} \frac{n(n-1)}{n!} + \dots + \frac{1}{n!} \right]$$

$$= \frac{1}{n!} \left[1 + \frac{n}{1!} + \frac{n(n-1)}{2!} + \dots + 1 \right]$$

$$= \frac{1}{n!} [1 + 1]^n = \frac{2^n}{n!}$$

Applications of z-transforms -

One of the most important application of z-transform is Solving Some linear difference Equations. z-transform is also one of the most effective methods for Solving difference Equation of convolution type.

Linear difference Equations with constant Coefficients

A linear difference Equation with constant coefficients has the form

$$y[n+k] + a_1 y[n+k-1] + a_2 y[n+k-2] + \dots + a_k y[n] = x[n]$$

where a_i 's are real or complex constants

for all ~~$n \geq 0$~~ , $i = 1, 2, \dots, k$

If $x(n) = 0$, then the Equation is called Homogeneous otherwise it is non-homogeneous.

→ The order of linear difference Equation is the difference between the largest and smallest

Indices of unknown Sequences $x(n)$.

→ If the initial values $y(0), y(1), y(2), \dots, y(a-1)$ are all given then Eq (1) is called initial value problem.

Working procedure to Solve difference Equations
The linear difference Equation with constant coefficients of order k is

$$y_{(n+k)} + a_1 y_{(n+k-1)} + a_2 y_{(n+k-2)} + \dots + a_k y_n = f(n)$$

Where $a_0, a_1, a_2, \dots, a_k$ are constants.

Step-①: Take z -Transform of the given difference Eqⁿ

Step-②: Use the formula

$$z(y_n) = Y(z)$$

$$z(y_{n+1}) = z[Y(z) - y_0]$$

$$z[y_{n+2}] = z^2[Y(z) - y_0 - y_1 z^{-1}]$$

$$z[y_{n+3}] = z^3[Y(z) - y_0 - y_1 z^{-1} - y_2 z^{-2}]$$

Step(3): Simplify and find $Y(z)$

Step(4): Take the Inverse z -transform of $Y(z)$
and find the solution y_n

* Solve $y_{n+1} - 2y_n = 0$

Given $y_{n+1} - 2y_n = 0$

Apply z -transform on both sides

$\Rightarrow z[y_{n+1} - 2y_n] = 0$

$\Rightarrow z[y_{n+1}] - 2z[y_n] = 0 \quad \text{--- (1)}$

We know that

$\Rightarrow z[y_n] = Y(z)$

$\Rightarrow z[y_{n+1}] = z[Y(z) - y_0]$

Sub these formula in Eq (1):

$\Rightarrow z[Y(z) - y_0] - 2Y(z) = 0$

$\Rightarrow zY(z) - zy_0 - 2Y(z) = 0$

$\Rightarrow Y(z)(z - 2) = zy_0$

$\Rightarrow Y(z) = \frac{zy_0}{z - 2}$

$\Rightarrow Y(z) = y_0 \left[\frac{z}{z - 2} \right]$

Apply z^{-1} on both sides

$$\Rightarrow Z^{-1}[Y(z)] = y_0 Z^{-1}\left[\frac{z}{z-2}\right]$$

$$\Rightarrow y_n = y_0 2^n$$

* Solve $y_{n+2} - 7y_{n+1} + 12y_n = 0$ using $y_0 = 1$ and $y_1 = 2$ using z -transforms.

Given,

$$y_{n+2} - 7y_{n+1} + 12y_n = 0 \text{ and } y_0 = 1 \text{ \& } y_1 = 2.$$

$$Z[y_{n+2} - 7y_{n+1} + 12y_n] = 0$$

$$Z[y_{n+2}] - 7Z[y_{n+1}] + 12Z[y_n] = 0$$

$$Z^2[y(z) - y_0 - y_1 z^{-1}] - 7[Z[y(z) - y_0]] + 12y(z) = 0$$

$$Z^2 y(z) - Z^2(1) - 2(z) - 7Z y(z) + 7(1)Z + 12y(z) = 0$$

$$Z^2 y(z) - Z^2 - 2z - 7Z y(z) + 7z + 12y(z) = 0$$

$$y(z)[Z^2 - 7Z + 12] - Z^2 + 5z = 0$$

$$y(z)[Z^2 - 7Z + 12] = Z^2 - 5z$$

$$y(z) = \frac{Z^2 - 5z}{Z^2 - 7Z + 12} \quad \text{--- (1)}$$

Now apply partial fraction method

$$\frac{y(z)}{z} = \frac{Z[z-5]}{Z[Z^2 - 7Z + 12]}$$

$$\frac{y(z)}{z} = \frac{z-5}{(z-3)(z-4)}$$

Now $\frac{z-5}{(z-3)(z-4)}$ resolve into P.F

$$\frac{z-5}{(z-3)(z-4)} = \frac{A}{z-3} + \frac{B}{z-4}$$

$$z-5 = A(z-4) + B(z-3) \quad \text{--- (2)}$$

Put $z=4$ in (2) for $z=3$ in (2)

$$\begin{array}{l|l} -1 = B(1) & -2 = A(3-4) \\ \hline \boxed{B = -1} & \boxed{A = 2} \end{array}$$

$$\frac{y(z)}{z} = \frac{z-5}{(z-3)(z-4)} = \frac{2}{z-3} - \frac{1}{z-4}$$

Multiply with z

$$z \frac{y(z)}{z} = \frac{2z}{z-3} - \frac{z}{z-4}$$

Apply on both

$$z^{-1}[y(z)] = z^{-1} \left[\frac{2z}{z-3} \right] - z^{-1} \left[\frac{z}{z-4} \right]$$

$$y_n = 2z^{-1} \left[\frac{z}{z-3} \right] - z^{-1} \left[\frac{z}{z-4} \right]$$

$$y_n = 2 \cdot 3^n - 4^n$$

$$* y_{n+2} - 4y_{n+1} + 3y_n = 5^n$$

$$\text{Given } y_{n+2} - 4y_{n+1} + 3y_n = 5^n$$

Apply Z transform on both sides

$$Z[y_{n+2} - 4y_{n+1} + 3y_n] = Z[5^n]$$

$$Z[y_{n+2}] - 4Z[y_{n+1}] + 3Z[y_n] = Z[5^n]$$

$$Z^2[y(z) - y_0 - y_1 Z^{-1}] - 4Z[y(z) - y_0] + 3y(z) = \frac{Z}{Z-5}$$

$$Z^2 y(z) - Z^2 y_0 - Z^2 y_1 Z^{-1} - 4Z y(z) + 4Z y_0 + 3y(z) = \frac{Z}{Z-5}$$

$$y(z)[Z^2 - 4Z + 3] - y_1(z) - y_0[Z^2 - 4Z] = \frac{Z}{Z-5}$$

$$y(z)[(Z-1)(Z-3)] = \frac{Z}{Z-5} + Z y_1 + y_0[Z^2 - 4Z]$$

$$y(z) = \frac{\frac{Z}{Z-5} + Z y_1 + y_0[Z^2 - 4Z]}{(Z-1)(Z-3)}$$

$$= \frac{Z + Z y_1(Z-5) + y_0(Z^2 - 4Z)(Z-5)}{(Z-1)(Z-3)(Z-5)}$$

$$y(z) = \frac{Z + Z^2 y_1 - 5Z y_1 + Z^3 y_0 - 4Z^2 y_0 - 5Z^2 y_0 + 20Z y_0}{(Z-1)(Z-3)(Z-5)}$$

$$\frac{y(z)}{Z} = \frac{Z[1 + Z y_1 - 5 y_1 + Z^2 y_0 - 4Z y_0 - 5Z y_0 + 20 y_0]}{Z[(Z-1)(Z-3)(Z-5)]}$$

$$= \frac{1 + 2y_1 - 5y_1 + 2^2 y_0 - 92y_0 + 20y_0}{(z-1)(z-3)(z-5)}$$

Resolve into partial fraction

$$\frac{1 + 2y_1 - 5y_1 + 2^2 y_0 - 92y_0 + 20y_0}{(z-1)(z-3)(z-5)} = \frac{A}{(z-1)} + \frac{B}{(z-3)} + \frac{C}{(z-5)}$$

$$1 + 2y_1 - 5y_1 + 2^2 y_0 - 92y_0 + 20y_0 = A(z-3)(z-5) + B(z-1)(z-5) + C(z-1)(z-3)$$

$$z = 3$$

$$1 + 3y_1 - 5y_1 + 9y_0 - 27y_0 + 20y_0 = -4B$$

$$1 - 2y_1 + 2y_0 = -4B$$

$$B = \frac{-(1 - 2y_1 + 2y_0)}{4}$$

$$B = \frac{2y_1 - 2y_0 - 1}{4}$$

$$z = 5$$

$$1 + 5y_1 - 5y_1 + 25y_0 - 45y_0 + 20y_0 = 8C$$

$$8C = 1$$

$$C = \frac{1}{8}$$

$$\text{If } z=1$$

$$1 + y_1 - 5y_1 + y_0 - 9y_0 + 20y_0 = 8A$$

$$1 - 4y_1 + 12y_0 = 8A$$

$$A = \frac{12y_0 - 4y_1 + 1}{8}$$

$$\frac{y(z)}{z} = \frac{1 + zy_1 - 5y_1 + zy_0 - 9zy_0 + 20y_0}{(z-1)(z-3)(z-5)} = \frac{1 - 4y_1 + 12y_0}{8(z-1)} +$$

$$\frac{2y_1 - 2y_0 - 1}{4(z-3)} + \frac{2y_1 - 2y_0 - 1}{8(z-5)}$$

multiply with z

$$z \frac{y(z)}{z} = \frac{1 - 4y_1 + 12y_0}{8} \cdot \frac{z}{(z-1)} + \frac{2y_1 - 2y_0 - 1}{4} \cdot \frac{z}{(z-3)} + \frac{2y_1 - 2y_0 - 1}{8} \cdot \frac{z}{(z-5)}$$

Apply z^{-1}

$$z^{-1}[y(z)] = \frac{1 - 4y_1 + 12y_0}{8} \cdot \frac{1}{z-1} + \frac{2y_1 - 2y_0 - 1}{4} \cdot \frac{1}{z-3} + \frac{2y_1 - 2y_0 - 1}{8} \cdot \frac{1}{z-5}$$

$$y_n = \frac{1 - 4y_1 + 12y_0}{8} (1^n) + \frac{2y_1 - 2y_0 - 1}{4} (3^n) + \frac{1}{8} (5^n)$$

substitute values for y_1, y_0

* Solve $y_{n+2} - 4y_{n+1} + 3y_n = 0$ using $y_0 = 2$ and $y_1 = 1$ using z-Transform

Given $y_{n+2} - 4y_{n+1} + 3y_n = 0$

Apply z transform

$$z[y_{n+2} - 4y_{n+1} + 3y_n] = 0$$

$$z[y_{n+2}] - 4z[y_{n+1}] + 3z[y_n] = 0$$

$$z^2[y(z) - y_0 - y_1 z^{-1}] - 4z[y(z) - y_0] + 3y(z) = 0$$

$$z^2 y(z) - z^2 y_0 - z^2 y_1 z^{-1} - 4z y(z) + 4z y_0 + 3y(z) = 0$$

$$z^2 y(z) - 2z^2 - 4z - 4z y(z) + 8z + 3y(z) = 0$$

$$y(z) [z^2 - 4z + 3] = 2z^2 + 4z - 8z$$

$$y(z) = \frac{2z^2 - 4z}{z^2 - 4z + 3}$$

$$\left[\begin{array}{c} z \\ z-3 \end{array} \right] \cdot \left[\begin{array}{c} z \\ z-1 \end{array} \right] \cdot \left[\begin{array}{c} z \\ z-3 \end{array} \right] \cdot \left[\begin{array}{c} z \\ z-1 \end{array} \right] \cdot \left[\begin{array}{c} z \\ z-3 \end{array} \right] \cdot \left[\begin{array}{c} z \\ z-1 \end{array} \right]$$

$$\frac{y(z)}{z} = \frac{z(2z-4)}{z(z^2-4z+3)}$$

$$\frac{2z-4}{z^2-4z+3} = \frac{2z-4}{(z-1)(z-3)}$$

Resolving into partial fractions

$$\frac{2z-4}{(z-1)(z-3)} = \frac{A}{z-1} + \frac{B}{z-3}$$

$$2z-4 = A(z-3) + B(z-1)$$

$$\text{If } z=3$$

$$6-4=2B$$

$$2B=2$$

$$\boxed{B=1}$$

$$\text{If } z=1$$

$$2-4=-2A$$

$$-2A=-2$$

$$\boxed{A=1}$$

$$\frac{Y(z)}{z} = \frac{2z-4}{(z-1)(z-3)} = \frac{1}{(z-1)} + \frac{1}{(z-3)}$$

Multiply with z

$$z, \frac{Y(z)}{z} = \frac{z}{(z-1)} + \frac{z}{z-3}$$

$$Y(z) = \frac{z}{(z-1)} + \frac{z}{(z-3)}$$

Apply z^{-1} on both sides

$$z^{-1} [Y(z)] = z^{-1} \left[\frac{z}{(z-1)} \right] + z^{-1} \left[\frac{z}{(z-3)} \right]$$

$$\boxed{Y_n = 1^n + 3^n}$$

UNIT - 4

MULTIPLE INTEGRALS

Definite Integral: The concept of Definite integral a to b

$$\int_a^b f(x) dx \quad \text{--- (1)}$$

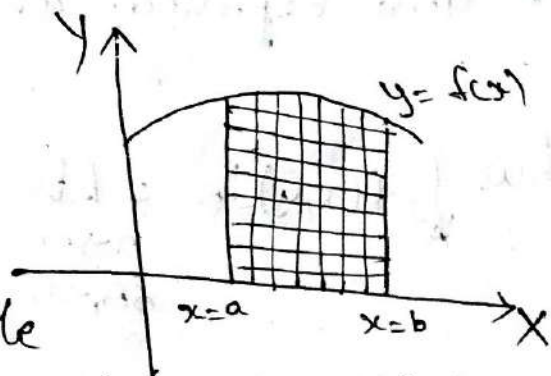
Is the area under the curve $y=f(x)$, the x-axis and 2-ordinates $x=a$ and $x=b$. It is defined as limit of the sum

$$f(x_1)\delta x_1 + f(x_2)\delta x_2 + \dots + f(x_n)\delta x_n$$

When $n \rightarrow \infty$ and Each of the length

$$\delta x_1, \delta x_2, \dots, \delta x_n \rightarrow 0; n \rightarrow \infty$$

Double Integral :-

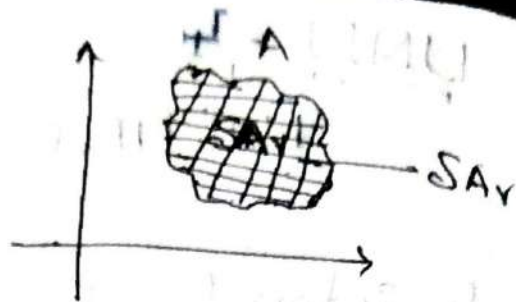


Let $f(x,y)$ be a single value and bounded function of 2 Independent variables. x and y defined in a closed region A in x,y plane.

Let A be divided into n Elementary areas $\delta A_1, \delta A_2, \dots, \delta A_n$. Let x_i, y_i

be any point inside the i th Elementary

area ΔA_r



* Consider the Sum

$$f(x_1, y_1) \Delta A_1 + f(x_2, y_2) \Delta A_2 + \dots + f(x_n, y_n) \Delta A_n \\ = \sum_{r=1}^n f(x_r, y_r) \Delta A_r$$

Then the Limit of the above Sum, exists as $n \rightarrow \infty$ and Each Sub Elementary area Approaches to Zero "0" is termed as Double Integral of $f(x, y)$ over the region A and Expressed as $\iint_A f(x, y) dA$

$$\text{Thus } \iint_A f(x, y) dA = \lim_{\substack{n \rightarrow \infty \\ \Delta A_r \rightarrow 0}} \sum_{r=1}^n f(x_r, y_r) \Delta A_r$$

* Evaluation of Double Integral.

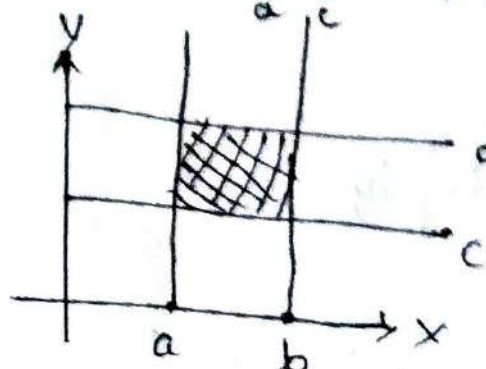
Evaluation of double integral.

$\iint_R f(x, y) dx dy$ the Bounds of given region may be following types,

* The Bounds x and y are constants

for given domain $x = a$ to b ; $y = c$ to d

$$\left| \iint_{xy} f(x,y) dx dy = \int_a^b \int_c^d f(x,y) dx dy \right|$$

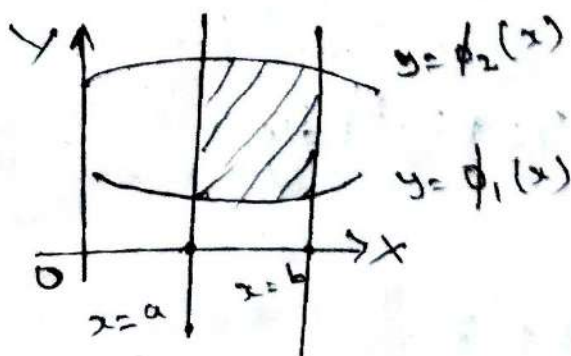


[Independent
co-ordinates]

* The Boundaries of x are constants the y are functions of x .

$$x = a \text{ to } b ; y = \phi_1(x) \text{ to } \phi_2(x)$$

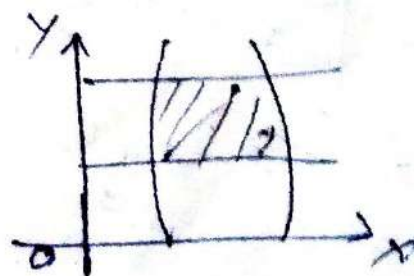
$$\left| \iint_{xy} f(x,y) dx dy = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x,y) dx dy \right| \left[\begin{array}{l} \text{dependent} \\ \text{co-ordinates} \end{array} \right]$$



* The Boundaries of y are constant and x are function of y .

$$x = \psi_1(y) \text{ to } \psi_2(y)$$

$$y = c \text{ to } d$$



$$\left| \iint_{yx} f(x,y) dy dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x,y) dx dy \right|$$

$$* \int_0^1 \int_0^2 (x^2 + y^2) dx dy$$

Given $\int_0^1 \int_0^2 (x^2 + y^2) dx dy$

$$\Rightarrow \int_{y=0}^1 \left[\int_0^2 (x^2 + y^2) dx \right] dy$$

$$= \int_{y=0}^1 \left[\int_0^2 x^2 dx + \int_0^2 y^2 dx \right] dy$$

$$= \int_{y=0}^1 \left[\frac{x^3}{3} + y^2 x \right]_0^2 dy$$

$$= \int_0^1 \left[\frac{8}{3} + 2y^2 - 0 \right] dy$$

$$= \int_0^1 \left(\frac{8}{3} + 2y^2 \right) dy$$

$$= \left[\frac{8}{3} y + 2 \cdot \frac{y^3}{3} \right]_0^1$$

$$= \left[\frac{8}{3} (1) + 2 \cdot \frac{(1)^3}{3} - 0 \right]$$

$$= \frac{8}{3} + \frac{2}{3}$$

$$= \frac{10}{3}$$

* Evaluate $\int_0^1 \int_x^{x^2} xy \, dx \, dy$

Given $\Rightarrow \int_{x=0}^1 \int_{y=x}^{x^2} xy \, dx \, dy$

$$= \int_{x=0}^1 \int_{y=x}^{x^2} xy \, dy \, dx$$

$$= \int_0^1 \left[x \cdot \frac{y^2}{2} \right]_x^{x^2} dx$$

$$= \int_0^1 \left[\frac{x^5}{2} - \frac{x^3}{2} \right] dx$$

$$= \left[\frac{x^6}{2 \times 6} - \frac{x^4}{2 \times 4} \right]_0^1$$

$$= \left[\frac{x^6}{12} - \frac{x^4}{8} \right]_0^1$$

$$= \left[\left(\frac{1}{12} - \frac{1}{8} \right) - 0 \right]$$

$$= \frac{8-12}{12 \times 8} = \frac{-4}{12 \times 8}$$

$$= -\frac{1}{24}$$

$$* \int_0^1 \int_0^y x \, dy \, dx$$

Given

$$\int_{y=0}^1 \int_{x=0}^y x \, dx \, dy$$

$$= \int_{y=0}^1 \left[\int_0^y x \, dx \right] dy$$

$$= \int_{y=0}^1 \left[\frac{x^2}{2} \right]_0^y dy$$

$$= \int_0^1 \frac{y^2}{2} dy$$

$$= \left[\frac{y^3}{6} \right]_0^1 = \left[\frac{1}{6} - 0 \right]$$

$$= \frac{1}{6}$$

$$* \int_0^2 \int_0^x e^{x+y} \, dy \, dx$$

Given

$$\int_{x=0}^2 \int_{y=0}^x e^{x+y} \, dy \, dx$$

$$\Rightarrow \int_{x=0}^2 \left[\int_{y=0}^x e^{x+y} \, dy \right] dx$$

$$\Rightarrow \int_{x=0}^2 \int_{y=0}^x e^x dy dx$$

$$\Rightarrow \int_0^2 \left[\int_0^x e^y dy \right] dx$$

$$= \int_0^2 e^x [e^y]_0^x dx$$

$$= \int_0^2 e^x [e^x - e^0] dx$$

$$= \int_0^2 [e^{2x} - e^x] dx$$

$$= \left[\frac{e^{2x}}{2} - \frac{e^x}{1} \right]_0^2$$

$$= \frac{e^4}{2} - \frac{e^2}{1} - \left[\frac{e^0}{2} - \frac{e^0}{1} \right]$$

$$= \frac{e^4}{2} - e^2 - \frac{1}{2} + 1$$

$$\therefore = \frac{e^4}{2} - e^2 + \frac{1}{2}$$

$$= \frac{1}{2} [e^4 - 2e^2 + 1]$$

$$= \frac{1}{2} [(e^2)^2 - 2e^2 + 1]$$

$$= \frac{1}{2} [(e^2 - 1)^2]$$

* $\iint_R xy(x+y) dx dy$, over the region bounded by the curves $x^2=y$ and $y=x$.

Sol Given $x^2=y$ and $y=x$

Here y varies from x^2 to x

Here $x^2=y$

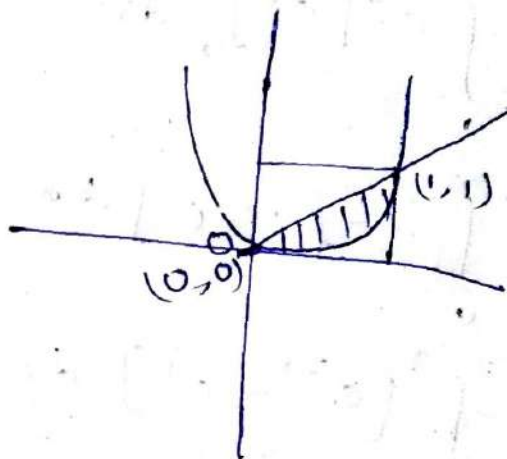
$$\Rightarrow x^2 - y = 0$$

$$\Rightarrow x^2 - x = 0$$

$$\Rightarrow x(x-1) = 0$$

$$\Rightarrow x=0, x-1=0$$

$$x=1$$



$\therefore x$ varies from 0 to 1

$$\text{Consider } \iint_R xy(x+y) dx dy = \int_{x=0}^1 \int_{y=x^2}^x xy(x+y) dy dx$$

$$= \int_{x=0}^1 \int_{y=x^2}^x (x^2y + xy^2) dy dx$$

$$= \int_{x=0}^1 \left[\int_{x^2}^x (x^2y + xy^2) dy \right] dx$$

$$= \int_0^1 \left[x^2 \frac{y^2}{2} + \frac{xy^3}{3} \right]_{x^2}^x dx$$

$$= \int_{x=0}^1 \left(\frac{x^4}{2} + \frac{x^4}{3} - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$$

$$= \int_0^1 \left(\frac{5}{6} x^4 - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$$

* Evaluate $\iint_R y \, dx \, dy$ where R is a region.

Bounded by $y^2 = 4x$ and $x^2 = 4y$.

Given $y^2 = 4x$ and $x^2 = 4y$

$$\Rightarrow x = \frac{y^2}{4} \Rightarrow x = 2\sqrt{y}$$

x varies from $\frac{y^2}{4}$ to $2\sqrt{y}$

Now $y^2 = 4x$

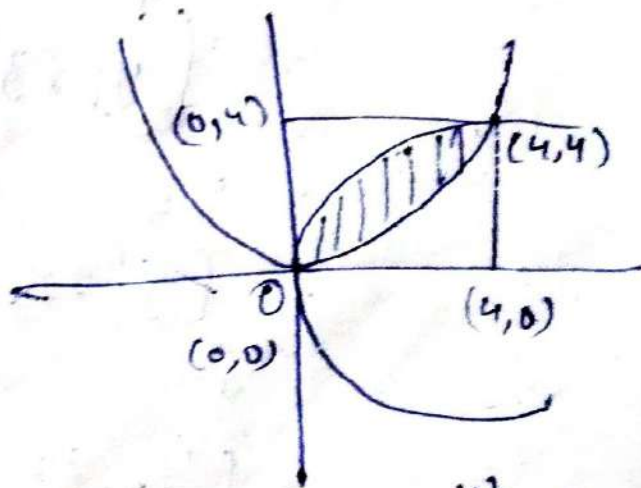
$$y^2 = 4x = 0$$

$$y^2 = 4 \cdot 2\sqrt{y} = 0$$

$$y^2 - 8\sqrt{y} = 0$$

$$y\sqrt{y} \cdot \sqrt{y} - 8\sqrt{y} = 0$$

$$\sqrt{y} (y\sqrt{y} - 8) = 0$$



$$y=0 \Rightarrow x = \frac{y^2}{4} = 0$$

$$y=4 \Rightarrow x = \frac{y^2}{4} = 4$$

$\therefore (x,y) = (0,0) \text{ and } (4,4)$ are two intersect

$$\Rightarrow \sqrt{y} = 0, \quad y\sqrt{y} - 8 = 0$$

$$\Rightarrow y = 0, \quad y\sqrt{y} = 8$$

$$\Rightarrow y^{3/2} = 2^3$$

$$= (y^{3/2})^2 = (2^3)^2$$

$$\Rightarrow y^3 = (2^2)^3$$

$$\Rightarrow y = 2^2$$

$$\Rightarrow y = 4$$

$\therefore y$ varies from 0 to 4

$$\text{Consider } \iint_R y \, dx \, dy = \int_{y=0}^4 \int_{x=\frac{y^2}{4}}^{2\sqrt{y}} y \, dx \, dy$$

$$= \int_{y=0}^4 \left[y \int_{x=\frac{y^2}{4}}^{2\sqrt{y}} dx \right] dy$$

$$= \int_0^4 y [x]_{y^2/4}^{2\sqrt{y}} dy$$

$$= \int_0^4 y \left[2\sqrt{y} - \frac{y^2}{4} \right] dy$$

$$= \int_0^4 \left[2y^{3/2} - \frac{y^3}{4} \right] dy$$

$$\rightarrow \left[2 \cdot \frac{y^{5/2}}{5/2} - \frac{1}{4} \cdot \frac{y^4}{4} \right]_0^4$$

$$= \left[\frac{4}{5} y^{5/2} - \frac{1}{16} y^4 \right]_0^4$$

$$= \left[\frac{4}{5} \cdot (4)^{5/2} - \frac{1}{16} (4)^4 - [0] \right]$$

$$= \frac{4}{5} (2^4)^{5/2} - \frac{4^2 \cdot 4^2}{16}$$

$$= \frac{4}{5} (32) - 16$$

$$= \frac{128}{5} - 16$$

$$= \frac{48}{5} = 9.6$$

* Evaluate $\iint_D xy \, dx \, dy$ over the region of positive co-ordinate of the circle $x^2 + y^2 = a^2$.

$$\Rightarrow x=0, x^2 + y^2 = a^2$$

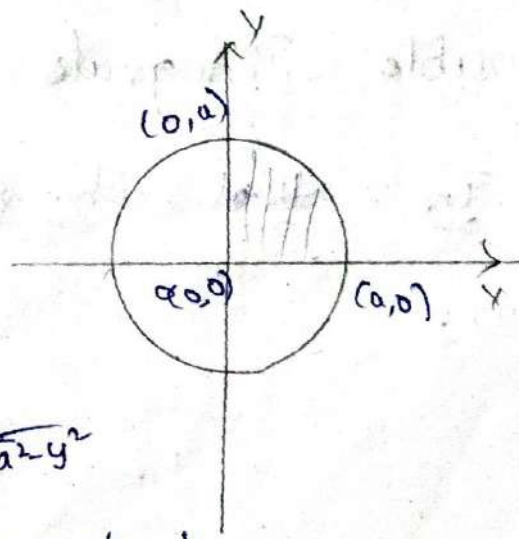
$$\Rightarrow x^2 = a^2 - y^2$$

$$\Rightarrow x = \sqrt{a^2 - y^2}$$

x varies from 0 to $\sqrt{a^2 - y^2}$

y varies from 0 to a

Consider $\iint_D xy \, dx \, dy = \int_{y=0}^a \int_{x=0}^{\sqrt{a^2-y^2}} xy \, dx \, dy$



$$= \int_{y=0}^a y \left[\frac{x^2}{2} \right]_0^{\sqrt{a^2-y^2}} dy$$

$$= \int_{y=0}^a \frac{y}{2} \left[(\sqrt{a^2-y^2})^2 - 0 \right] dy$$

$$= \int_{y=0}^a \frac{y}{2} [a^2 - y^2] dy$$

$$= \frac{1}{2} \int_{y=0}^a [ya^2 - y^3] dy$$

$$= \frac{1}{2} \left[\frac{a^2 y^2}{2} - \frac{y^4}{4} \right]_0^a$$

$$= \frac{1}{2} \left[\frac{a^4}{2} - \frac{a^4}{4} - (0) \right]$$

$$= \frac{a^4}{2} \left[\frac{1}{2} - \frac{1}{4} \right]$$

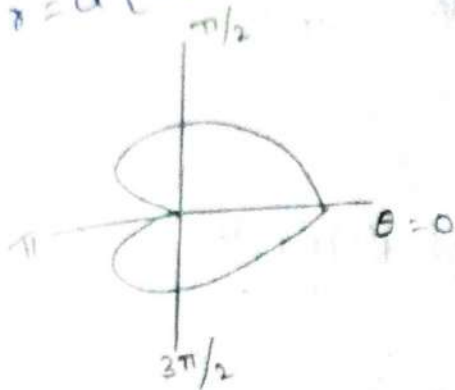
$$= \frac{a^4}{8} \text{ Squares}$$

Double Integrals in polar form:

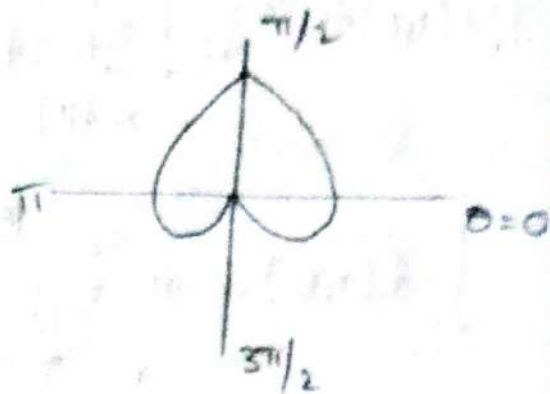
* In that shapes of curves in polar form.

i) cardioid

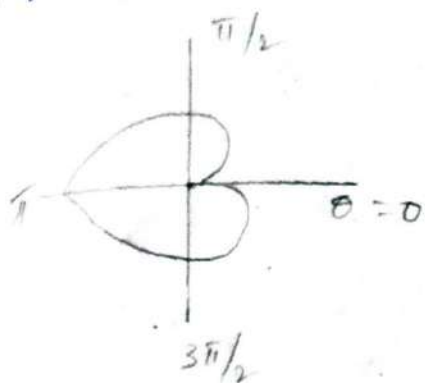
(a) $r = a(1 + \cos\theta)$



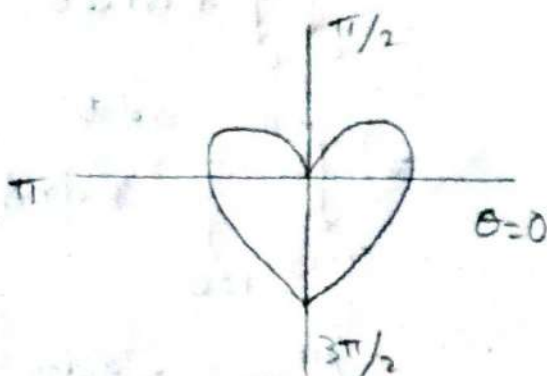
(b) $r = a(1 + \sin\theta)$



(c) $r = a(1 - \cos\theta)$

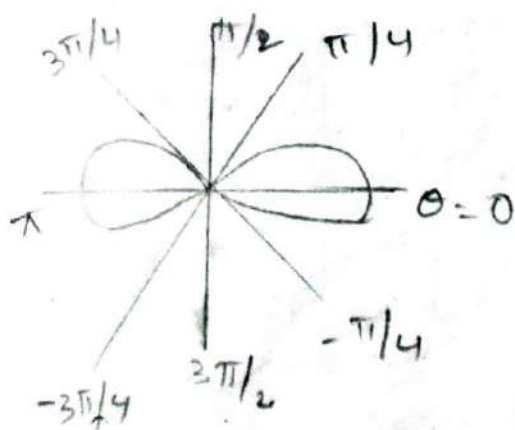


(d) $r = a(1 - \sin\theta)$

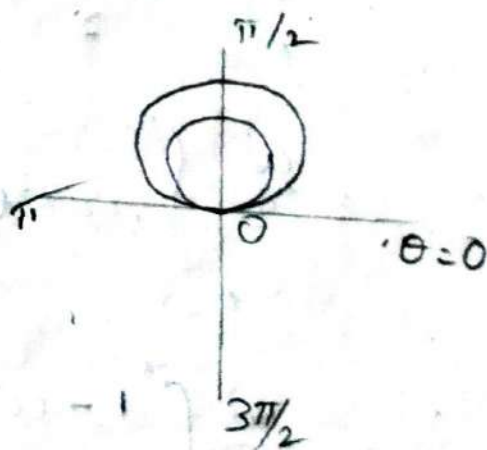


ii) Lemniscate:

(a) $r^2 = a^2 \cos 2\theta$



(b) $r = 2 \sin\theta$; $r = 4 \sin\theta$



$$\iint f(r, \theta) dr d\theta = \int_{\theta=\alpha}^{\beta} \int_{r=f(\theta)}^{f_2(\theta)} f(r, \theta) dr d\theta \quad \text{and}$$

$$\iint f(r, \theta) dr d\theta = \int_{r=\alpha}^{\beta} \int_{\theta=f(r)}^{f_2(r)} f(r, \theta) dr d\theta$$

* Evaluate: $\int_0^{\pi} \int_0^{a \sin \theta} r \cdot dr d\theta$

$$= \int_{\theta=0}^{\pi} \int_{r=0}^{a \sin \theta} r \cdot dr d\theta$$

$$= \int_{\theta=0}^{\pi} \left[\frac{r^2}{2} \right]_0^{a \sin \theta} d\theta$$

$$= \int_{\theta=0}^{\pi} \frac{a^2 \sin^2 \theta}{2} \cdot d\theta$$

$$= \frac{a^2}{2} \int_{\theta=0}^{\pi} \sin^2 \theta \cdot d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= \frac{a^2}{4} \int_0^{\pi} [1 - \cos 2\theta] d\theta$$

$$= \frac{a^2}{4} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi}$$

$$= \frac{a^2}{4} \left[\pi - \frac{\sin 2\pi}{2} - \left[0 - \frac{\sin 0}{2} \right] \right]$$

$$= \frac{a^2}{4} [\pi - 0]$$

$$= \frac{\pi a^2}{4}$$

* Evaluate $\iint r \sin \theta \, dr \, d\theta$ over the cardioid
 $r = a(1 - \cos \theta)$ above the initial line

Sol = Given $r = a(1 - \cos \theta)$

$$r = 0 \text{ to } a(1 - \cos \theta)$$

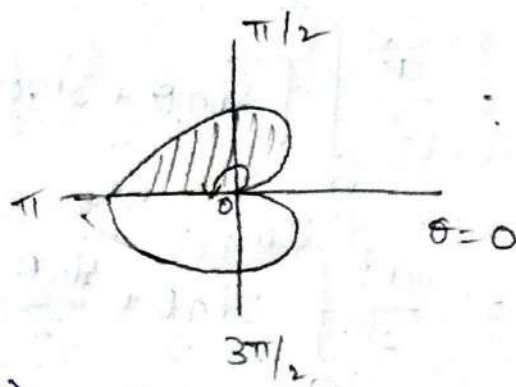
$$\theta = 0 \text{ to } \pi$$

consider

$$\iint r \sin \theta \, dr \, d\theta = \int_{\theta=0}^{\pi} \int_{r=0}^{a(1-\cos \theta)} r \sin \theta \, dr \, d\theta$$

$$= \int_{\theta=0}^{\pi} \sin \theta \left[\int_{r=0}^{a(1-\cos \theta)} r \, dr \right] d\theta$$

$$= \int_{\theta=0}^{\pi} \sin \theta \left[\frac{r^2}{2} \right]_0^{a(1-\cos \theta)} d\theta$$



$$= \int_{\theta=0}^{\pi} \frac{\sin \theta}{2} [a^2(1-\cos \theta)^2 - 0] d\theta$$

$$= \frac{1}{2} \int_0^{\pi} \sin \theta [a^2(1-\cos \theta)^2] d\theta$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi} \sin \theta [a^2(1 + \cos^2 \theta - 2\cos \theta)] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} [\sin \theta + \sin \theta \cos^2 \theta - 2\cos \theta \sin \theta] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \left[\sin \theta + \sin \theta \left[\frac{1 + \cos 2\theta}{2} \right] - \sin 2\theta \right] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \left[\sin \theta + \frac{\sin \theta}{2} + \frac{\sin \theta \cos 2\theta}{2} - \sin 2\theta \right] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \left[\frac{3}{2} \sin \theta + \frac{2\sin \theta \cos 2\theta}{4} - \sin 2\theta \right] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \left[\frac{3}{2} \sin \theta + \frac{\sin(\theta+2\theta) + \sin(\theta-2\theta)}{4} - \sin 2\theta \right] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \left[\frac{3}{2} \sin \theta + \frac{\sin 3\theta}{4} + \frac{\sin(-\theta)}{4} - \sin 2\theta \right] d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi} \left[\frac{3}{2} \sin \theta + \frac{\sin 3\theta}{4} - \frac{\sin \theta}{4} - \sin 2\theta \right] d\theta$$

$$= \frac{a^2}{2} \left[\frac{3}{2}(-\cos\theta) + \frac{1}{4} \left[-\frac{\cos 3\theta}{3} \right] - \frac{1}{4} \left[-\cos\theta \right] - \left[-\frac{\cos 2\theta}{2} \right] \right]_0^\pi$$

$$= \frac{a^2}{2} \left[-\frac{3}{2} \cos\theta - \frac{1}{12} \cos 3\theta + \frac{1}{4} \cos\theta + \frac{\cos 2\theta}{2} \right]_0^\pi$$

$$= \frac{a^2}{2} \left[-\frac{3}{2} \cos\pi - \frac{1}{12} \cos 3\pi + \frac{1}{4} \cos\pi + \frac{\cos 2\pi}{2} - \left[-\frac{3}{2} \cos\theta - \frac{1}{12} \cos 3\theta + \frac{1}{4} \cos\theta + \frac{\cos 2\theta}{2} \right] \right]$$

$$= \frac{a^2}{2} \left[\frac{3}{2} + \frac{1}{12} - \frac{1}{4} + \frac{1}{2} + \frac{3}{2} + \frac{1}{12} - \frac{1}{4} - \frac{1}{2} \right]$$

$$= \frac{a^2}{2} \left[3 + \frac{1}{6} - \frac{1}{2} \right]$$

$$= \frac{a^2}{2} \times \frac{8}{3} \Rightarrow \frac{8a^2}{3}$$

$$\Rightarrow \frac{4a^2}{3}$$

* Evaluate $\iint \frac{r}{\sqrt{a^2+r^2}} dr d\theta$ over one loop of

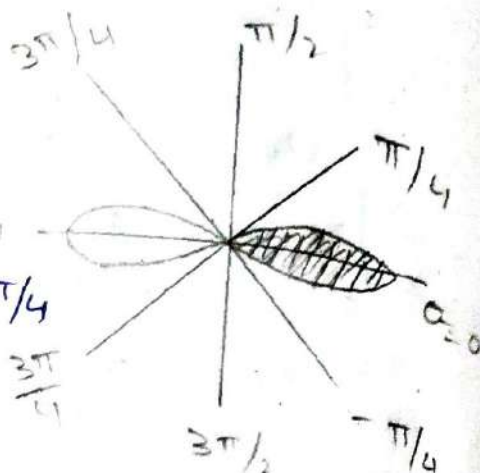
the ~~cardioid~~ ^{limaçon} _{scale} $r^2 = a^2 \cos 2\theta$
 Lemniscate

Given

$$r^2 = a^2 \cos 2\theta$$

Here θ varies from $-\pi/4$ to $\pi/4$
 and $r^2 = a^2 \cos 2\theta$

$$\Rightarrow r = a \sqrt{\cos 2\theta}$$



r varies from 0 to $a \sqrt{\cos 2\theta}$

consider
$$\iint \frac{r}{\sqrt{a^2+r^2}} dr d\theta = \int_{\theta=-\pi/4}^{\pi/4} \int_{r=0}^{a\sqrt{\cos 2\theta}} \frac{r}{\sqrt{a^2+r^2}} dr d\theta$$

$$= \frac{1}{2} \int_{\theta=-\pi/4}^{\pi/4} \int_{r=0}^{a\sqrt{\cos 2\theta}} \frac{2r}{\sqrt{a^2+r^2}} dr d\theta$$

$$= \frac{1}{2} \int_{\theta=-\pi/4}^{\pi/4} \left[2\sqrt{a^2+r^2} \right]_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$= \frac{1}{2} \int_{-\pi/4}^{\pi/4} \left[\sqrt{a^2 + (a\sqrt{\cos 2\theta})^2} - \sqrt{a^2+0} \right] d\theta$$

$$= \int_{-\pi/4}^{\pi/4} [\sqrt{a^2 + a^2 \cos 2\theta} - a] d\theta$$

$$= \int_{-\pi/4}^{\pi/4} [\sqrt{a^2(1 + \cos 2\theta)} - a] d\theta$$

$$= \int_{-\pi/4}^{\pi/4} [a\sqrt{2\cos^2\theta} - a] d\theta$$

$$= a \int_{-\pi/4}^{\pi/4} [\sqrt{2}\cos\theta - 1] d\theta$$

$$= a \left[\sqrt{2} \frac{\sin\theta}{1} - \theta \right]_{-\pi/4}^{\pi/4}$$

$$= a \left[\sqrt{2} \sin \pi/4 - \frac{\pi}{4} - \left[\sqrt{2} \sin(-\pi/4) - \left(-\frac{\pi}{4}\right) \right] \right]$$

$$= a \left[\sqrt{2} \cdot \frac{1}{\sqrt{2}} - \frac{\pi}{4} + \sqrt{2} \cdot \frac{1}{\sqrt{2}} - \frac{\pi}{4} \right]$$

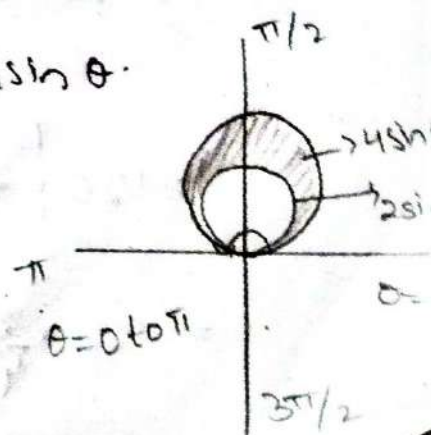
$$= a \left[2 - \frac{\pi}{2} \right]$$

* Evaluate $\iint r^3 dr d\theta$ over the area bounded by the circles $r = 2\sin\theta$ and $r = 4\sin\theta$.

Given $r = 2\sin\theta$

$$r = 4\sin\theta$$

r varies from $2\sin\theta$ to $4\sin\theta$



θ varies from 0 to π

consider
$$\int \int r^3 dr d\theta = \int_{\theta=0}^{\pi} \int_{r=2\sin\theta}^{4\sin\theta} r^3 dr d\theta$$

$$= \int_{\theta=0}^{\pi} \left[\frac{r^4}{4} \right]_{2\sin\theta}^{4\sin\theta} d\theta$$

$$= \int_{\theta=0}^{\pi} \left(\frac{(4\sin\theta)^4}{4} - \frac{(2\sin\theta)^4}{4} \right) d\theta$$

$$= \int_{\theta=0}^{\pi} \frac{1}{4} [256 \sin^4\theta - 16 \sin^4\theta] d\theta$$

$$= \int_0^{\pi} \frac{1}{4} [240 \sin^4\theta] d\theta$$

$$= 60 \int_0^{\pi} \sin^4\theta d\theta$$

$$= 60 \times 2 \int_0^{\pi/2} \sin^4\theta d\theta$$

$\because \int_0^{\pi/2} \sin^n\theta d\theta = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2}\theta d\theta$

$\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$

$$= 120 \left[\frac{3}{4} \int_0^{\pi/2} \sin^2\theta d\theta \right]$$

$$= 120 \times \frac{3}{4} \times \frac{1}{2} \int_0^{\pi/2} \sin\theta d\theta$$

$$= 120 \times \frac{3}{8} \int_0^{\pi/2} d\theta$$

$$= \frac{360}{8} [\theta]_0^{\pi/2}$$

$$= \frac{360}{8} \left[\frac{\pi}{2} \right]$$

$$= \frac{45\pi}{2}$$

~~Cartesian~~ form and polar form :-
Cartesian

We have $x = r \cos \theta$, $y = r \sin \theta$ (to convert cartesian form to polar form)

$$x = r \cos \theta, y = r \sin \theta$$

$$\frac{dx}{dr} = \cos \theta, \frac{dx}{d\theta} = -r \sin \theta$$

$$\frac{dy}{dr} = \sin \theta, \frac{dy}{d\theta} = r \cos \theta$$

$$J = \begin{vmatrix} dx/dr & dx/d\theta \\ dy/dr & dy/d\theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$= r [1]$$

$$= r$$

$$\iint f(x, y) dx dy = \iint f(r, \theta) \cdot J dr d\theta$$

$$= \iint f(r, \theta) \cdot r \cdot dr \cdot d\theta$$

* Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar

Co-ordinates Hence show that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$

Sol We know that

$$x = r \cos \theta, y = r \sin \theta \text{ and } dx dy = r dr d\theta$$

$$x^2 + y^2 = (r \cos \theta)^2 + (r \sin \theta)^2$$

$$= r^2 \cos^2 \theta + r^2 \sin^2 \theta$$

$$= r^2 [\cos^2 \theta + \sin^2 \theta]$$

$$= r^2$$

consider $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta$

$$\theta = 0 \text{ to } \pi/2$$

$$r = 0 \text{ to } \infty$$

Put $r^2 = t$

$$2r dr = dt$$

$$r dr = \frac{dt}{2}$$

$$r = 0 \text{ to } \infty$$

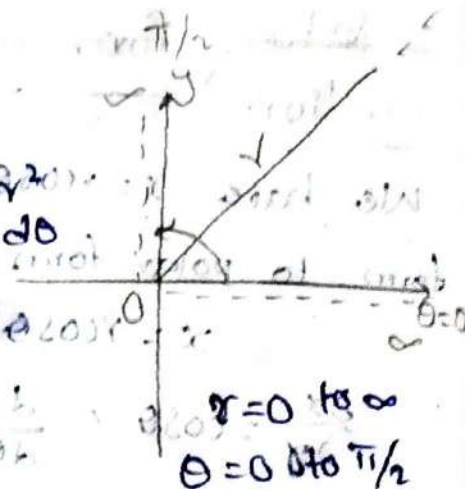
$$t = 0 \text{ to } \infty$$

$$= \int_0^{\pi/2} \int_0^\infty e^{-t} \frac{dt}{2} d\theta$$

$$= \int_0^{\pi/2} \frac{1}{2} \left[\int_0^\infty e^{-t} dt \right] d\theta$$

$$= \int_0^{\pi/2} \frac{1}{2} \left[\frac{e^{-t}}{-1} \right]_0^\infty d\theta = -\frac{1}{2} \int_0^{\pi/2} [e^{-\infty} - e^{-0}] d\theta$$

$$= -\frac{1}{2} \int_0^{\pi/2} [0 - 1] d\theta$$



$$= \frac{1}{2} \int_0^{\pi/2} 1 d\theta \Rightarrow \frac{1}{2} [\theta]_0^{\pi/2} = \frac{1}{2} [\pi/2 - 0]$$

$$= \frac{\pi}{4} \quad \text{--- (1)}$$

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \int_0^\infty \int_0^\infty e^{-x^2} e^{-y^2} dx dy$$

$$= \int_0^\infty e^{-x^2} dx \cdot \int_0^\infty e^{-y^2} dy$$

$$= \left[\int_0^\infty e^{-x^2} dx \right]^2 \quad \text{--- (2)}$$

From (1) & (2)

$$\left[\int_0^\infty e^{-x^2} dx \right]^2 = \frac{\pi}{2}$$

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

By changing into polar co-ordinates. Evaluate $\iint \frac{x^2 y^2}{x^2 + y^2} dx dy$ over the angular region b/w the circles $x^2 + y^2 = a^2$ & $x^2 + y^2 = b^2$ ($b > a$)

Sol we know that

$$x = r \cos \theta, \quad y = r \sin \theta \quad \text{and} \quad dx dy = r dr d\theta$$

$$x^2 + y^2 = (r \cos \theta)^2 + (r \sin \theta)^2$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta)$$

$$x^2 + y^2 = r^2$$

Given $x^2 + y^2 = a^2$

$$r^2 = a^2$$

$$\boxed{r = a}$$

$$x^2 + y^2 = b^2$$

$$r^2 = b^2$$

$$\boxed{r = b}$$

r varies from a to b .

$$\text{Consider } \iint \frac{x^2 y^2}{x^2 + y^2} dx dy = \int_{\theta=0}^{\pi} \int_{r=a}^b \frac{r^2 \cos^2 \theta r^2 \sin^2 \theta}{r^2} r dr d\theta$$

$$= \int_0^{2\pi} \int_a^b r^3 (\cos^2 \theta \sin^2 \theta) dr d\theta$$

$$= \int_0^{2\pi} (\cos^2 \theta \sin^2 \theta) \left[\frac{r^4}{4} \right]_a^b d\theta$$

$$= \int_0^{2\pi} (\cos^2 \theta \sin^2 \theta) \left[\frac{b^4 - a^4}{4} \right] d\theta$$

$$= \frac{b^4 - a^4}{4} \int_0^{2\pi} \frac{2^2 \cos^2 \theta \sin^2 \theta}{4} d\theta$$

$$= \frac{b^4 - a^4}{16} \int_0^{2\pi} (2 \sin^2 \theta \cos^2 \theta)^2 d\theta$$

$$= \frac{b^4 - a^4}{16} \int_0^{2\pi} (\sin 2\theta)^2 d\theta$$

$$= \frac{b^4 - a^4}{16} \int_0^{2\pi} \sin^2 2\theta d\theta$$

$$= \frac{b^4 - a^4}{16} \int_0^{2\pi} \frac{1 - \cos 4\theta}{2} d\theta$$

$$= \frac{b^4 - a^4}{32} \int_0^{2\pi} (1 - \cos 4\theta) d\theta$$

$$= \frac{b^4 - a^4}{32} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{2\pi}$$

$$= \frac{b^4 a^4}{32} \left[\theta - \frac{\sin 4\theta}{4} \right]_0^{2\pi}$$

$$= \frac{b^4 a^4}{32} \left[2\pi - \frac{\sin 8\pi}{4} - \left[0 - \frac{\sin 0}{4} \right] \right]$$

$$= \frac{b^4 a^4}{32} [2\pi]$$

$$= \frac{[b^4 a^4] \pi}{16}$$

By change into polar co-ordinates evaluate

$$\int_0^a \int_y^a \frac{x}{x^2+y^2} dy dx$$

We know that

$$x = r \cos \theta, y = r \sin \theta, dx dy = r dr d\theta.$$

$$x^2 + y^2 = r^2 [\cos^2 \theta + \sin^2 \theta] = r^2$$

x limits are y to a

i.e., $x = y$ & $x = a$

$$r \cos \theta = r \sin \theta$$

$$r \cos \theta - r \sin \theta = 0$$

$$r [\cos \theta - \sin \theta] = 0$$

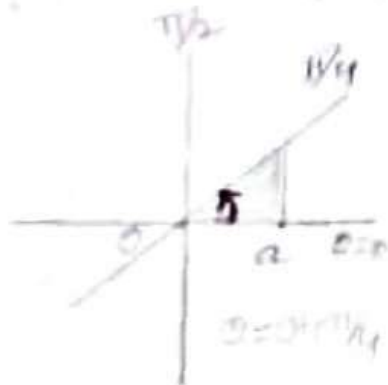
$$r = 0$$

r varies from 0 to $\frac{a}{\cos \theta}$

θ varies from 0 to $\pi/4$

$$r \cos \theta = a$$

$$r = \frac{a}{\cos \theta}$$



consider

$$\iint \frac{x}{x^2+y^2} dy dx = \int_0^{\pi/4} \int_0^{a/\cos \theta} \frac{x \cos \theta}{r^2} r dr d\theta$$

$$\begin{aligned}
&= \int_{\theta=0}^{\pi/4} \int_0^{a/\cos\theta} \cos\theta \, dr \, d\theta \\
&= \int_{\theta=0}^{\pi/4} \left[\cos\theta \left[r \right]_0^{a/\cos\theta} \right] d\theta \\
&= \int_{\theta=0}^{\pi/4} \cos\theta \left[r \right]_0^{a/\cos\theta} d\theta \\
&= \int_{\theta=0}^{\pi/4} \cos\theta \left[\frac{a}{\cos\theta} - 0 \right] d\theta \\
&= \int_{\theta=0}^{\pi/4} \cos\theta \cdot \frac{a}{\cos\theta} d\theta \\
&= a \int_{\theta=0}^{\pi/4} 1 d\theta \\
&= a \left[\theta \right]_0^{\pi/4} \\
&= a \left[\frac{\pi}{4} \right].
\end{aligned}$$

⇒ Triple Integrations :

*) Evaluate $\int_0^x \int_0^x \int_0^{x+y} (x+y+z) \, dz \, dy \, dx$

Sol: Given: $\int_0^x \int_0^x \int_0^{x+y} (x+y+z) \, dz \, dy \, dx$

$$\begin{aligned}
&= \int_0^x \int_0^x \int_0^{x+y} (x+y+z) \, dz \, dy \, dx \\
&= \int_0^x \int_0^x \left[xz + yz + \frac{z^2}{2} \right]_0^{x+y} dy \, dx \\
&= \int_0^x \int_0^x \left[x(x+y) + y(x+y) + \frac{(x+y)^2}{2} \right] dy \, dx
\end{aligned}$$

$$= \int_0^1 \int_0^x \left[x^2 + xy + yx + y^2 + \frac{x^2 + y^2 + 2xy}{2} \right] dy dx$$

$$= \frac{1}{2} \int_0^1 \int_0^x [3x^2 + 6xy + 3y^2] dy dx$$

$$= \frac{1}{2} \int_0^1 \left[3x^2 y + 3x \frac{y^2}{2} + \frac{3y^3}{3} \right]_0^x dx$$

$$= \frac{1}{2} \int_0^1 [3x^2 + 3xy^2 + y^3]_0^x dx$$

$$= \frac{1}{2} \int_0^1 [3x^2 + 3x^3 + x^3] dx$$

$$= \frac{1}{2} \int_0^1 7x^3 dx$$

$$= \frac{7}{2} \int_0^1 x^3 dx$$

$$= \frac{7}{2} \left[\frac{x^4}{4} \right]_0^1$$

$$= \frac{7}{8} [1^4 - 0]$$

$$= \underline{\underline{\frac{7}{8}}}$$

† Evaluate $\int_0^1 \int_0^1 \int_0^{1-x} x dz dx dy$.

Sol:- Given: $\int_0^1 \int_0^1 \int_0^{1-x} x dz dx dy$

$$= \int_{y=0}^1 \int_{x=y}^1 \int_{z=0}^{1-x} x dz dx dy$$

$$= \int_{y=0}^1 \int_{x=y}^1 x [xz]_0^{1-x} dx dy$$

$$= \int_{y=0}^1 \int_{x=y}^1 x(1-x) dx dy$$

$$= \int_{y=0}^1 \int_{x=y}^1 (x-x^2) dx dy$$

$$= \int_{y=0}^1 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_y^1 dy$$

$$= \int_{y=0}^1 \left[\frac{1}{2} - \frac{1}{3} - \left[\frac{y^2}{2} - \frac{y^3}{3} \right] \right] dy$$

$$= \int_{y=0}^1 \left[\frac{1}{2} - \frac{1}{3} - \frac{y^2}{2} + \frac{y^3}{3} \right] dy$$

$$= \int_{y=0}^1 \left[\frac{1}{6} - \frac{y^2}{2} + \frac{y^3}{3} \right] dy$$

$$= \frac{1}{6} \int_{y=0}^1 \left[\frac{y^3}{3} - \frac{y^2}{2} \right] dy$$

$$= \frac{1}{6} \left[\frac{y^4}{12} - \frac{y^3}{6} \right]_0^1$$

$$= \frac{1}{6} \left[\frac{1}{12} - \frac{1}{6} - 0 \right]$$

$$= \frac{1}{12}$$

2) Evaluate $I = \int_0^1 \int_1^2 \int_2^3 xyz dz dy dx$

Sol Given $= \int_0^1 \int_1^2 \int_2^3 xyz dz dy dx$

$$= \int_{x=0}^1 \int_{y=1}^2 \int_{z=2}^3 xyz dz dy dx$$

3) Evaluate $I = \int_0^1 \int_1^2 \int_2^3 x y z \, dx \, dy \, dz$

Sol

Given;

$$I = \int_0^1 \int_1^2 \int_2^3 x y z \, dx \, dy \, dz$$

$$= \int_{z=0}^1 \int_{y=0}^2 y z \left[\frac{x^2}{2} \right]_2^3 dy \, dz$$

$$= \int_{z=0}^1 \int_{y=1}^2 \frac{y z}{2} [5] dy \, dz$$

$$= \frac{5}{2} \int_{z=0}^1 \int_{y=1}^2 y z \, dy \, dz$$

$$= \frac{5}{2} \int_{z=0}^1 z \left[\frac{y^2}{2} \right]_1^2 dz$$

$$= \frac{15}{4} \int_{z=0}^1 [4-1] z \, dz$$

$$= \frac{15}{4} \int_{z=0}^1 z \, dz$$

$$= \frac{15}{4} \left[\frac{z^2}{2} \right]_0^1 = \frac{15}{4} \left[\frac{1}{2} - 0 \right]$$

$$= \frac{15}{8}$$

$$4 \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$$

$$\text{Sol} \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx$$

$$= \int_0^a \int_0^x \left[e^{x+y+z} \right]_0^{x+y} dy dx$$

$$= \int_0^a \int_0^x \left[e^{x+y+x+y} - e^{x+y} \right] dy dx = \int_0^a \int_0^x e^{2(x+y)} - e^{x+y} dy dx$$

$$\Rightarrow \int_0^a \left[\frac{e^{2(x+y)}}{2} - e^{x+y} \right]_0^x dx$$

$$\Rightarrow \int_0^a \left[\frac{e^{2(x+x)}}{2} - e^{x+x} \right] - \left[\frac{e^{2(x+0)}}{2} - e^{x+0} \right] dx$$

$$= \int_0^a \left[\frac{e^{4x}}{2} - \frac{e^{2x}}{2} - e^{2x} + e^x \right] dx = \left[\frac{e^{4x}}{8} - \frac{e^{2x}}{4} - \frac{e^{2x}}{2} + e^x \right]_0^a$$

$$= \left[\frac{e^{4a}}{8} - \frac{e^{2a}}{4} - \frac{e^{2a}}{2} + e^a - \left[\frac{e^0}{8} - \frac{e^0}{4} - \frac{e^0}{2} + e^0 \right] \right]$$

$$= \frac{e^{4a}}{8} + \frac{e^{2a}}{4} - \frac{e^{2a}}{2} + e^a - \left[\frac{1}{8} - \frac{1}{4} - \frac{1}{2} + 1 \right]$$

$$= \frac{e^{4a}}{8} + \frac{(1-2)e^{2a}}{4} + e^a - \left[\frac{1-2-4+8}{8} \right]$$

$$\Rightarrow \frac{e^{4a}}{8} - \frac{3e^{2a}}{4} + e^a - \frac{3}{8}$$

* change of order of Integration

* change the order of Integration and Evaluate

$$\int_0^a \int_{x/a}^{\sqrt{x/a}} (x^2 + y^2) dx dy$$

Given y varies from $\frac{x}{a}$ to $\sqrt{x/a}$

x varies from 0 to a

$$y = x/a \text{ and } y = \sqrt{x/a}$$

$$y = x/a \text{ and } y^2 = x/a$$

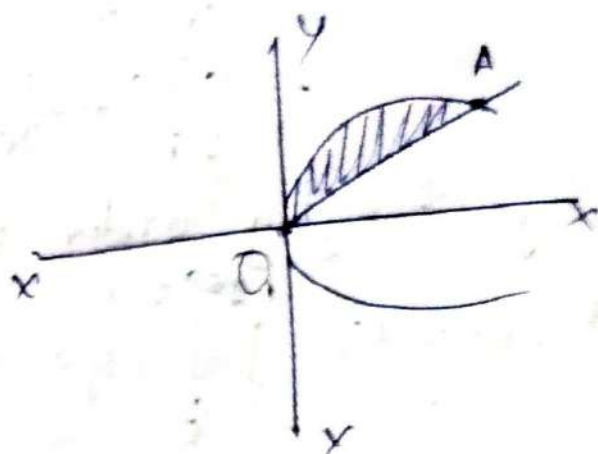
$$x = ay \text{ and } x = ay^2$$

$$x = 0 \text{ and } x = a$$

$$y = x/a ; y = \sqrt{x/a}$$

$$y = a/a ; y = a/a$$

$$y = 0 ; y = 1$$



y varies from 0 to 1

x varies from ay^2 to ay

$$\int_0^1 \int_{ay^2}^{ay} (x^2 + y^2) dx dy$$

$$= \int_0^1 \left[\frac{x^3}{3} + y^2 x \right]_{ay^2}^{ay} dy$$

$$= \int_0^1 \left[\frac{a^3 y^3}{3} + a y^3 - \left(\frac{a^3 y^6}{3} + a y^4 \right) \right] dy$$

$$\int_0^1 \left[\frac{ax^3}{3} + ay^3 - \frac{axy^2}{2} - ay^4 \right] dx$$

$$= \left[\frac{ax^3}{3} \frac{y^4}{4} - \frac{axy^4}{4} - \frac{ax}{2} \frac{y^3}{3} - \frac{ay^4}{5} \right]_0^1$$

$$= a \left[\frac{1}{12} - \frac{1}{24} \right] + a \left[\frac{1}{4} - \frac{1}{5} \right]$$

$$= a \left[\frac{1}{24} \right] + a \left[\frac{1}{20} \right]$$

$$= \frac{a}{24} + \frac{a}{20}$$

By changing order of integration:

• Evaluate $\int_0^1 \int_0^{2-x} 2y \, dx \, dy$

Given x varies from 0 to 1

y varies from 0 to $2-x$

$y=0$ to $2-x$

$y=0$ and $y=2-x$

Now $y=2-x$

$x=2-y$

x varies from 0 to $2-y$

Now $x=0$ and $x=1$

$$y = 2 - x$$

$$x = 0, y = 2$$

$$x = 1, y = 1$$

Consider $\int_{x=0}^1 \int_{y=0}^{2-x} 2y \, dx \, dy = \int_{y=1}^2 \int_{x=0}^{2-y} 2y \, dx \, dy$

$$= \int_{y=1}^2 \left[y \frac{x^2}{2} \right]_0^{2-y} dy \Rightarrow \int_1^2 \left[\frac{y(2-y)^2}{2} - 0 \right] dy$$

$$= \int_1^2 \frac{y(4 - y^2 - 4y)}{2} dy$$

$$= \frac{1}{2} \int_1^2 [4y + y^3 - 4y^2] dy$$

$$= \frac{1}{2} \left[\frac{4y^2}{2} + \frac{y^4}{4} - \frac{4y^3}{3} \right]_1^2 = \frac{1}{2} \left[2y^2 + \frac{y^4}{4} - \frac{4y^3}{3} \right]_1^2$$

$$= \frac{1}{2} \left(8 + 4 - \frac{32}{3} - 2 - \frac{1}{4} + \frac{4}{3} \right)$$

$$= \frac{1}{2} \left[\frac{100 - 112 - 3}{12} \right]$$

$$= \frac{5}{24}$$

Applications of Multiply Integrals.

* Area enclosed by plane curve

Consider area enclosed by the curves

$y = f(x)$ and $y = g(x)$, x from a to b in

the xy plane. The area of the region R

Bounded by the given curves is given by

$$\iint_R dx dy = \iint_R dy dx = \int_{x=a}^b \int_{y=f(x)}^{y=g(x)} dy dx$$

If the region is represented through polar co-ordinate then the area is given by

$$\iint_R r dr d\theta$$

Formulae

① Area of a circle $x^2 + y^2 = r^2$ having radius R

$$\Rightarrow A = \pi R^2 \text{ Sq units.}$$

Area of an Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\Rightarrow A = \pi ab \text{ Sq units}$$

(3) Area bounded by the parabola

$$y^2 = 4ax \text{ \& } x^2 = 4ay$$
$$\Rightarrow A = \frac{16a^2}{3} \text{ Sq units.}$$

(4) Area Enclosed by the parabola $y^2 = 4ax$ and its latus rectum

$$A = \frac{8a^2}{3} \text{ Sq units}$$

* find the area of the plate plate in the co-ordinate

of Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

x varies from 0 to a

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

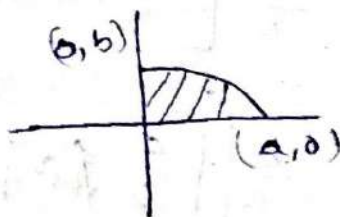
$$\Rightarrow \frac{y}{b} = \sqrt{1 - \frac{x^2}{a^2}}$$

$$\Rightarrow y = b \sqrt{1 - \frac{x^2}{a^2}}$$

$$= \frac{b}{a} \sqrt{a^2 - x^2}$$

y varies from 0 to $\frac{b}{a} \sqrt{a^2 - x^2}$

$$\text{Area} = \int_x \int_y dy dx = \int_{x=0}^a \int_{y=0}^{\frac{b}{a} \sqrt{a^2 - x^2}} dy dx$$



$$= \int_{x=0}^a [y]_0^{\frac{b}{a} \sqrt{a^2-x^2}} dx$$

$$= \int_0^a \frac{b}{a} \sqrt{a^2-x^2} dx$$

$$= \frac{b}{a} \int_0^a \sqrt{a^2-x^2} dx \left[\begin{array}{l} \int \sqrt{a^2-x^2} = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \end{array} \right]$$

$$= \frac{b}{a} \left[\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a$$

$$= \frac{b}{a} \left[\left(\frac{a}{2} \sqrt{a^2-a^2} + \frac{a^2}{2} \sin^{-1} \frac{a}{a} \right) - \left(0 + \frac{a^2}{2} \sin^{-1} 0 \right) \right]$$

$$= \frac{b}{a} \left[\frac{a^2}{2} \sin^{-1}(1) - \frac{a^2}{2} \sin^{-1}(0) \right] = \frac{b}{a} \left[\frac{a^2}{2} \sin^{-1}(\sin \frac{\pi}{2}) - \frac{a^2}{2} \sin^{-1}(\sin 0) \right]$$

$$= \frac{b}{a} \left[\frac{a^2}{2} \frac{\pi}{2} \right]$$

$$= \frac{\pi ab}{4}$$

2. Find the area of a region bounded by the curves $x = y^2$ and $y = x^2$

Given $x = y^2$ and $y = x^2$

$$y = x^2 \text{ and } x = y^2$$

$$\Rightarrow y = \sqrt{x}$$

y varies from x^2 to $\sqrt{x}$

$$\text{let } x = y^2$$

$$x = (x^2)^2$$

$$x - x^4 = 0$$

$$x(1 - x^3) = 0$$

$$x = 0, x = 1$$

x varies from 0 to 1

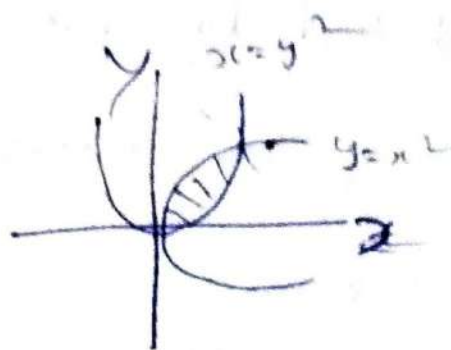
$$\text{Area} = \int_0^1 \int_{y=x^2}^{\sqrt{x}} dy dx = \int_0^1 \int_{y=x^2}^{\sqrt{x}} dy dx$$

$$= \int_0^1 [y]_x^{\sqrt{x}} dx$$

$$= \int_0^1 [\sqrt{x} - x^2] dx \Rightarrow \int_0^1 (x^{1/2} - x^2) dx$$

$$= \left[\frac{x^{3/2}}{3/2} - \frac{x^3}{3} \right]_0^1 = \frac{2}{3} - \frac{1}{3} - [0 - 0]$$

$$= \frac{1}{3}$$



to find by double integration the area enclosed
by the curves $y = 2 - x$ and $y^2 = 2(2 - x)$

Sol Given

$$y = 2 - x \quad \text{--- (1)} \quad y^2 = 2(2 - x) \quad \text{--- (2)}$$
$$\Rightarrow y = \sqrt{2(2 - x)}$$

y varies from $2 - x$ to $\sqrt{2(2 - x)}$

$$y^2 = 2(2 - x)$$

$$\Rightarrow (2 - x)^2 = 4 - 2x$$

$$\Rightarrow 4 + x^2 - 4x = 4 - 2x$$

$$\Rightarrow x^2 - 2x = 0$$

$$\Rightarrow x(x - 2) = 0$$

$$\Rightarrow x = 0, 2$$

x varies from 0 to 2

$$\text{Area} = \int_{x=0}^2 \int_{y=2-x}^{\sqrt{4-2x}} dy \, dx$$

$$\Rightarrow \int_0^2 \left[y \right]_{2-x}^{\sqrt{4-2x}} dx$$

$$\Rightarrow \int_0^2 \left[\sqrt{4-2x} - (2-x) \right] dx$$

$$= \int_0^2 \left[(4-2x)^{1/2} - (2-x) \right] dx$$

$$= \left[\frac{(4-2x)^{3/2}}{3/2(-2)} - 2x + \frac{x^2}{2} \right]_0^2$$

$$= \left[\left(\frac{(4-4)^{3/2}}{-3} - 2(2) + \frac{4}{2} \right) - \left(\frac{(4)^{3/2}}{-3} - 2(0) + \frac{0}{2} \right) \right]$$

$$= \left[-4 + 2 \right] + \frac{2^{3/2}}{3}$$

$$= -2 + \frac{8}{3}$$

$$= \underline{\underline{2/3}}$$

* use double integration find the area of cardioid $r = a(1 - \cos\theta)$

Given $r = a(1 - \cos\theta)$

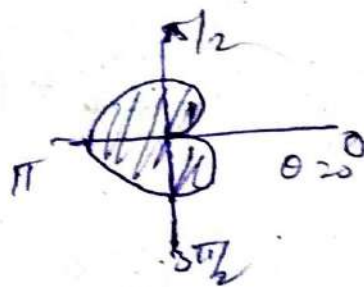
r varies from 0 to $a(1 - \cos\theta)$

θ varies from 0 to π

$$\text{Area} = 2 \iint r \, dr \, d\theta$$

$$= 2 \int_0^\pi \int_{r=0}^{a(1-\cos\theta)} r \, dr \, d\theta$$

$$= 2 \int_0^\pi \left[\frac{r^2}{2} \right]_0^{a(1-\cos\theta)} d\theta$$



$$= \frac{2}{2} \int_0^{\pi} a^2 (1 - \cos \theta)^2 d\theta$$

$$= a^2 \int_0^{\pi} (1 - \cos \theta)^2 d\theta$$

$$\Rightarrow a^2 \int_0^{\pi} (2 \sin^2 \theta/2)^2 d\theta$$

$$= 4a^2 \int_0^{\pi} \sin^4 \theta/2 d\theta$$

$$= 8a^2 \int_0^{\pi/2} \sin^4 \frac{\theta}{2} d\theta$$

Put $\theta/2 = \phi$

$$= 8a^2 \int_0^{\pi/2} \sin^4 \phi \cdot 2 d\phi$$

$$\frac{d\theta}{2} = d\phi$$

$$d\theta = 2d\phi$$

$$= 16a^2 \int_0^{\pi/2} \sin^4 \phi d\phi$$

$$= 16a^2 \left(\frac{4-1}{4} \right) \int_0^{\pi/2} \sin^2 \phi d\phi$$

$$= 16a^2 \cdot \frac{3}{4} \cdot \frac{(2-1)}{2} \int_0^{\pi/2} \sin^0 \phi d\phi$$

$$= \cancel{16} a^2 \cdot \frac{3}{4} \times \frac{1}{2} \int_0^{\pi/2} d\phi$$

$$= 6a^2 [\phi]_0^{\pi/2}$$

$$= 6a^2 \left[\frac{\pi}{2} - 0 \right]$$

$$= 2a^2 \left[\frac{\pi}{2} \right]$$

volume of triple integral.
The total volume of a solid is $\iiint_V dv = \iiint_V dx dy dz$

$$\iiint_V dx dy dz$$

Find the volume common to the cylinders

$$x^2 + y^2 = a^2 \text{ and } x^2 + z^2 = a^2$$

$$\text{from } x^2 + y^2 = a^2 \text{ --- (1)}$$

$$\text{and } x^2 + z^2 = a^2 \text{ --- (2)}$$

$$\text{from (1) } x^2 + y^2 = a^2$$

$$y^2 = a^2 - x^2 \Rightarrow y = \pm \sqrt{a^2 - x^2}$$

y varies from $-\sqrt{a^2 - x^2}$ to $\sqrt{a^2 - x^2}$

$$\text{from (2) } x^2 + z^2 = a^2$$

$$z^2 = a^2 - x^2$$

$$z = \pm \sqrt{a^2 - x^2}$$

z varies from $-\sqrt{a^2 - x^2}$ to $\sqrt{a^2 - x^2}$

x varies from $-a$ to a

$$\text{Volume} = \iiint_V dz dy dx = \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dz dy dx$$

$$= 2 \int_{x=-a}^a \int_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2}} dz dy dx$$

$$= 2 \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \left[z \right]_0^{\sqrt{a^2-x^2}} dy dx$$

$$= 2 \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2} dy dx$$

$$= 4 \int_{-a}^a \int_0^{\sqrt{a^2-x^2}} dy dx$$

$$= 4 \int_{-a}^a \sqrt{a^2-x^2} \left[y \right]_0^{\sqrt{a^2-x^2}} dx$$

$$= 4 \int_{-a}^a \sqrt{a^2-x^2} \left[\sqrt{a^2-x^2} - 0 \right] dx$$

$$= 4 \int_{-a}^a (a^2-x^2) dx = 8 \int_0^a (a^2-x^2) dx$$

$$= 8 \left[a^2x - \frac{x^3}{3} \right]_0^a$$

$$= 8 \left[2a^3 - \frac{a^3}{3} \right]$$

$$= \frac{16a^3}{3} \text{ cubic units}$$

* Find the volume bounded by the xy plane
 $x^2 + y^2 = 1$ and the plane $x + y + z = 3$

Given $x^2 + y^2 = 1$ — (1) & $x + y + z = 3$ — (2)

from (1) $x^2 + y^2 = 1$

$$y = \pm \sqrt{1-x^2}$$

y varies from $-\sqrt{1-x^2}$ to $\sqrt{1-x^2}$

x varies from —

from (2), $x + y + z = 3$

$$\Rightarrow z = 3 - x - y$$

z varies from 0 to $3 - x - y$

x varies from -1 to 1

Volume = $\iiint_V dz dy dx$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_0^{3-x-y} dz dy dx$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left[z \right]_0^{3-x-y} dy dx$$

$$= \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} [3-x-y] dy dx$$

$$= 2 \int_{-1}^1 \int_0^{\sqrt{1-x^2}} [3-x-y] dy dx$$

$$= 2 \int_{-1}^1 \left[3y - xy - \frac{y^2}{2} \right]_0^{\sqrt{1-x^2}} dx$$

$$= 2 \int_{-1}^1 \left[3\sqrt{1-x^2} - x\sqrt{1-x^2} - \frac{(\sqrt{1-x^2})^2}{2} \right] dx$$

$$= 2 \int_{-1}^1 \left[3\sqrt{1-x^2} - x\sqrt{1-x^2} - \frac{(1-x^2)}{2} \right] dx$$

$$= \frac{2}{2} \int_{-1}^1 [6\sqrt{1-x^2} - 2x\sqrt{1-x^2} - (1-x^2)] dx$$

$$= 2 \left[\int_0^1 6\sqrt{1-x^2} dx - 2 \int_0^1 x\sqrt{1-x^2} dx - \int_0^1 (1-x^2) dx \right]$$

$$= 2 \left[6 \int_0^1 \sqrt{1-x^2} dx - \int_0^1 (1-x^2) dx \right]$$

$$= 2 \left[6 \left(\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} \left(\frac{x}{1} \right) \right) - \left[x - \frac{x^3}{3} \right] \right]_0^1$$

$$= 2 \left[6 \left[\frac{1}{2} \sqrt{1-1^2} + \frac{1}{2} \sin^{-1} \left(\frac{1}{1} \right) \right] - \left[\frac{0}{2} \sqrt{1-0^2} - \frac{1}{2} \sin^{-1} \left(\frac{0}{1} \right) \right] \right]$$

$$- \left[\left[1 - \frac{1}{3} \right] - \left[0 - \frac{0}{3} \right] \right]$$

$$= 2 \left[\sqrt[3]{\frac{1}{2} \sin^{-1} \left(\sin \frac{\pi}{2} \right)} - \frac{2}{3} \right]$$

$$= 2 \left[3 \frac{\pi}{2} - \frac{2}{3} \right]$$

$$= \frac{6\pi}{2} - \frac{4}{3}$$

$$= 3\pi - \frac{4}{3} \text{ cubic units}$$

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UNIT-5

Vector Differentiation and Integration

Vector Differentiation

* Scalar: A quantity which is completely specified by its magnitude is called scalar

Ex: distance, speed.

* Vector: A quantity which is completely specified by its magnitude as well as direction is called a vector.

Ex: Velocity, acceleration.

* Unit vector: A vector whose magnitude is unity is called unit vector. If $\vec{a}$ is any vector then $\hat{n}$ is unit vector in the direction of $\vec{a}$.

$$\rightarrow \hat{n} = \frac{\vec{a}}{|\vec{a}|}$$

* Scalar point function: The function

Consider a region in 3 dimensional

Space. To each point $P(x, y, z)$, Suppose we associate a unique number say ϕ (Scalar). This $\phi(x, y, z)$ is called scalar differential operator

Differential operator :-

The vector differential operator ' ∇ ' called as 'del'

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Q Gradient of a Scalar point function :-

Let $\phi(x, y, z)$ be a scalar point function defined in a region R . Then the gradient ϕ is denoted by $\text{grad } \phi$ or $\nabla \phi$

$$\text{grad } \phi = \nabla \phi = \left[\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right] \cdot \phi$$

$$= \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

Properties :-

* The gradient of scalar point function is a vector point function

* Suppose $\phi(x, y, z) = c$ be the Equation of the given Surface. Then $\text{grad } \phi$ or $\nabla \phi$ represents normal to the Surface.

* If $\phi(x, y, z)$ be the Equation of given Surface then unit normal to the Surface is

$$\hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$$

* Find grad ϕ at a point $(1, 1, 1)$ is

$$\phi(x, y, z) = x^2y + xy^2 + z^2$$

Sol Given $\phi(x, y, z) = x^2y + xy^2 + z^2$

Point $P = (1, 1, 1)$

We know that

$$\text{grad } \phi = \nabla \phi = \left[\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right] \phi$$

$$= \left[\frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k} \right]$$

— (1)

Given $\phi = x^2y + xy^2 + z^2$

Diff with resp to x, y, z

$$\frac{\partial \phi}{\partial x} = 2xy + y^2$$

$$\frac{\partial \phi}{\partial y} = x^2 + 2xy$$

$$\frac{\partial \phi}{\partial z} = 2z$$

Sub (1)

we get

$$\nabla \phi = (2xy + y^2)\vec{i} + (x^2 + 2xy)\vec{j} + 2z\vec{k}$$

$$\nabla \phi(1, 1, 1) = 3\vec{i} + 3\vec{j} + 2\vec{k} //$$

* Find unit normal vector to the given surface
 $\phi(x, y, z) = 3x^2y - y^3z^2$ at a point $(1, 2, -1)$

Given, $\phi = 3x^2y - y^3z^2$

Point $P = (1, 2, -1)$

Unit normal vector is $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$

We know that

$$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$$

Given $\phi = 3x^2y - y^3z^2$

Diff with rep to x, y, z

$$\frac{\partial\phi}{\partial x} = 6xy \quad ; \quad \frac{\partial\phi}{\partial y} = 3x^2 - 3y^2z^2 \quad ; \quad \frac{\partial\phi}{\partial z} = -y^3 \cdot 2z$$

$$\frac{\partial\phi}{\partial y} = 3x^2 - 3y^2z^2$$

Sub in Eq ① we get

$$\nabla\phi = (6xy) \hat{i} + (3x^2 - 3y^2z^2) \hat{j} + (-y^3 \cdot 2z) \hat{k}$$

$$\nabla\phi(1, 2, -1) = 12\hat{i} - 9\hat{j} + 16\hat{k}$$

$$\nabla\phi(1, 2, -1) = 12\hat{i} - 9\hat{j} + 16\hat{k}$$

Unit normal vector $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$

$$\hat{n} = \frac{12\hat{i} - 9\hat{j} + 16\hat{k}}{\sqrt{12^2 + 9^2 + 16^2}}$$

$$\hat{n} = \frac{12\hat{i} - 9\hat{j} + 16\hat{k}}{\sqrt{481}} \Rightarrow \frac{12\hat{i} - 9\hat{j} + 16\hat{k}}{\sqrt{481}}$$

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Directional Derivative: The directional derivative of a scalar point function at a point (x, y, z) in the direction of the unit normal vector is given by $\hat{n} \text{ grad } \phi$ (or) $\hat{n} \cdot \nabla \phi$

$$\therefore D.D = \hat{n} \cdot \nabla \phi$$

$$= \frac{\vec{a}}{|\vec{a}|} \cdot \nabla \phi \quad (\text{or}) = \frac{\nabla \phi}{|\nabla \phi|} \cdot \nabla \phi$$

The maximum value of DD of scalar point function ϕ at point P is $|\nabla \phi|$

* Find the maximum value of DD of function

$$f = x^2 y z^3 \text{ at } (2, 1, -1)$$

Given $f = x^2 y z^3$, at $(2, 1, -1)$

$$\text{Now } \nabla f = \left[\frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k} \right] = \vec{i} (2yz^3) + \vec{j} (x^2 z^3) + \vec{k} (3x^2 y z^2)$$

$$= \left[\frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k} \right] = (2yz^3) \vec{i} + (x^2 z^3) \vec{j} + (3x^2 y z^2) \vec{k}$$

diff f with res to x, y, z

$$f = x^2 y z^3$$

$$\frac{\partial f}{\partial x} = 2xy z^3 ; \frac{\partial f}{\partial y} = x^2 z^3 ; \frac{\partial f}{\partial z} = 3x^2 y z^2$$

Sub in (1) we get

$$\nabla f = (2xyz^3)\vec{i} + (x^2z^3)\vec{j} + (3x^2yz^2)\vec{k}$$

$$\nabla f_{(2,1,-1)} = -4\vec{i} + (-4)\vec{j} + 12\vec{k}$$

The maximum value is $|\nabla f|$

$$|\nabla f| = \sqrt{(-4)^2 + (-4)^2 + (12)^2}$$

$$|\nabla f| = \sqrt{16 + 16 + 144}$$

$$|\nabla f| = \sqrt{176}$$

* Find the D.D. of $f = xy + yz + zx$ in direction of vector $\vec{i} + 2\vec{j} + 2\vec{k}$ at a point $(1, 2, 0)$

Given $f = xy + yz + zx$; $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k}$; $P(1, 2, 0)$

The D.D. is $\frac{\vec{a} \cdot \nabla f}{|\vec{a}|}$

$$D.D \Rightarrow \frac{\vec{a} \cdot \nabla f}{|\vec{a}|}$$

Given $f = xy + yz + zx$

$$\frac{\partial f}{\partial x} = y + z ; \frac{\partial f}{\partial y} = x + z ; \frac{\partial f}{\partial z} = y + x$$

$$\nabla f = (y+z)\vec{i} + (x+z)\vec{j} + (y+x)\vec{k}$$

$$\nabla f_{(1,2,0)} = 2\vec{i} + \vec{j} + 3\vec{k}$$

Given $\vec{a} = \vec{i} + 2\vec{j} + 2\vec{k} \Rightarrow |\vec{a}| = \sqrt{1^2 + 2^2 + 2^2}$

$$|\bar{a}| = \sqrt{9}$$

$$|\bar{a}| = 3$$

$$\therefore D.D = \frac{\bar{a}}{|\bar{a}|} \nabla f$$

$$= \frac{\bar{i} + 2\bar{j} + 2\bar{k}}{3} \cdot [2\bar{i} + \bar{j} + 3\bar{k}]$$

$$= \frac{2+2+6}{3}$$

$$\left[\begin{array}{l} \bar{i} \cdot \bar{i} = 1; \bar{j} \cdot \bar{j} = 1 \\ \bar{k} \cdot \bar{k} = 1 \end{array} \right]$$

$$= \frac{10}{3}$$

$$[\bar{i} \cdot \bar{j} = \bar{j} \cdot \bar{k} = \bar{i} \cdot \bar{k} = 0]$$

* Find the Directional Derivate $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $\bar{a} = 2\bar{i} + \bar{j} - 2\bar{k}$

$$\phi = x^2yz + 4xz^2, \bar{a} = 2\bar{i} + \bar{j} - 2\bar{k}; \text{ Point } (1, -2, -1)$$

$$D.D = \hat{n} \nabla \phi$$

$$= \frac{\bar{a}}{|\bar{a}|} \nabla \phi$$

Given $\phi = x^2yz + 4xz^2$

$$\frac{\partial \phi}{\partial x} = 2xy z + 4z^2$$

$$\frac{\partial \phi}{\partial z} = x^2 y + 8x$$

$$\frac{\partial \phi}{\partial y} = x^2 z$$

$$\nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\nabla \phi = (6xyz + 4z^2) \vec{i} + (x^2z) \vec{j} + (x^2y + 8z) \vec{k}$$

$$\nabla \phi_{(1,-2,-1)} = 18\vec{i} + 6(-\vec{j}) + 6\vec{k}$$

$$\vec{a} = 8\vec{i} - \vec{j} + 2\vec{k} \Rightarrow |\vec{a}| = \sqrt{4+1+4} = \sqrt{9} = 3$$

$$D.D = \frac{\vec{a}}{|\vec{a}|} \cdot \nabla \phi$$

$$= \frac{2\vec{i} - \vec{j} - 2\vec{k}}{3} \cdot (8\vec{i} - \vec{j} + 6\vec{k})$$

$$= \frac{16\vec{i} + \vec{j} - 12\vec{k}}{3} = \frac{16+1-12}{3} = \frac{5}{3}$$

Find the D.D of $\phi = xyz$ along the direction of the normal to the surface, $f = x^2z + y^2x + yz^2 = 3$ at a point $(1,1,1)$

Q) Given $\phi = xyz$ & $f = x^2z + y^2x + z^2y = 3$
 $P = (1,1,1)$

The directional derivation is $\hat{n} \cdot \nabla \phi$

$$= \frac{\nabla f}{|\nabla f|} \cdot \nabla \phi$$

Given $\phi = xyz$

$$\frac{\partial \phi}{\partial x} = yz ; \frac{\partial \phi}{\partial y} = xz ; \frac{\partial \phi}{\partial z} = xy$$

$$\nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\nabla \phi = y\bar{i} + x\bar{j} + z\bar{k}$$

$$\nabla \phi(1,1,1) = \bar{i} + \bar{j} + \bar{k}$$

Here $f = x^2z + y^2z + z^2y - 3$

$$\frac{\partial f}{\partial x} = 2xz + y^2$$

$$\frac{\partial f}{\partial z} = x^2 + y^2 + 2zy$$

$$\frac{\partial f}{\partial y} = 2yx + z^2$$

$$\nabla f = \frac{\partial f}{\partial x} \bar{i} + \frac{\partial f}{\partial y} \bar{j} + \frac{\partial f}{\partial z} \bar{k}$$

$$= (2xz + y^2)\bar{i} + (2yx + z^2)\bar{j} + (x^2 + y^2 + 2zy)\bar{k}$$

$$\nabla f(1,1,1) = 3\bar{i} + 3\bar{j} + 3\bar{k}$$

$$|\nabla f| = \sqrt{3^2 + 3^2 + 3^2} = \sqrt{27} = 3\sqrt{3}$$

The D.O is $\frac{\nabla f}{|\nabla f|} \cdot \nabla \phi$

$$= \frac{3\bar{i} + 3\bar{j} + 3\bar{k}}{3\sqrt{3}} \cdot (\bar{i} + \bar{j} + \bar{k})$$

$$= \frac{9}{3\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

UNIT-V
Vector Differentiation
and Integration.

S. V. C. ENGG. COLLEGE
MIDDELMAN

NAME

CLASS

BRANCH

- 1) Find the directional derivative of $f(x, y, z) = xy^3 + yz^3$ at the point $(2, -1, 1)$ in the direction of vector $\bar{i} + 2\bar{j} + 3\bar{k}$.

Sol: Given $f(x, y, z) = xy^3 + yz^3$

$$\nabla f = \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] [xy^3 + yz^3]$$

$$= \frac{\partial}{\partial x}(xy^3 + yz^3) \bar{i} + \frac{\partial}{\partial y}(xy^3 + yz^3) \bar{j} + \frac{\partial}{\partial z}(xy^3 + yz^3) \bar{k}$$

$$\therefore \nabla f = y^3 \bar{i} + (3xy^2 + z^3) \bar{j} + 3yz^2 \bar{k}$$

$$\text{Now } \nabla f_{(2, -1, 1)} = (-1)^3 \bar{i} + [3(2)(-1)^2 + (1)^3] \bar{j} + 3(-1)(1)^2 \bar{k}$$
$$= -\bar{i} + 7\bar{j} - 3\bar{k}$$

Given the direction of vector, $\bar{n} = \bar{i} + 2\bar{j} + 3\bar{k}$

$$\text{Then } \hat{e} = \frac{\bar{n}}{|\bar{n}|} = \frac{\bar{i} + 2\bar{j} + 3\bar{k}}{\sqrt{(1)^2 + (2)^2 + (3)^2}} = \frac{\bar{i} + 2\bar{j} + 3\bar{k}}{\sqrt{14}}$$

$$\begin{aligned} \text{The directional derivative} &= \hat{e} \cdot \nabla f \\ &= \frac{(\bar{i} + 2\bar{j} + 3\bar{k}) \cdot (-\bar{i} + 7\bar{j} - 3\bar{k})}{\sqrt{14}} \\ &= \frac{-1 + 14 - 9}{\sqrt{14}} \\ &= \frac{14 - 10}{\sqrt{14}} \\ &= \frac{4}{\sqrt{14}} \end{aligned}$$

$\approx$

2. Find the divergence and curl of $\vec{F}$, where
 $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$.

Sol:- Given $\phi = x^3 + y^3 + z^3 - 3xyz$

$$\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$$

$$= \nabla \phi$$

$$= \left[\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right] (x^3 + y^3 + z^3 - 3xyz)$$

$$= (3x^2 - 3yz)\vec{i} + (3y^2 - 3xz)\vec{j} + (3z^2 - 3xy)\vec{k}$$

$$\text{Now } \text{div } \vec{F} = \nabla \cdot \vec{F}$$

$$= \left[\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right] \cdot [(3x^2 - 3yz)\vec{i} + (3y^2 - 3xz)\vec{j} + (3z^2 - 3xy)\vec{k}]$$

$$= \frac{\partial}{\partial x}(3x^2 - 3yz) + \frac{\partial}{\partial y}(3y^2 - 3xz) + \frac{\partial}{\partial z}(3z^2 - 3xy)$$

$$= 6x + 6y + 6z$$

$$\therefore \nabla \cdot \vec{F} = 6x + 6y + 6z$$

$$\text{Now } \text{Curl } \vec{F} = \nabla \times \vec{F}$$

$$= \left[\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right] \times [(3x^2 - 3yz)\vec{i} + (3y^2 - 3xz)\vec{j} + (3z^2 - 3xy)\vec{k}]$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (3x^2 - 3yz) & (3y^2 - 3xz) & (3z^2 - 3xy) \end{vmatrix}$$

$$= \vec{i} \left[\frac{\partial}{\partial y}(3z^2 - 3xy) - \frac{\partial}{\partial z}(3y^2 - 3xz) \right] - \vec{j} \left[\frac{\partial}{\partial x}(3z^2 - 3xy) - \frac{\partial}{\partial z}(3x^2 - 3yz) \right] + \vec{k} \left[\frac{\partial}{\partial x}(3y^2 - 3xz) - \frac{\partial}{\partial y}(3x^2 - 3yz) \right]$$

$$= \vec{0}$$

$$\begin{aligned}
 &= \bar{i}[-3x+3x] - \bar{j}[-3y+3y] + \bar{k}[-3z+3z] \\
 &= 0\bar{i} + (0\cdot\bar{j}) + 0\cdot\bar{k} = 0 \\
 &= \bar{0} \\
 \therefore \nabla \times \bar{F} &= \underline{\underline{0}}
 \end{aligned}$$

3. Show that the vector field $\bar{F} = 2xz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$ is irrotational. Find the potential function.

sol

Given $\bar{F} = 2xz^2\bar{i} + (x^2z^2 + z\cos yz)\bar{j} + (2x^2yz + y\cos yz)\bar{k}$

$$\text{curl } \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz^2 & (x^2z^2 + z\cos yz) & (2x^2yz + y\cos yz) \end{vmatrix}$$

$$\begin{aligned}
 &= \bar{i}[2x^2z - y\sin(yz) + \cos yz - 2x^2z + z\cos yz(y) - \cos yz] \\
 &\quad + \bar{j}[4xyz - 4xyz] + \bar{k}[2xz^2 - 2xz^2] \\
 &= 0\bar{i} + 0\bar{j} + 0\bar{k} = \bar{0}
 \end{aligned}$$

$\therefore$ The function is irrotational

There exists ϕ such that $\text{grad } \phi = \bar{F}$

To find the scalar potential function

$$\text{We have } d\phi = \frac{\partial \phi}{\partial x} \bar{i} + \frac{\partial \phi}{\partial y} \bar{j} + \frac{\partial \phi}{\partial z} \bar{k}$$

$$\vec{F} = \text{grad } \phi$$

$$= (2x^2yz^2)\vec{i} + (x^2z^2 + z\cos yz)\vec{j} + (2x^2yz + y\cos yz)\vec{k}$$

Here $\frac{\partial \phi}{\partial x} = 2xyz^2$, $\frac{\partial \phi}{\partial y} = x^2z^2 + z\cos yz$

$$\frac{\partial \phi}{\partial z} = 2x^2yz + y\cos yz$$

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

$$\Rightarrow d\phi = 2xyz^2 dx + (x^2z^2 + z\cos yz) dy + (2x^2yz + y\cos yz) dz$$

Integrating on b.s ,

$$\phi = 3x^2yz^2 + 2\sin yz$$

4 Find the values of a and b such that the surface $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1, -1, 2)$.

Sol Let the given surfaces be

$$f(x, y, z) = ax^2 - byz - (a+2)x \quad \text{--- (1)}$$

$$g(x, y, z) = 4x^2y + z^3 - 4 \quad \text{--- (2)}$$

Given two surfaces meet at the point $(1, -1, 2)$
Substituting the point in (1)

$$a + 2b - (a+2) = 0$$

$$\Rightarrow 2b = 2$$

$$\Rightarrow \boxed{b=1}$$

$$f = ax^2 - byz - (a+2)z$$

$$\text{Now } \frac{\partial f}{\partial x} = 2ax - (a+2), \quad \frac{\partial f}{\partial y} = -bz, \quad \frac{\partial f}{\partial z} = -by$$

$$\therefore \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$= [2ax - (a+2)]\hat{i} - bz\hat{j} - by\hat{k}$$

$$\nabla f_{(1,-1,2)} = (a-2)\hat{i} - 2\hat{j} + \hat{k} = \vec{n}_1 \quad \left[\begin{array}{l} \text{normal vector} \\ \text{to surface 1} \end{array} \right]$$

$$g = 4x^2y + z^3 - 4$$

$$\frac{\partial g}{\partial x} = 8xy, \quad \frac{\partial g}{\partial y} = 4x^2, \quad \frac{\partial g}{\partial z} = 3z^2$$

$$\therefore \nabla g = 8xy\hat{i} + 4x^2\hat{j} + 3z^2\hat{k}$$

$$\nabla g_{(1,-1,2)} = -8\hat{i} + 4\hat{j} + 12\hat{k} = \vec{n}_2 \quad \left[\begin{array}{l} \text{normal vector} \\ \text{to surface 2} \end{array} \right]$$

Given the surfaces f, g are orthogonal at the point $(1, -1, 2)$

$$\therefore \nabla f \cdot \nabla g = 0$$

$$\Rightarrow [(a-2)\hat{i} - 2\hat{j} + \hat{k}] [-8\hat{i} + 4\hat{j} + 12\hat{k}] = 0$$

$$\Rightarrow -8a + 16 - 8 + 12 = 0$$

$$\Rightarrow -8a + 20 = 0$$

$$\Rightarrow -8a = -20$$

$$\Rightarrow a = \frac{20}{8} = \frac{5}{2}$$

$$\Rightarrow \boxed{a = 5/2}$$

$$\therefore a = 5/2 \quad \text{and} \quad b = 1$$

5. Show that $r^\alpha \bar{r}$ is any irrotational vector for any value of α but is solenoidal if $\alpha + 3 = 0$ where $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$ and r is the magnitude of $\bar{r}$.

Sol First we have to show that $r^\alpha \bar{r}$ is solenoidal if $\alpha + 3 = 0$.

S.T $r^\alpha \bar{r}$ is any irrotational vector.

$$\begin{aligned}\nabla \times (r^\alpha \bar{r}) &= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] \times [r^\alpha \bar{r}] \\&= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] \times [r^\alpha (x\bar{i} + y\bar{j} + z\bar{k})] \\&= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] \times [r^\alpha x\bar{i} + r^\alpha y\bar{j} + r^\alpha z\bar{k}] \\&= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^\alpha x & r^\alpha y & r^\alpha z \end{vmatrix} \\&= \bar{i} \left[\frac{\partial}{\partial y} (r^\alpha z) - \frac{\partial}{\partial z} (r^\alpha y) \right] - \bar{j} \left[\frac{\partial}{\partial x} (r^\alpha z) - \frac{\partial}{\partial z} (r^\alpha x) \right] \\&\quad + \bar{k} \left[\frac{\partial}{\partial x} (r^\alpha y) - \frac{\partial}{\partial y} (r^\alpha x) \right] \\&= \bar{i} \left[z \alpha r^{\alpha-1} \frac{\partial r}{\partial y} - y \alpha r^{\alpha-1} \frac{\partial r}{\partial z} \right] - \bar{j} \left[z \alpha r^{\alpha-1} \frac{\partial r}{\partial x} - x \alpha r^{\alpha-1} \frac{\partial r}{\partial z} \right] \\&\quad + \bar{k} \left[y \alpha r^{\alpha-1} \frac{\partial r}{\partial x} - x \alpha r^{\alpha-1} \frac{\partial r}{\partial y} \right] \\&= \bar{i} \left[\alpha y z r^{\alpha-2} - y z \alpha r^{\alpha-2} \right] - \bar{j} \left[\alpha z x r^{\alpha-2} - x z \alpha r^{\alpha-2} \right] \\&\quad + \bar{k} \left[\alpha y x r^{\alpha-2} - x y \alpha r^{\alpha-2} \right] \\&= 0\bar{i} + 0\bar{j} + 0\bar{k} = \bar{0}\end{aligned}$$

$\therefore r^\alpha \bar{r}$ is irrotational vector.

Now we have to s.t $r^n \bar{r}$ is solenoidal if
 $\alpha + 3 = 0$ & $\alpha = -3$

$$\text{i.e., } \text{div}(r^n \bar{r}) = \nabla \cdot (r^n \bar{r})$$

$$= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] \left[r^n (x\bar{i} + y\bar{j} + z\bar{k}) \right]$$

$$= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] \cdot \left[r^n x \bar{i} + r^n y \bar{j} + r^n z \bar{k} \right]$$

$$= \frac{\partial}{\partial x} (r^n x) + \frac{\partial}{\partial y} (r^n y) + \frac{\partial}{\partial z} (r^n z)$$

$$= r^\alpha + x \alpha r^{\alpha-1} \frac{\partial r}{\partial x} + r^\alpha + y \alpha r^{\alpha-1} \frac{\partial r}{\partial y} + r^\alpha + z \alpha r^{\alpha-1} \frac{\partial r}{\partial z}$$

$$= 3r^\alpha + \alpha r^{\alpha-1} \left[x \cdot \frac{z}{r} + y \cdot \frac{y}{r} + z \cdot \frac{z}{r} \right]$$

$$= 3r^\alpha + \alpha r^{\alpha-1} \cdot \frac{x^2 + y^2 + z^2}{r}$$

$$= 3r^\alpha + \alpha r^{\alpha-1} \cdot \frac{r^2}{r} = 3r^\alpha + \alpha r^\alpha$$

$$= r^\alpha (\alpha + 3)$$

$r^n \bar{r}$ is solenoidal if $\nabla \cdot (r^n \bar{r}) = 0$

$$\text{i.e., } r^\alpha (\alpha + 3) = 0$$

$$\alpha + 3 = 0 \quad \& \quad \alpha = -3$$

6. Show that the vectors $(x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational.

Sol Let $\bar{f} = (x^2 - yz)\bar{i} + (y^2 - xz)\bar{j} + (z^2 - xy)\bar{k}$

$$\text{Curl } \bar{f} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2 - yz) & (y^2 - xz) & (z^2 - xy) \end{vmatrix}$$

$$= \bar{i}[-x+x] - \bar{j}[-y+y] + \bar{k}[-z+z]$$

$$= 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$$= \bar{0}$$

$\therefore \bar{f}$ is irrotational

[There exists ϕ such that $\bar{f} = \nabla \phi$]

7. Prove that $\text{div}(r^n \bar{r}) = (n+3)r^n$. Hence show that $\frac{\bar{r}}{r^3}$ is solenoidal.

sol. Let $\bar{r}$ be the position vector

$$\text{i.e., } \bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$$

$$r = |\bar{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow r^2 = x^2 + y^2 + z^2$$

differentiate partially w.r.t x, y, z

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\bar{F} = r^n \bar{r} = \nabla (r^n \bar{r})$$

$$= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] [r^n (x\bar{i} + y\bar{j} + z\bar{k})]$$

$$= \left[\frac{\partial}{\partial x} \bar{i} + \frac{\partial}{\partial y} \bar{j} + \frac{\partial}{\partial z} \bar{k} \right] \cdot [r^n x \bar{i} + r^n y \bar{j} + r^n z \bar{k}]$$

$$= \frac{\partial}{\partial x} [r^n x] + \frac{\partial}{\partial y} [r^n y] + \frac{\partial}{\partial z} [r^n z]$$

$$= \sum \frac{\partial}{\partial x} (r^n x)$$

$$= \sum \left[r^n (1) + x n r^{n-1} \frac{\partial r}{\partial x} \right]$$

$$= \sum [r^n + x^2 n r^{n-2}]$$

$$= 3r^n + n r^{n-2} [x^2 + y^2 + z^2] = r^n (n+3)$$

$$\therefore \text{div}(r^n \bar{r}) = (n+3)r^n$$

Now we have to show that $\frac{\vec{r}}{r^3}$ is solenoidal

$$\text{i.e., } \nabla \cdot \vec{F} = 0$$

Here $\vec{F} = \frac{\vec{r}}{r^3}$

$$\nabla \cdot \vec{F} = \left[\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right] \cdot \left[\frac{x\vec{i} + y\vec{j} + z\vec{k}}{r^3} \right]$$

$$= \frac{\partial}{\partial x} \left[\frac{x}{r^3} \right] + \frac{\partial}{\partial y} \left[\frac{y}{r^3} \right] + \frac{\partial}{\partial z} \left[\frac{z}{r^3} \right]$$

$$= \sum \frac{\partial}{\partial x} \left[\frac{x}{r^3} \right]$$

$$= \sum \left[\frac{r^3(1) - x \cdot 3r^2 \frac{\partial r}{\partial x}}{r^6} \right]$$

$$= \sum \frac{r^3 - 3x^2 r}{r^6} = \frac{3r^3 - 3r^3}{r^6} = 0$$

$$\therefore \nabla \cdot \vec{F} = 0$$

8. Evaluate $\nabla^2(\log r)$

sol $\nabla^2(\log r) = \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] (\log r)$

$$= \frac{\partial^2}{\partial x^2} \log r + \frac{\partial^2}{\partial y^2} \log r + \frac{\partial^2}{\partial z^2} \log r$$

$$= \sum \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} \log r \right]$$

$$= \sum \frac{\partial}{\partial x} \left[\frac{1}{r} \frac{\partial r}{\partial x} \right] = \sum \frac{\partial}{\partial x} \left[\frac{x}{r^2} \right]$$

$$= \sum \frac{r^2(1) - 2x^2}{r^4}$$

$$= \frac{1}{r^4} [r^2 - 2x^2 + r^2 - 2y^2 + r^2 - 2z^2]$$

$$= \frac{1}{r^4} [3r^2 - 2r^2] = \frac{1}{r^2}$$

$$\therefore \nabla^2(\log r) = \frac{1}{r^2}$$

9 Show that $\nabla^2(r^n) = n(n+1)r^{n-2}$

sol

$$\nabla^2(r^n) = \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right] r^n$$

$$= \frac{\partial^2}{\partial x^2} r^n + \frac{\partial^2}{\partial y^2} r^n + \frac{\partial^2}{\partial z^2} r^n$$

$$= \frac{\partial}{\partial x} \left[\frac{\partial}{\partial x} r^n \right]$$

$$= \frac{\partial}{\partial x} \left[n r^{n-1} \frac{x}{r} \right]$$

$$= \frac{\partial}{\partial x} \left[n x r^{n-2} \right]$$

$$= n \left[r^{n-2} + x(n-2)r^{n-3} \frac{\partial r}{\partial x} \right]$$

$$= n \left[r^{n-2} + (n-2)r^{n-3} \frac{x^2}{r} \right]$$

$$= n \left[r^{n-2} + (n-2)x^2 r^{n-4} + r^{n-2} + (n-2)y^2 r^{n-4} + r^{n-2} + (n-2)z^2 r^{n-4} \right]$$

$$= n \left[3r^{n-2} + (n-2)r^{n-2} \right]$$

$$= n r^{n-2} [3 + n - 2]$$

$$\therefore \nabla^2(r^n) = n(n+1)r^{n-2}$$

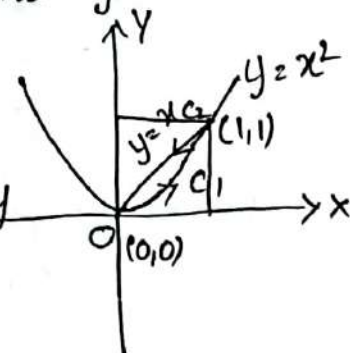
10. Verify Green's theorem for $\int_C [(xy+y^2)dx + x^2dy]$, where C is bounded by $y=x$ and $y=x^2$.

sol.

By Green's theorem,

$$\oint_C Mdx + Ndy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy$$

$$\text{Here } M = xy + y^2, \quad N = x^2$$



The line $y=x$ and the parabola $y=x^2$ intersect at $O(0,0)$ and $A=(1,1)$

$$\text{Now } \int_C Mdx + Ndy = \int_{C_1} Mdx + Ndy + \int_{C_2} Mdx + Ndy \quad \text{--- (1)}$$

Along C_1 (ie., $y=x^2$) from $(0,0)$ to $(1,1)$ the line integral is

$$\begin{aligned} \int_{C_1} Mdx + Ndy &= \int_{C_1} [xy + y^2]dx + x^2dy \\ &= \int_0^1 [x(x^2) + (x^2)^2]dx + x^2d(x^2) \\ &= \int_0^1 (x^3 + x^4 + 2x^3)dx = \int_0^1 (x^4 + 3x^3)dx \\ &= \left[3 \cdot \frac{x^4}{4} + \frac{x^5}{5} \right]_0^1 = \frac{3}{4} + \frac{1}{5} = \frac{19}{20} \quad \text{--- (2)} \end{aligned}$$

Along C_2 (ie., $y=x$) from $(1,1)$ to $(0,0)$

$$\begin{aligned} \int_{C_2} Mdx + Ndy &= \int_{C_2} [xy + y^2]dx + x^2dy \\ &= \int_{C_2} [x(x) + x^2]dx + x^2d(x) \\ &= \int_{C_2} 2x^2dx + x^2dx \\ &= \int_{C_2} 3x^2dx = 3 \left[\frac{x^3}{3} \right]_1^0 = 0 - 1 = -1 \quad \text{--- (3)} \end{aligned}$$

from (1), (2), (3) we have

$$\int_C Mdx + Ndy = \frac{19}{20} - 1 = -\frac{1}{20} \quad \text{--- (4)}$$

$$\begin{aligned}
 \text{Now } \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy &= \iint_R [2x - x - 2y] dx dy \\
 &= \int_{x=0}^1 \int_{y=x^2}^x (x-2y) dx dy \\
 &= \int_0^1 \left[x^2 - 2y \right]_{y=x^2}^x dy \\
 &= \int_0^1 \left[xy - 2 \frac{y^2}{2} \right]_{x^2}^x dx = \int_0^1 \left[xy - y^2 \right]_{x^2}^x dx \\
 &= \int_0^1 [(x^2 - x^2) - (x^3 - x^4)] dx \\
 &= \int_0^1 (x^4 - x^3) dx = \left[\frac{x^5}{5} - \frac{x^4}{4} \right]_0^1 \\
 &= \frac{1}{5} - \frac{1}{4} = -\frac{1}{20} \quad \text{--- (5)}
 \end{aligned}$$

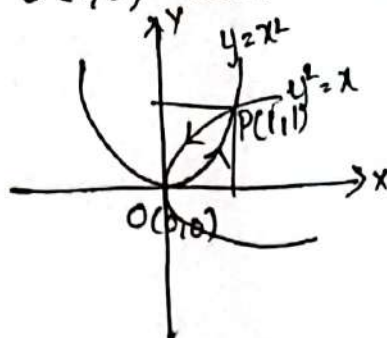
From (4) & (5), we have

$$\int_C M dx + N dy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy$$

Hence Verified

11. Using Green's theorem Evaluate $\int_C (2xy - x^2) dx + (x^2 + y^2) dy$ where C is the closed curve in xy -plane bounded by the curves $y = x^2$ and $y^2 = x$.

Sol The two parabolas $y^2 = x$ and $y = x^2$ are intersecting at $O(0,0)$ and $P(1,1)$



Here $M = 2xy - x^2$ and $N = x^2 + y^2$

$$\therefore \frac{\partial M}{\partial y} = 2x, \quad \frac{\partial N}{\partial x} = 2x$$

Hence $\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} = 2x - 2x = 0$

By Green's Theorem, $\int_C Mdx + Ndy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy$

$$\text{i.e., } \int_C (2xy - x^2)dx + (x^2 + y^2)dy = \int_{x=0}^1 \int_{y=x^2}^{\sqrt{x}} (0) dx dy = 0$$

12 By using Green's Theorem, Evaluate $\oint_C (3x^2 - 8y^2)dx + (4y - 6xy)dy$, where C is made up of the curves $y = \sqrt{x}$ and $y = x^2$.

Sol By Green's Theorem, we have

$$\int_C Mdx + Ndy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy$$

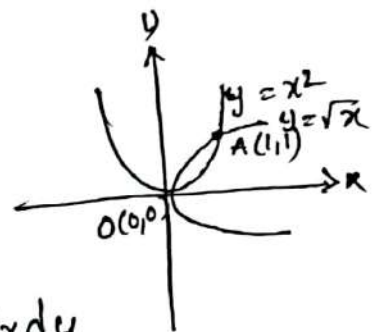
Here $M = 3x^2 - 8y^2$, $N = 4y - 6xy$

$$\frac{\partial M}{\partial y} = -16y, \quad \frac{\partial N}{\partial x} = -6y$$

$$\text{Now } \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy = \iint_R (-6y - (-16y)) dx dy$$

$$= \iint_R (16y - 6y) dx dy$$

$$= \iint_R 10y dx dy = 10 \int_{x=0}^1 \int_{y=x^2}^{\sqrt{x}} y dy dx$$



$$\begin{aligned}
 &= 10 \int_0^1 \left[\frac{y^2}{2} \right]_{x^2}^{\sqrt{x}} dx \\
 &= 5 \int_0^1 (x - x^4) dx \\
 &= 5 \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 = 5 \left[\frac{1}{2} - \frac{1}{5} \right] = \frac{3}{2} \quad \text{--- (1)}
 \end{aligned}$$

Verification

The line integral along c .

$$\begin{aligned}
 &= \left[\text{line integral along } y=x^2 \text{ from } (0 \text{ to } A) \right] + \left[\text{line integral along } y=x \text{ from } A \text{ to } 0 \right] \\
 &= I_1 + I_2
 \end{aligned}$$

$$\begin{aligned}
 I_1 &= \int_0^1 \left\{ [3x^2 - 8(x^2)^2] dx + [4x^2 - 6x(x^2)] 2x dx \right\} \quad \left\{ \begin{array}{l} y=x^2 \\ \frac{dy}{dx} = 2x \end{array} \right. \\
 &= \int_0^1 [3x^2 + 8x^3 - 20x^4] dx
 \end{aligned}$$

$$\begin{aligned}
 &= -1 \\
 \text{and } I_2 &= \int_1^0 \left[(3x^2 - 8x) dx + (4\sqrt{x} - 6x^{3/2}) \frac{1}{2\sqrt{x}} dx \right] \\
 &= \int_1^0 (3x^2 - 11x + 2) dx = \frac{5}{2}
 \end{aligned}$$

$$\therefore I_1 + I_2 = -1 + \frac{5}{2} = \frac{3}{2} \quad \text{--- (2)}$$

From (1) & (2), we have $\int_C Mdx + Ndy = \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right] dx dy$

Hence Verified

13 Evaluate $\iiint_S [x dy dz + y dz dx + z dx dy]$ using divergence Theorem where S is the sphere $x^2 + y^2 + z^2 = a^2$

Sol By Gauss divergence Theorem,

$$\int_S \vec{F} \cdot \vec{n} ds = \int_V \text{div } \vec{F} dv$$

L.H.S can be written as $\int (F_1 dy dz + F_2 dz dx + F_3 dx dy)$ in Cartesian form.

Comparing with the given Expression

we have $F_1 = x, F_2 = y, F_3 = z$

$$\text{Then } \text{div } \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$= 1 + 1 + 1$$

$$= 3$$

$$\therefore \int_V \text{div } \vec{F} dv = \int_V 3 dv = 3V = 3 \cdot \frac{4}{3} \pi a^3 = 4\pi a^3$$

where V is the volume of the sphere with 'a'

$$\text{Hence } \iiint_S [x dy dz + y dz dx + z dx dy] = 4\pi a^3$$

14. If $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4z\vec{k}$ Then Evaluate $\int \nabla \cdot \vec{F} dv$ where V is the closed region bounded by $x=0, y=0, z=0, 2x+2y+z=4$

Sol Given $\vec{F} = (2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}$

$$\nabla \cdot \vec{F} = \left[\frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right] \cdot [(2x^2 - 3z)\vec{i} - 2xy\vec{j} - 4x\vec{k}]$$

$$= \frac{\partial}{\partial x} (2x^2 - 3z) + \frac{\partial}{\partial y} (-2xy) + \frac{\partial}{\partial z} (-4x)$$

$$= 4x - 2z$$

$$\nabla \cdot \vec{F} = 2x$$

The limits are:

$$z = 0 \text{ to } 4 - 2x - 2y$$

$$y = 0 \text{ to } \frac{4-2x}{2} = 2-x.$$

$$\text{and } x = 0 \text{ to } \frac{4}{2} = 2$$

$$\therefore \int_V \nabla \cdot \vec{F} dv = \int_{x=0}^2 \int_{y=0}^{2-x} \int_{z=0}^{4-2x-2y} 2x dz dy dx$$

$$= \int_{x=0}^2 \int_{y=0}^{2-x} (2x)(z)_0^{4-2x-2y} dy dx$$

$$= 4 \int_{x=0}^2 \int_{y=0}^{2-x} (2x - x^2 - xy) dy dx = 4 \int_{x=0}^2 \left[2xy - x^2y - \frac{xy^2}{2} \right]_0^{2-x} dx$$

$$= 4 \int_0^2 \left[(2x - x^2)(2-x) - \frac{x}{2}(2-x)^2 \right] dx$$

$$= 4 \int_0^2 \frac{2-x}{2} [(2x - x^2) - (2x - x^2)] dx$$

$$= 2 \int_0^2 (x^3 - 4x^2 + 4x) dx$$

$$= 2 \left[\frac{x^4}{4} - 4\frac{x^3}{3} + 4\frac{x^2}{2} \right]_0^2$$

$$= \left[\frac{x^4}{2} - \frac{8x^3}{3} + 4x^2 \right]_0^2$$

$$= -8$$

$$\therefore \int_V \nabla \cdot \vec{F} dv = -8$$

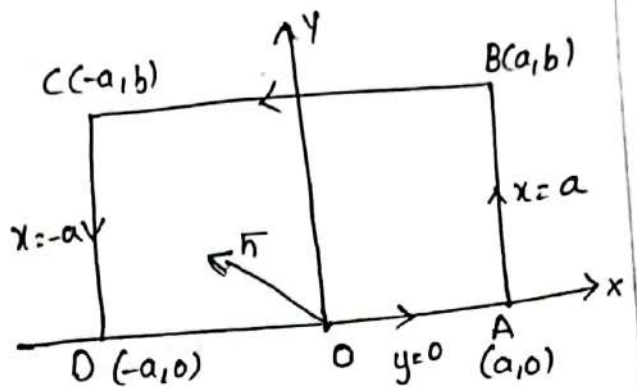
15. Verify Stoke's theorem for $\vec{f} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken around the rectangle bounded by the lines $x = \pm a, y = 0, y = b$.

Sol Let ABCD be the rectangle whose vertices are

$(a, 0), (a, b), (-a, b)$ and $(-a, 0)$

Equations of AB, BC, CD and DA are

$x = a, y = b, x = -a$ and $y = 0$



We have to prove that $\oint_C \vec{F} \cdot d\vec{r} = \int_C \text{curl } \vec{F} \cdot \vec{n} \, ds$

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C \{ (x^2 + y^2)\vec{i} - 2xy\vec{j} \} \cdot \{ \vec{i}dx + \vec{j}dy \}$$

$$= \oint_C (x^2 + y^2)dx - 2xydy$$

$$= \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA} \quad \text{--- (1)}$$

(i) Along AB, $x = a, dx = 0$

$$\text{From (1)} \int_{AB} = \int_{y=0}^b -2ay \, dy = -2a \left[\frac{y^2}{2} \right]_0^b = -ab^2$$

(ii) Along BC, $y = b, dy = 0$

$$\text{From (1)} \int_{BC} = \int_{x=a}^{-a} (x^2 + b^2)dx = \left[\frac{x^3}{3} + b^2x \right]_{x=a}^{-a} = -\frac{2a^3}{3} - 2ab^2$$

(iii) Along CD, $x = -a$, $dx = 0$

$$\text{From (1), } \int_{CD} = \int_{y=b}^0 2axy \, dy = a[y^2]_b^0 = -ab^2$$

(iv) Along DA, $y = 0$, $dy = 0$

$$\text{From (1) } \int_{DA} = \int_{x=-a}^a x^2 \, dx = \left[\frac{x^3}{3} \right]_{-a}^a = \frac{2a^3}{3}$$

(i) + (ii) + (iii) + (iv) gives

$$\oint_C \vec{F} \cdot d\vec{r} = -ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 + \frac{2a^3}{3} = -4ab^2 \quad \text{--- (2)}$$

Now consider $\int_S \text{curl } \vec{F} \cdot \vec{n} \, ds$

Vector perpendicular to the xy -plane is $\vec{n} = \vec{k}$

$$\therefore \text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2+y^2) & -2xy & 0 \end{vmatrix}$$

Since the rectangle lies in the xy -plane

$\therefore \vec{n} = \vec{k}$ and $ds = dxdy$

$$\begin{aligned} \therefore \int_S \text{curl } \vec{F} \cdot \vec{n} \, ds &= \int_S -4y \vec{k} \cdot \vec{k} \, dxdy = \int_{x=-a}^a \int_{y=0}^b -4y \, dxdy \\ &= \int_{y=0}^b \int_{x=-a}^a -4y \, dxdy = -4 \int_{y=0}^b y [x]_{-a}^a \, dy \\ &= -4 \int_{y=0}^b 2ay \, dy = -4a[y^2]_0^b = -4ab^2 \quad \text{--- (3)} \end{aligned}$$

Hence from (2) & (3), Stoke's Theorem is Verified